

INFORMATION AND COMMUNICATION TECHNOLOGY

S1

STUDENT BOOK

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FOREWORD

Dear Student,

Rwanda Basic Education Board is honored to present to you this ICT Student's Book for Senior One which serves as a guide to competence-based teaching and learning to ensure consistency and coherence in the learning of ICT. The Rwandan educational philosophy is to ensure that you achieve full potential at every level of education which will prepare you to be well integrated in society and exploit employment opportunities.

The government of Rwanda emphasizes the importance of aligning teaching and learning materials with the syllabus to facilitate your learning process. Many factors influence what you learn, how well you learn and the competences you acquire. Those factors include the instructional materials available among others. In this book, special attention was paid to the activities that facilitate the learning process in which you can develop your ideas and make new discoveries during concrete activities carried out individually or with peers.

In competence-based curriculum, learning is considered as a process of active building by the learner where concepts are mainly introduced by an activity, a situation or a scenario that helps the learner to construct knowledge, develop skills and acquire positive attitudes and values.

For effective use of this textbook, your role is to:

- Work on given activities which lead to the development of knowledge and skills;
- Share relevant information with other learners through presentations, discussions, group work and other active learning techniques such as field visit, laboratory work, investigation, role play, case studies and research in the library, from the Internet or from your community;
- Participate and take responsibility for your own learning;
- Draw conclusions based on the findings from the learning activities.

I wish to sincerely extend my appreciation to REB staff who organized the editing process of this book. Special gratitude goes to the lecturers, teachers, illustrators and designers who diligently worked to successful completion of this book. Any comment or contribution would be welcome for the improvement of this textbook for the next edition.

Dr. MBARUSHIMANA Nelson
Director General, REB



ACKNOWLEDGEMENT

I wish to express my appreciation to all the people who played a major role in editing process of ICT Student's Book for Senior One. It would not have been successful without their active participation.

Special thanks are given to those who gave their time to read and refine this textbook to meet the needs of competence-based curriculum. I owe gratitude to different Universities and secondary schools in Rwanda that allowed their staff to work with REB to edit this book. I therefore, wish to extend my sincere gratitude to lecturers, teachers, illustrators, designers and all other individuals whose efforts in one way or the other contributed to the completion of this edition.

Finally, my word of gratitude goes to the Rwanda Basic Education Board staff particularly those from Curriculum, Teaching and Learning Resources Department who were involved in the whole process of editorial work.



Joan MURUNGI,

Head of Department of Curriculum, Teaching and Learning Resources

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UNIT 1: ICT FOUNDATION



Key Unit Competence

Describe common ICT equipment and the role of ICT in various sectors

Learning objectives

At the end of this unit, you should be able to:

Knowledge and understanding

- Identify ways of communication.
- Mention different types of technologies.
- List ICT tools and their use in daily life.
- Explain the importance of ICT
- Describe the use of ICT in different sectors.

Skills

- Use ICT related terms appropriately.
- Classify types of communication and technologies using real-life cases.
- Demonstrate simple ICT use in everyday contexts
- Use ICT equipment, namely tablet, projector and computers in their relevant context

Attitudes and values

- Value accuracy and reliability of information
- Appreciate the importance of sharing knowledge with peers during ICT-related activities.
- Show curiosity and interest in learning and using ICT tools.
- Appreciate the role played by ICT in the evolution of society.

1.0. Introduction

The purpose of this unit is to help learners understand the basic concepts of Information and Communication Technology (ICT), it enables learners to identify and differentiate data and information, understand how communication works, recognize different types of technology, and become familiar with common ICT equipment. It also aims to show the importance, advantages, and disadvantages of ICT in daily life and in key sectors such as education, health, business, and communication, with examples from Rwanda.

Introductory activity

Read the scenario below and answer the questions:

SHEMA and his mother visited Nyanza Market. His mother checked mobile money messages on her smartphone while the seller used a billing machine to print tax receipts. Nearby, a bank agent used a POS device for cash withdrawals, and a truck driver used a GPS app for directions. At home, they watched a TV news report on how ICT provides weather updates to farmers.

- 1) List three different electronic devices used by the people in the scenario.
- 2) In the context of the truck driver using GPS, identify the sender and the receiver of the directions.
- 3) Name two specific sectors that benefited from technology in this scenario.
- 4) What is the main function of the billing machine and what is one safety precaution SHEMA's mother should take when using mobile money?

1.1. Introduction to ICT concepts

Activity 1.1

Your class is choosing a new class monitor. Students must vote for their preferred candidate. The teacher wants to collect and process the votes fairly.

- 1) Identify the raw data that need to be collected.
- 2) State whether this data is analog or digital.
- 3) Describe how to turn the votes into clear information that shows the winner.
- 4) Mention one factor affecting the quality of information.

1.1.1. Data and information

a. Data

Data can be defined as raw facts that have been entered into a computer but have not been processed. It refers to statistics, symbols, basic facts, figures, numbers or simply records. Note that the term **data** is the plural form of **datum**. However in modern usage, the term data is used both as a singular and plural term. Examples of data include:

- Number of students in a class
- Number of ICT textbooks for Senior One ordered by a school
- Number of computers in a school computer laboratory.

There are two main types of data in a computer. They include analog data and digital data.

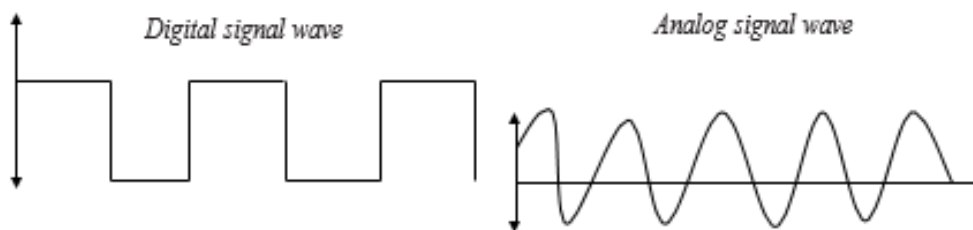
- i. **Analog data:** It is in continuous form. It is processed using computers which process the continuously changing variable quantities. Analog data can be

represented as a wave form. Examples of analog data include temperature, pressure and humidity.

ii. **Digital data:** It is the kind of data that does not vary continuously and is discrete.

The Possible values (states) are only two: 0 or 1; on or off; and true or false.

Digital data is commonly found in electronic devices like computers and cannot be viewed or directly understood by humans.



Picture 1. 1: Analog and digital waves



Picture 1. 2: Examples of devices that measure analog data



Picture 1. 3: Examples of devices that measure digital data

b. Methods of data gathering

This is the systematic collection of data from various sources for a specific purpose.

Examples of data collection methods are: observation, interviews, questionnaires, data logging, direct data capture, studying existing documents, and use of other technologies.

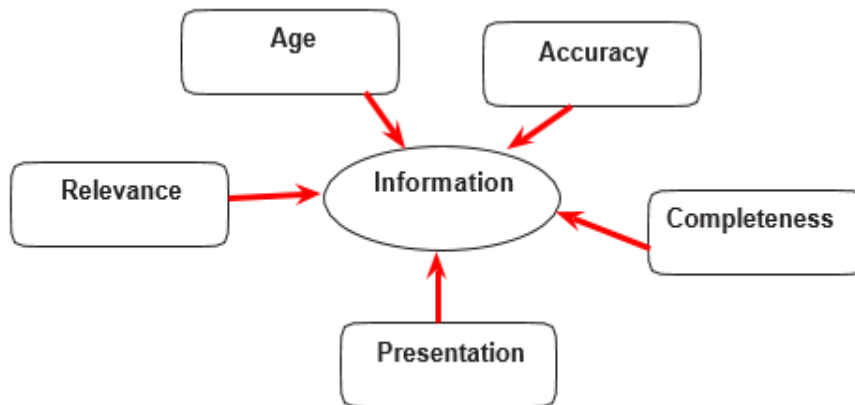
- **Observation:** This is the careful study and recording of data as it occurs in the environment.
- **Interviews:** This is gathering of data by asking questions orally. The person asking the questions is the interviewer. The respondent is the interviewee.
- **Questionnaires:** This involves the use of a set of written questions.
- **Data logging:** This is the automatic collection of data over a period of time using sensors connected to a computer.
- **Direct data capture:** This involves the use of devices such as Optical Mark Recognition\Reader (OMR), Optical Card Recognition\Reader (OCR) and Optical Bar Recognition\ Reader (OBR) among others.
- **Study of existing documents:** This involves going through documents such as reports, journals, articles, and records of previous interviews.

c. Information

Information can be defined as processed facts or data that is meaningful and useful to the user.

There are many factors affecting the quality of information. They includes:

- **Accuracy:** This is the measure of how close information is to the original data. For example, take the number 5.263 as the original data. If this number is recorded as 5.26, it will be more accurate than if it is rounded off and recorded as 5.3.
- **Relevance:** Information is considered to be relevant if it can be used for what it was intended. Information that is not accurate or timely is therefore not relevant. Relevant information should therefore be appropriate to the user needs. For example, an academic assignment that is received late may not be relevant because it may not be used to process the average mark for the student.
- **Timeliness/Age:** Information changes over time. This means that out of date information can be irrelevant; therefore information needs to be up-to-date to avoid cases of misleading or giving the wrong picture of what is happening.
- **Completeness:** Information should be comprehensive. Partial information may only represent a small part of the whole picture. If part of the information is missing, then one will not be able to make use of it, or make accurate decisions.
- **Presentation:** Information should be presented in an organized manner to make it more useful. Presenting information using the best method, possibly by sorting or organizing it makes it much easier to understand and to be more useful.



Picture 1. 4: Factors affecting the quality of information

1.1.2. Communication

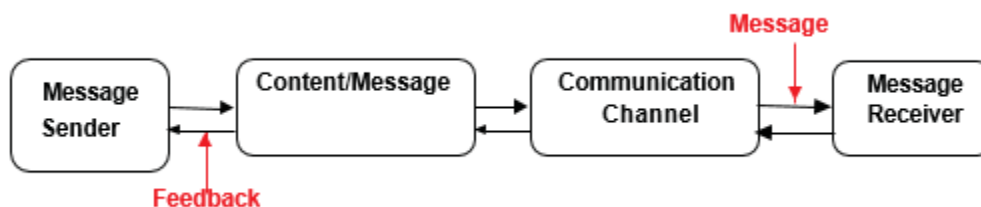
Communication refers to the exchange of information. In computing, it refers to the process of transmitting information or data signals from the source (**sender**) to the destination (**receiver**) through a transmission channel that use various technologies.

A. Elements of communication

Communication involves the following elements.

- (i) **Source /Sender:** This refers to the communicator or originator of the message. Sometimes, the sender becomes the receiver of the message.
- (ii) **Content:** This refers to the message being communicated. It may be an attitude, opinion, views, orders, or suggestions.
- (iii) **Communication channel:** This refers to the medium of transmission used to deliver the message.
- (iv) **Receiver:** This is the person who receives the message.
- (v) **Feedback:** This is the reaction given after the message is received. It is the output from the receiver in order to improve performance by providing self-corrective action. Feedback gives the communication progress. It can be negative or positive.

The figure below shows how communication takes place between the message sender and the message receiver.



Picture 1. 5: The elements of communication

B. Types of communicating

There are two basic types of communication which include verbal and non-verbal communication.

- i. **Verbal communication:** This is the use of spoken words to send a message. It can also be oral or written. It involves both speaking and listening.
- ii. **Non-verbal communication:** It includes the use of body language, facial expressions, gestures, and posture. Non-verbal responses can sometimes contradict verbal communication. This changes the effectiveness of the message.

1.1.3. Technology

Technology refers to the application of scientific knowledge to the practical purposes in the human environment. Technology is used in industries and learning institutions among others. Technology includes the use of materials, tools, techniques, and sources of power.

Some types of technology are technology of teaching, assistive technology, medical technology and Information Technology.

A. Technology of teaching

Systematic instructional approaches are carefully planned teaching methods with clear objectives, structured lessons, active teacher involvement, high student participation, and close monitoring of performance.

B. Assistive technology

Assistive technology refers to services, devices, and special materials that help people with disabilities perform tasks and function better in their environment. It helps people with disabilities work and compete their work successfully.

C. Medical technology

It is the use of scientific knowledge in healthcare to diagnose, monitor and treat diseases or medical conditions. For example, artificial limbs, knee and hip implants can help people to function.

D. Information Technology

Information technology (IT) is the use of computers, storage devices, networking devices, and other electronic devices to create, process, store, secure, and exchange all forms of data. Electronic gadgets such as a washing machine and a microwave oven, among other examples, use information technology to operate.

Application Activity 1.1

- 1) Classify each item as either *analog data* or *digital data*.
 - a) 25°C
 - b) ON/OFF
 - c) Humidity level
- 2) Your school wants to know how students use computers in the library. Suggest two suitable methods of data collection and briefly explain how each method works.
- 3) The school management received some information about student computer usage, but

some of it was incomplete, old, and not well presented. Identify three factors of information quality that have been violated.

4) Look at the communication process below:

Student → Ask a question → Teacher → gives the answer → Student nods

Identify the following elements:

- a) Sender: _____
- b) Receiver: _____
- c) Message/content: _____
- d) Feedback: _____
- e) Communication channel: _____

5) List four types of technology and give one example for each.

1.2. Information and Communication Technology

Activity 1.2

1) Answer with True or False

- a) ICT stands for Information and Communication Technology.
 - b) Mobile phones can be used to withdraw money.
 - c) Tablets cannot be used for learning.
 - d) Computers are used only for games.
 - e) Projectors help teachers show lessons to many students at once.
- 2) Give three examples of ICT devices and their uses.
- 3) Explain how ICT helps the school run smoothly.

1.2.1. Definition of ICT

ICT is the abbreviation for Information and Communication Technology. ICT refers to the application of information technology and telecommunication that includes the use of computer hardware, software, and telecommunications equipment for transmission and presentation of information.

1.2.2. ICT equipment

The following are some of the equipment used in ICT

A. Mobile phones

Mobile phones, also called cellular phones, are electronic devices that connect to wireless networks using radio waves or satellites. They allow communication through voice calls, SMS, and MMS. A smartphone is an advanced mobile phone that performs many computer functions using a touch screen, and includes features such as GPS, camera, internet browsing, email, gaming, and content downloading, with connectivity through Bluetooth and Wi-Fi.



Picture 1. 6. Smart phones

B. Computer

Computers have diverse uses across almost every sector, including communication (email, video calls), education (online learning, research), business (data management, analytics), entertainment (gaming, movies, music), and healthcare (patient records, diagnostics).

C. Billing machines

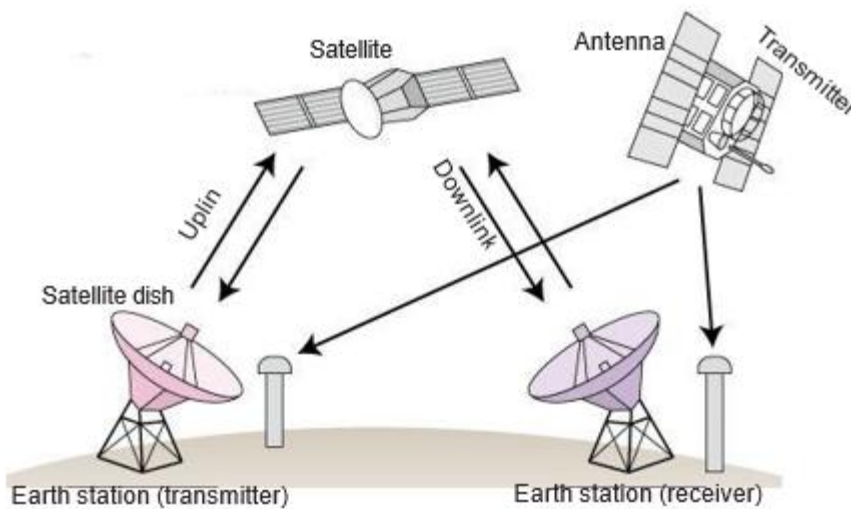
A billing machine is used in business to itemize, generate bills, provide customer invoices, and send account records.



Picture 1. 7: examples of billing machines

D. Satellite

A satellite is a self-contained communications system. It revolves around a planet in a circular or elliptical path. It has the ability to receive signals from the Earth and to retransmit those signals back. An example of an earth station that communicates with orbiting satellites is Very Small Aperture Terminal (VSAT).



Picture 1. 8. How a satellite communication system works

E. Projector

It is an output device that uses light and lenses to take images generated by a computer and reproduce them on a larger flat screen or wall (usually lightly coloured). For example, projectors are used in meetings to help ensure that all participants can view the information being presented.



Picture 1. 9: Projector connected to a laptop

Application Activity 1.2

1) Identify the ICT equipment described below:

- a) A device that receives radio signals and converts them into sound is: _____
- b) A device that displays moving images and sound from broadcasting stations is: _____
- c) A device used to make calls, send SMS, browse the Internet, and take photos is: _____
- d) An electronic device used for typing documents, searching information, and sending emails is: _____

2) Match each ICT equipment with its correct use:

ICT Equipment	Use
a) Radio	1) Used to project images from a computer to a large screen
b) Television	2) Used to relay news, educational programs, and advertisements
c) Projector	3) Used to make voice calls and send messages
d) Mobile phone	4) Used to receive audio broadcasts from stations

3) State the steps you would follow to interact with a television when watching your favourite program.

4) Answer by True or False.

- Mobile phones can support SMS, calls, internet browsing, and photography.
- A smartphone cannot connect to Wi-Fi.

1.3. Importance of ICT

Activity 1.3

- Write two sentences about the benefits of ICT in education.
- Explain how ICT improves business and communication in Rwanda.
- KAYITESI lives in a rural area in Rwanda. She studies online courses from the University of Rwanda, consults doctors via video calls, shops online using her phone, and receives alerts about market prices for her crops.
 - List three uses of ICT in KAYITESI's daily life.
 - Explain how ICT makes learning easier for Kayitesi.
 - Identify one challenge she may face using ICT.
 - How does ICT help her make better decisions about her crops?

1.3.1. General use of ICT

A. Education

ICT has facilitated education through Computer Aided Learning (CAL), Computer Aided Instruction (CAI), Computer Aided Assessment (CAA), E-learning and computer-based simulation among others.

- Computer Aided Learning (CAL):** This is the integration of ICT into the learning process. It is also known as Computer Based Training (CBT). It involves the use of computers in education and training.
- Computer Aided Instruction (CAI):** CAI is used to develop communication and language skills. It includes computer modelling and computer tutors, for example, the typing tutor.
- Computer Aided Assessment (CAA):** It is a computerized innovation used in marking multiple choice examination papers where a box is marked in pencil to indicate the candidate's preferred answer. An Optical Mark Reader (OMR) can identify the box having a mark and award a score appropriately.
- E-learning:** It is the delivery of educational programs, training or learning materials by electronic means. It involves the use of electronic devices such as

mobile phones, tablets, computers and the internet.

- **Computer simulation:** This is the modelling of a real situation using a computer. It is used to model industrial processes enabling better understanding and improvement of production systems. Simulation can be used to train workers such as pilots. Simulation is also applied in the learning of mathematics and biological sciences. In the latter, it is used to simulate body systems such as the digestive, respiratory, and circulatory systems.

B. Health

The following are some of the uses of ICT in healthcare:

- ICT is used in the medical sector to keep records of patients.
- In the Intensive Care Unit (ICU), ICT is used to control life support machines.
- Patients also receive medical diagnosis and advice from foreign medical experts through the use computerized expert systems.
- Knowledge enrichment, research, and education: Many doctors, pharmacists and medical practitioners use the internet to keep themselves updated. They do research on specific questions and areas of interest.
- ICT-based equipment such as the Magnetic Resonance Imaging (MRI), X-Rays and CT-Scan (Computed Tomography Scan) are used to diagnose complicated medical conditions.

C. Business

Business refers to commercial activities carried out for profit. Electronic business (e-business) is the application of ICT in support of business activities. Examples of ICT-based business activities include online shopping, advertising and marketing and use of electronic billboards:

D. Communication

ICT is used to transmit textual, visual and audio information from one point to another through the use of telecommunication equipment at a very high speed. This is achieved through the use of electronic mails (e-mails), live chatting systems, online meetings (via Skype), short messages services (SMS), facsimile (fax), and video-conferencing among others.

E. Banking

ICT is used in banking where transactions, internet banking, mobile banking, use of Automated Teller Machine (ATM) are made possible. Customers are also able to access their bank details through SMS alerts.

F. Manufacturing and production

ICT-based systems such as Computer Integrated Manufacturing (CIM), Computer Aided Design (CAD) and Computer Aided Engineering (CAE) are used to monitor and control production processes in factories.

1.3.2. Current use of ICT in Rwanda

a. ICT use in education

- **Learning in schools:** In schools, learners use computers to learn different subjects, do assignments and present using a computer and a projector. Also the distribution of computer in schools enable students to use computers for learning without needing hard copy textbooks.
- **Distance Learning:** ICT supports distance education whereby learners use their computer and internet to follow a lesson being delivered by someone far away. They can also do assignments and submit.

b. ICT use in the health sector

- **Record keeping:** There are specialized software that are developed to support the management of patient information. Systems like OpenMRS and TRACnet help manage and track that information.
- **Mobile e-Health systems:** Community health workers use ICT tools to collect and manage health data .
- **Telemedicine:** ICT allows hospitals to share medical information and offer specialized treatment remotely.

c. ICT use in business

- **Mobile banking and financial services:** Mobile service providers have introduced money transfer and other financial services.
- **E-business:** ICT is a great medium of transacting businesses. Online payments, advertising, and banking allow businesses to operate locally and internationally
- **Market survey:** Rwandans have real time updates about the Stock Exchange daily market status. This enables them to make informed decisions about investments.
- **E-Soko:** This project enables farmers to make informed market pricing decisions on their produces.

d. Communication

- **Video conferencing:** This refers to communication between individuals in different locations by use of simultaneous two-way audio and video transmissions. The users are able to communicate with each other as though they were in the same room. In Rwanda, video conferencing is used in holding meetings, facilitating distance learning and interviewing employees remotely.
- **Television and video broadcasting:** ICT enables live reporting and global interviews.
- **E-mail:** Electronic mail is used for fast, affordable communication with attachments. It has enabled people to communicate with others outside Rwanda. Examples of e-mail software include Yahoo! Mail, gmail, and hotmail.

d. Internet

The following are some of the application areas of the Internet:

- **Research:** Internet has a lot of information which can be used for research materials. This information can be accessed using search engines such as google or AI tools such as ChatGPT.
- **Online books:** internet has enabled the availability of books online. A book can be accessed easily by many readers. For example learners can access books from the REB elearning platform (elearning.reb.rw)
- **Socializing:** The internet enables access to social media platforms such as whatsapp, youtube, etc.

e. Service delivery

ICT enables online public services through platforms like **www.irembo.gov.rw**, allowing access to services such as birth certificates, driving licences, and criminal record checks.

1.3.3. Advantages and disadvantages of ICT

a. Advantages of ICT

- **Access to information and learning:** ICT provides easy access information online. Students, teachers, and professionals can use the internet, digital libraries, and educational software to learn new skills, conduct research, and stay updated with the latest knowledge.
- **Improved communication:** ICT allows people to communicate quickly and easily across long distances.
- **Efficiency and productivity:** Computers, software, and automated systems make tasks faster and more accurate.
- **Supports innovation and creativity:** ICT tools like design software, multimedia applications, and programming platforms help people create new products, designs, and solutions.
- **Facilitates government services (e-government):** Governments can use ICT to provide services more effectively, such as online tax payment, e-health systems, and digital identification.

b. Disadvantages of ICT

- Content on internet is available to everyone while some of this is not appropriate.
- ICT and automation can lead to loss of jobs, especially for computer-illiterate persons.
- **Cybersecurity risks:** ICT systems are vulnerable to security threats such as hacking, viruses, and malware. Users may also face identity theft or online scams, leading to the loss of personal or sensitive information.
- **Social issues:** Excessive use of ICT can affect social interactions. People may communicate less face-to-face, become addicted to social media or online games, and experience isolation from family and friends. This can weaken relationships and reduce community engagement.

- Dependence on technology: Over reliance on ICT can be risky if systems fail due to power outages, network issues, or software problems.
- Environmental impact: ICT devices contribute to electronic waste (e-waste), which can be harmful if not disposed of properly.

Application Activity 1.3

- 1) Explain the meaning of Computer Aided Learning (CAL).
- 2) Give two advantages of using ICT in education.
- 3) Explain how ICT is used in the health sector.
- 4) Explain how ICT can help students improve their learning performance.
- 5) Suggest ways ICT can help farmers improve agricultural production.
- 6) Explain why responsible use of ICT is important in society.
- 7) Describe the importance of protecting personal information when using ICT.

End of unit assessment 1

- 1) Answer by True or False
 - a) Data refers to raw facts that have not been processed.
 - b) Information is the same as data.
 - c) Communication involves a sender, message, channel, receiver, and feedback.
 - d) Analog data is continuous, while digital data is discrete.
 - e) Computers, radios, and mobile phones are examples of ICT devices.
- 2) Which of the following is an example of non-verbal communication?

a) Speaking on the phone	b) Giving a speech
c) Facial expressions	d) Sending an email
- 3) Which of these is a type of technology?

a) Assistive technology	b) Handwriting
c) Chalkboard	d) Pen and paper
- 4) Give three examples of public services that are offered using online platforms.
- 5) Explain how mobile phones are used in business in Rwanda.
- 6) Explain how a responsible student should manage ICT tools.
- 7) ICT can expose users to harmful online content. Describe how you should behave responsibly while using the internet.

UNIT 2: COMPUTER SYSTEM



Key unit competence

Manipulate the main parts of the computer system

Learning Objectives

At the end of this unit, you should be able to:

Knowledge and understanding

- Explain the functions of a computer system
- Describe the history and evolution of computers
- Describe different types of external and internal hardware components
- Describe different steps for the installation of different types of software

Skills

- Differentiate between generations and types of computers.
- Connect internal and external hardware components correctly.
- Apply troubleshooting techniques for hardware and software problems.
- Follow safety procedures when handling computer parts and installing software.

Attitudes and values

- Demonstrate proper use of computer hardware and software.
- Show curiosity and interest in understanding how computer systems work.
- Respect safety rules and promote good maintenance culture for hardware and software.
- Value the use of legal and licensed software to promote digital integrity.

2.0. Introduction

This unit will help you understand how modern computers work, what components they are made of, and how they have changed over time. You will explore the main computer hardware and software components and how computers are classified into different families. The unit also covers basic troubleshooting, safe handling of hardware, and basic computer software concepts, including installation and common software issues.

Introductory activity

Read this scenario and answer the questions below:

One afternoon, SIMBI visited a community ICT room where people are learning computer skills. She notices one learner typing using a keyboard while the computer displays words on the monitor. Another learner is repairing a computer that shows no display, checking the RAM and cables. On another desk, a technician is installing new software on a computer using a flash disk drive. A poster on the wall shows pictures of old computers from the first generation to modern microcomputers, mainframes, and supercomputers. SIMBI also saw storage devices like HDDs and SSDs, and a chart showing how a computer performs input, processing, output and storage. She becomes curious about how computers work and how they have changed over time.

- 1) Identify two hardware devices SIMBI saw in the ICT room.
- 2) What computer system function is represented by typing on a keyboard?
- 3) Which function is about displaying words on a monitor?
- 4) Mention one storage device SIMBI observed.
- 5) What may be the reason why the computer had no display?
- 6) What is one example of software installation you saw in the scenario?

2.1.Functions of computer system

Activity 2.1

- 1) What device do you use to enter information into a computer?
- 2) When you type on a keyboard, what do you think happens inside the computer?
- 3) After you enter data, how does the computer give you the answer?
- 4) Where does a computer keep saved files like documents and pictures?
- 5) What happens when you calculate a sum using a computer?

A computer performs four main functions. These are known as the Input, Processing, Output, Storage (IPOS) cycle:

- **Input:** This is data entered into a computer for processing. A computer has input devices that are used for that purpose.
- **Processing:** This is the conversion of data into information. Once data is accepted, the computer takes it to the processing unit where it is converted into information based on the instructions given by the user.
- **Output:** It is the information produced after processing of data is complete. This information can be used as data in another stage or it could be the last result required.
- **Storage:** A computer can store a large amount of data in it. It has primary and secondary storage devices that are used for this purpose.

The IPOS cycle (Input, Process, Output, Storage) illustrates how computers work: Taking in raw **Input** (keyboard, mouse), the CPU **Processes** it into data, generates **Output** (screen, printer), and uses **Storage** (RAM, HDD) for temporary or long-term data keeping, repeating this flow continuously for any task, like calculating a sum or editing a document.

Data → (Input) → Processing → (Output) → Storage

Examples of each function

Function	Examples
Input	Typing on keyboard, clicking with mouse, scanning with scanner
Processing	CPU calculating results, opening apps
Output	Display on monitor, printed documents, listening using speakers
Storage	Saving files on HDD, SSD, or flash drive

Table 2. 1: Examples of each function

Application Activity 2.1

- 1) Open a simple application on a computer or phone.
- 2) Write a short explanation of how the IPOS cycle works when you use that app.

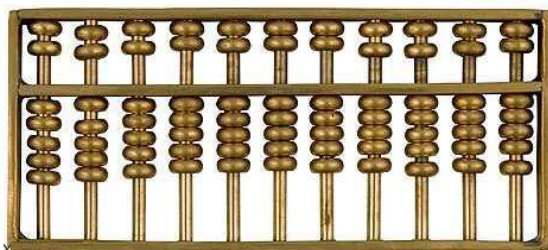
2.2. Evolution of computers

Activity 2.2

- 1) How did people calculate numbers before modern computers were invented?
- 2) Were the first computers small like today's laptops? Explain?
- 3) Why have computers become smaller and faster over time?
- 4) What is the difference between old computers and the computers we use today?
- 5) Name different types of computers you know?

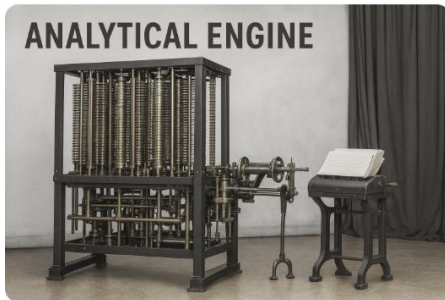
2.2.1. History of computers

Computers did not start as machines. Long ago, people used simple tools to help them calculate. The **abacus** was one of the first tools used for counting.



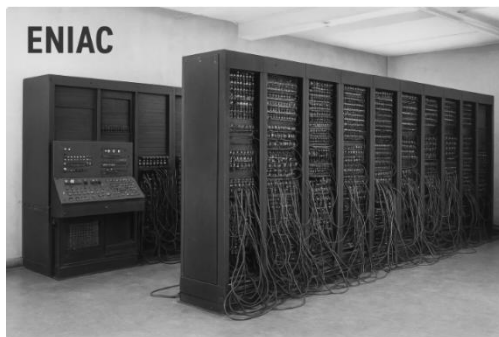
Picture 2. 1: Abacus

- In the 1800s, a man named **Charles Babbage** designed a machine called the *Analytical Engine*. It is known as the first idea of a real computer.



Picture 2. 2. The analytical engine could be programmed using punched cards

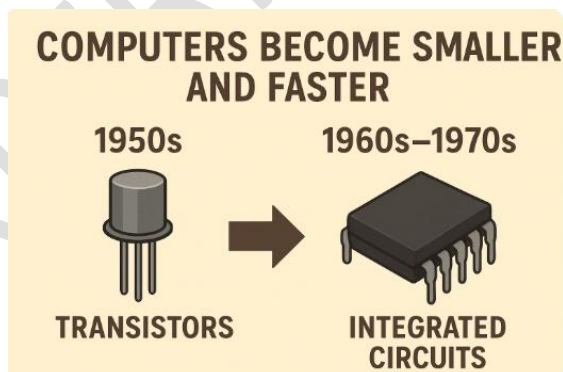
- In the 1940s, the first true computers were built. They were very big, used a lot of electricity and had *vacuum tubes*. One of them was called **ENIAC**.



Picture 2. 3. The ENIAC computer

Later, computers became smaller and faster:

- In the 1950s, **transistors** made computers smaller.
- In the 1960s and 1970s, **integrated circuits (chips)** made them even smaller and more powerful.



Picture 2. 4. The computers with transistors and integrated circuits

In the 1980s, personal computers were created. Companies like Apple and IBM made computers that people could use at home and in offices.

Today, computers are everywhere. We use laptops, tablets, and smartphones, which are very powerful and much smaller than old computers.

2.2.2. Generation of computers

a. First generation computers (1940's to mid-1950's)

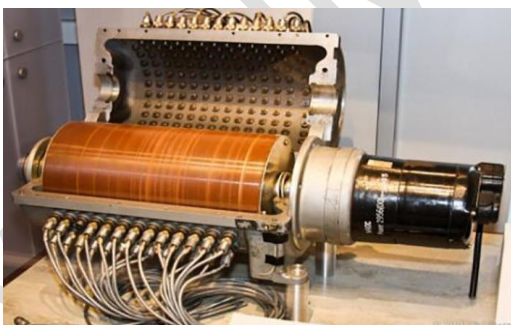
The following are the characteristics of the first generation computers:

- They used vacuum tubes or **thermionic valves** technology for amplification and as switching devices. A thermionic valve is an electronic valve in which electrons are emitted from a heated cathode.

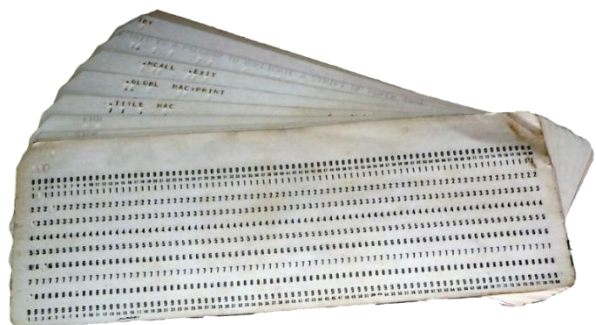


Picture 2. 5. Vacuum tubes (thermionic valves)

- These computers used the machine language for programming they used **magnetic drums** for internal memory. **Punched cards** were used for data input and the output was given in form of printouts.



Picture 2. 6. Magnetic drum



Picture 2. 7. Punched cards were used for data input

Some examples of computers in the First Generation are: Electronic Discrete Variable Automatic Computer (EDVAC), Universal Automatic Computer (UNIVAC), Electronic Numerical Integrator and Calculator (ENIAC) and Mark 1.

i. Disadvantages of the first generation computers

- The vacuum tubes were huge, therefore they made the computers heavy and bulky.
- They consumed a lot of electrical power.
- They generated a lot of heat. This resulted in frequent breakdowns.

b. Second generation computers (mid 1950's to early 1960's)

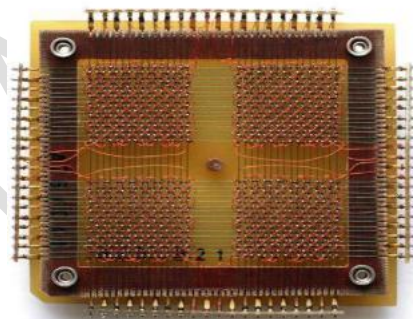
The following are the characteristics of the Second Generation computers:

- They used **transistor technology**. A transistor is a semiconductor device with three terminals. It allows amplification and switching of signals.
- They used the magnetic core for internal memory and magnetic tape was introduced as a secondary memory.
- The assembly language was used for programming.
- They used punched cards for input of data and printouts were used for output of information.

Some examples of computers in the Second Generation are: Honeywell 1800, IBM140 and IBM 709.



Picture 2. 9. Transistor



Picture 2. 10. The Magnetic core memory

i. Advantages of Second Generation computers

Compared to the first generation computers:

- They were smaller due to the use of transistors;
- They produced less heat; and
- They were less prone to hardware failure.

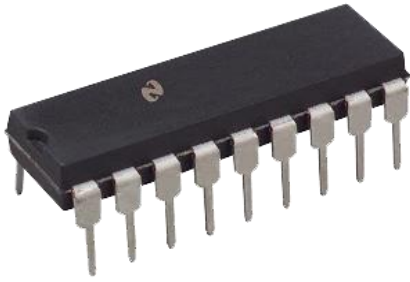
ii. Disadvantages of Second Generation computers

- They still produced a lot of heat. Therefore, they required a special cooling system.
- They required frequent maintenance since they easily broke down.

c. Third generation computers (early 1960's to early 1970's)

The following are the characteristics of the Third Generation computers:

- They used **Integrated Circuits (IC's)** technology.



Picture 2. 11. Integrated Circuits (ICs)

- Keyboards were used for data input while monitors were used for data output.
- Common Business Oriented Language (COBOL) and Formula Translator (FOTRAN) programming languages were used for programming.

Some examples of computers in this generation are Burroughs B6500, Honeywell-115 and IBM 360.

i. Advantages of Third Generation computers

- They were smaller and less expensive than the previous two generations.
- They had a higher processing speed and storage capacity compared to the previous generations.
- They consumed lesser electrical power compared to the previous generations.

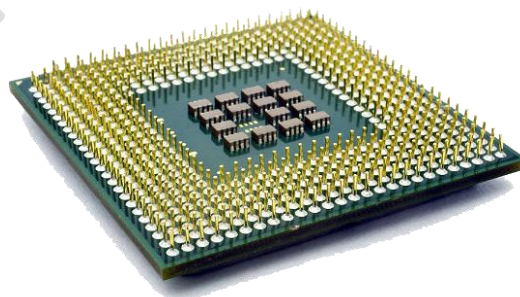
ii. Disadvantages of Third Generation computers

- The initial cost was high due to the sophisticated technology required in the development of Integrated Circuits (ICs).
- Air conditioning was required as they still produced a lot of heat.

d. Fourth generation computers (early 1970's to late 1980's)

The following are the characteristics of the Fourth Generation computers:

- They used Large Scale Integration (LSI) and Very Large Scale Integrated (VLSI) Circuits technology commonly known as **Microprocessor**.



Picture 2. 14. Microprocessor

- Optical disks, magnetic disks, and magnetic tapes were used as secondary memory and virtual memory was introduced. Virtual memory refers to the use of part of the hard disk as main memory for storing tasks held in the processor.
- Fourth Generation computers include IBM 370, Macintosh, Apple II and Honeywell DPS-88.

i. Advantages of the fourth generation computers

- They are smaller because of high level of circuit integration making them more portable.
- They are cheaper since they are readily available in the market.
- They have very high processing speeds compared to the previous generations.

ii. Advantages of the fourth generation computers

- High cost: Expensive to buy and maintain.
- Limited memory: Could not store large programs or data.
- Heat generation: Chips got very hot and needed cooling.
- Software dependence: Needed complex programs to work properly.

e. Fifth generation computers (Early 1990's to present)

The following are the characteristics of the fifth generation computers:

- They use parallel architecture and super conductor technology.
- They use Artificial Intelligence (AI). This is the science of developing machines that mimic human behaviour such as reasoning, learning, hearing, seeing, and communicating.

Their goal is to have a machine that can use and process natural language, respond to inputs, learn about its surrounding, and organize itself to adapt to the environment. These computers are still under development and further advancement is still going on.

Generation	Technology	Characteristics
1 st Gen (1940–1956)	Vacuum tubes	Very large, produced heat, slow
2 nd Gen (1956–1963)	Transistors	Smaller, faster, more reliable
3 rd Gen (1964–1971)	Integrated Circuits (ICs)	Powerful, cheaper, smaller
4 th Gen (1971–Present)	Microprocessors	Very fast, small, widely used in pcs
5 th Gen (Present–Future)	AI & advanced processors	Smart systems, robotics, machine learning

Table 2. 2: Computer generations based on their use of technologies

2.2.3. Computer family

The term computer family refers to category of computers with the same operational design and microprocessors that are compatible with each other. Examples of computer families are the IBM or PC versus the Apple or the Mac family. Computers in these families can be classified as **Microcomputers**, **Minicomputers**, **Mainframe** and **Supercomputers**.

a. Microcomputers

They are popularly known as Personal Computers (PCs). They are designed to be used by one person at a time. They are the smallest, cheapest, and most portable. They are commonly used in offices, for training, entertainment, communication, business enterprises, and in homes among other places.

Examples of microcomputers are desktops, laptops, tablets, and Personal Digital Assistants (PDAs) or palmtops.

- **Desktop computers:** These are computers designed to be used while placed on top of a desk or a table. They cannot be powered from an internal battery and therefore must remain connected to a wall power outlet. In Rwanda, these are commonly used in offices, schools, and in businesses among other areas
- **Laptop computers:** These are portable computers. They have all input/output, processing and memory devices inside the system unit. This leaves only a few external accessories to be connected to the ports.

These computers are designed to be used while placed on the user's lap. These are commonly used in offices. The XO laptops are also used in schools with the introduction of One Laptop Per Child Program in Rwanda



Picture 2. 15. Laptop computer

- **Tablets:** These are small, portable computing devices, similar to a notebook. They are primarily used for internet access and email. They come with limited features and are relatively cheaper and much lighter compared to laptops..
- **Palmtops:** They are also referred to as **handheld computers**, ors. They are designed to fit in the palm of the hand and easily fit in a pocket. Due to their small size, they do not have disk drives. However, many contain Personal Computer Memory Card International Association (PCMCIA) slots for expansion of memory and insertion of disk drives and modems among others. They are commonly used in communication.



Picture 2. 16. Examples of microcomputers

b. Minicomputers

They are also referred to as **mid-range**

Computers. They are small in size and cheaper.

They have a low processing speed compared to super and mainframe computers.

Their storage capacity is higher than that of microcomputers.

They are popularly used to computerize processes in manufacturing industries and production lines as well as in autopilots in aeroplanes.



Picture 2. 17. Minicomputer

c. Mainframe computers

They are the second largest in physical size. They are very large and expensive. They can support thousands of users in a computer network simultaneously.

They mostly use operating systems which are based on UNIX and Linux.

Mainframes are used for centralized processing, for example, in large commercial organizations such as government agencies, banks, hospitals, airlines, manufacturing companies, and telecommunication agencies among others.



Picture 2. 18. A mainframe computer

d. Supercomputers

They are the largest in size, fastest, most powerful and most expensive computers. Supercomputers are used to process large amounts of data at very high speed and they solve problems requiring complex calculations within a fraction of a second.

Due to their large size and weight, they are not portable and are housed in a special room that has special cooling system. These computers are mainly applied in scientific research work, for example, in weather forecasting, advanced military applications, and telecommunication companies among others.



picture 2. 19: A supercomputer

Here is a summary of computers classification according to size, power, and purpose:

Family	Description	Examples
Microcomputers	Small, used at home and school	Laptops, desktops
Minicomputers	Medium-sized, used by institutions	Department servers
Mainframe computers	Large, powerful, support many users	Banks, government data centers
Supercomputers	Extremely powerful, fastest computers	Weather forecasting, scientific research

Table 2. 3: Computers according to size, power, and purpose

Application activity 2.2

- 1) Why was the invention of transistors important in computer evolution?
- 2) How are today's computers different from the first generation computers?
- 3) Give two examples of microcomputers.
- 4) Mention two areas where supercomputers are used.
- 5) Which computer family does a tablet belong to?
- 6) Complete the statements:
 - a) The analytical engine was designed by _____.
 - b) The first generation used _____ tube.
 - c) Transistors are used in the _____ generation.
 - d) Microprocessor were introduced in the _____ generation.
 - e) _____ are the most powerful computers.

2.3. Computer hardware

2.3.1. External hardware

Activity 2.3.1

- 1) a) You are setting up a new computer lab. Which input and output devices would you need?
b) Why would you need those devices?
- 2) What is the difference between a port and a connector?
- 3) What is the difference between internal and external hardware components?
- 4) Give two examples of external hardware.

Computer hardware refers to the physical parts of a computer that you can see and touch. It is divided into two parts namely external hardware and internal hardware.

a. Input devices

These are hardware components that are used to enter data and instructions into a computer. They are classified into keying devices, pointing devices, scanning devices and voice input devices

i. Keying devices

Those are input devices which use keys to enter data into a computer. They allow users to type letters, numbers, symbols, and commands by pressing keys. Examples include the keyboard and keypad.

The keyboard, also called an alphanumeric keyboard, is used for typing letters, numbers, and special symbols, and comes in types such as traditional, flexible, Braille, and ergonomic keyboards.

- **Traditional keyboard:** They are large and rigid keyboards. They are named based on the arrangement of the first six characters in the first row of the alphabet keys. They include QWERTY keyboards, AZERTY, QWERTZ, and HCESAR.



Picture 2. 20: The keyboard is used to key in data.

- **Ergonomic keyboard:** This is designed to minimize wrist injury. It is connected

to the system unit using the Universal Serial Bus (USB) cable. Some ergonomic keyboards are split.



Picture 2. 21: The ergonomic keyboard is designed to minimize wrist injury

- **Flexible keyboard:** This is a portable keyboard which can be folded and carried in a bag.



Picture 2. 22: A flexible keyboard can easily be folded and carried in a bag

- **Braille keyboard:** This is specifically designed for the visually impaired. The keys have raised dots enabling the user to feel and identify the character or symbol represented by the key



Picture 2. 23: The braille keyboard is designed for the visually impaired

- **The keypad:** This is a small keyboard, which is mostly used on devices such as palmtops, calculator, microwaves and washing machines among others.



Picture 2. 24. The keypad is a small keyboard used on devices such as palmtops and mobile phones.

ii. Pointing devices

These are devices which control the movement of a cursor or a pointer on the computer screen. Examples of pointing devices are the mouse, trackball, joystick, touch screen, light pen, touch pad and graphic tablet among others.

- **Mouse:** There are three types of computer mouse. Those are mechanical mouse that contains a ball underneath which is rotated when the mouse is moved hence controlling a pointer on the screen, optical mouse that uses digital image processing technology to track the position of the mouse and wireless mouse which does not use cables.



Picture 2. 25. : A mouse is a pointing device with sensors that execute instructions by pointing

b. Output devices

These are devices used for displaying data and information. They are classified into two categories, namely softcopy output devices and hardcopy output devices.

(i) Monitors

A monitor is also referred to as **Video Display Unit (VDU)** or the screen. It is used to display data and information. They come in different types the common ones being the Cathode Ray Tubes (CRT) and Liquid Crystal Display (LCD) commonly known as flat screens.



Picture 2. 26. CRT monitor



Picture 2. 27. LCD monitor

(ii) Printers

A printer is a peripheral device that produces hardcopy output by transferring digital information from a computer on the paper using ink. When a document or image is sent from a computer to a printer, the printer receives the digital data and converts it into a physical form. The printer then places ink on the paper in the shape of letters, numbers, or images corresponding to the digital information received.



picture 2. 28: A thermal printer

Picture 2. 29. Different types of printers

c. Ports and Connectors

A **port** is an interface on a computer on which a device can be connected. A **connector** is a part of a cable that plugs into a port or interface to connect two devices. Connectors can either be male (containing one or more exposed pins) or female (containing holes into which the male connector can be inserted).

Types of ports

Port name	Connector	Port	Colour code	Use
PS2/ Mouse			Green	Mouse

PS2/ Keyboard			Purple	Keyboard
Power Desktop computer			None	Desktop computer
USB			White	Flash drives/ Keyboards
Fire wire			Grey	Camcorders
Serial Port			Teal/ Turquoise	Projector
Ethernet LAN /NIC/ RJ45			None	LAN/Hub/ Switch
MODEM RJ11			None	Telephone

Table 2. 4: Power and General Communication Ports in a Computer System

Application activity 2.3.1

- 1) Name two examples of external hardware.
- 2) Which device is used to type letters and numbers into a computer?
- 3) A monitor is an input device. (True or False)
- 4) Fill in the blank: A _____ keyboard is designed for visually impaired users.
- 5) Why are ergonomic keyboards important for people who type for long hours?
- 6) How do ports and connectors improve the usability of a computer system?

2.3.2. Internal hardware

Activity 2.3.2

- 1) Write down the names of at least six internal components
- 2) For each component listed, explain its main function in two sentences.

a. Internal components

i. CPU Central Processing Unit (CPU)

CPU is the brain of the computer which executes instructions, performs calculations, and controls how data moves between parts. Without the CPU, the computer cannot process information.



Picture 2. 30. Central Processing Unit

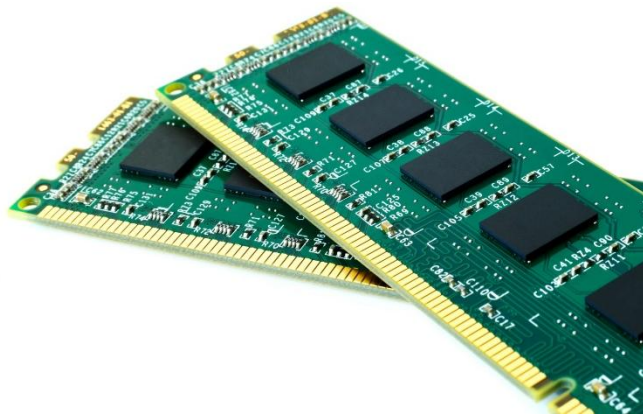
Function of the CPU: The CPU executes instructions, performs calculations, and manages data flow. It contains the **Control Unit (CU)**, **Arithmetic Logic Unit (ALU)**, and **Registers** for processing tasks.

ii. Storage devices

They store permanently information and are not volatile. Some like hard disk and SSD are internal to the computer while some others like flash disks are external and are used to move information from one computer to another. Details on storage devices will be seen in the next section.

iii. Random Access Memory (RAM)

RAM (Random Access Memory) is the computer's *short-term memory* that stores data and programs currently in use so the CPU can access them quickly.

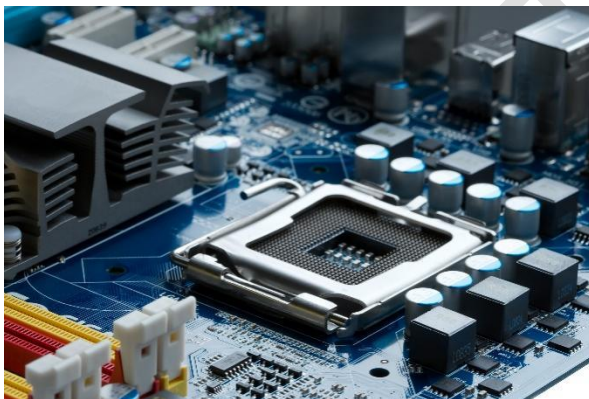


Picture 2. 31. Random Access Memory

The RAM stores data and instructions currently in use for quick access. It is said to be “volatile” because its contents are lost when power is off.

iv. Motherboard

The motherboard is the main circuit board that connects all components (CPU, RAM, storage, GPU, PSU, etc.) so that they can work together. It’s the backbone of the computer everything plugs into it. It provides communication pathways (buses) and houses the chipset



Picture 2. 32. The computer motherboard

v. Power Supply Unit (PSU)

The Power Supply Unit (PSU) is the computer component that supplies electrical power to all parts of the computer. It converts electricity from the wall socket into the correct voltage that the computer components can use.



Picture 2. 33. Power Supply Unit

It converts AC power from the wall into DC power and supplies stable voltage to CPU, motherboard, drives, and peripherals.

vi. Graphics Processing Unit (GPU)

The Graphics Processing Unit (GPU) is a specialized electronic processor that creates and processes images, videos, and animations for display on a computer screen.



Picture 2. 34. Graphics Processing Unit

vii. Cooling system

Cooling system is a Fan, heat sinks, or liquid cooling systems which prevents overheating of CPU and GPU. It maintains stable performance and extends lifespan of components.



Picture 2. 35. Computer cooling system

viii. Optical drive

A computer optical drive is a hardware device that reads and or writes data on optical discs using laser light. It can read CDs, DVDs or Blu ray.



Picture 2. 36. Optical drive

ix. Network Interface Card (NIC)

A Network Interface Card (NIC) is a hardware component that allows a computer to connect to a network. It provides wired (Ethernet) or wireless (Wi-Fi) connectivity. And enables internet access and communication between computers.



Picture 2. 37. Network Interface Card (NIC)

x. Sound card

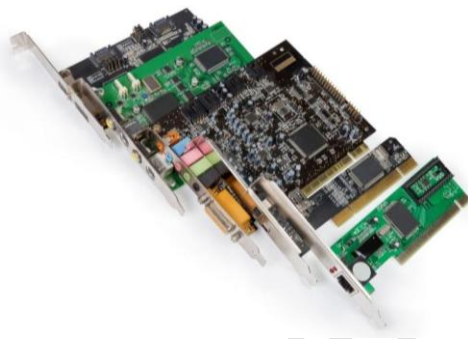
A sound card is a device in a computer that processes audio so you can hear sound through speakers or headphones. It converts digital audio signals from the computer into analog signals that speakers or headphones can play or converts analog sound from a microphone into digital data that the computer can store or process.



Picture 2. 38. Sound card

xi. Expansion cards

Expansion cards are extra cards installed into the motherboard which add new features like USB ports, video capture, or TV tuners.



Picture 2. 39. Expansion cards

xii. BIOS/CMOS battery

A BIOS/CMOS battery is a small battery that keeps the computer's date, time, and system settings stored even when the computer is off. It keeps the system clock (date and time) running, saves important hardware settings in the computer and allows the BIOS to remember configuration settings when the computer is not connected to the power. The BIOS (Basic Input Output System) initializes hardware during startup and loads the operating system.



Picture 2. 40. CMOS battery

Application activity 2.3.2

- 1) Choose four internal components and fill a table with this information:
Component name, main function, importance in daily use, possible problems if missing.
- 2) Which component do you think is the most important for everyday computer use.
Explain.

2.3.3. Storage devices

Activity 2.3.3

- 1) Which storage device uses spinning magnetic platters to store data?
a) SSD b) HDD c) Flash Drive d) Memory Card
- 2) Which storage device is faster and more durable because it has no moving parts?
a) HDD b) SSD c) Flash Drive d) Memory Card
- 3) Which storage device is small, portable, and plugs directly into a USB port?
a) HDD b) SSD c) Flash Drive d) Memory Card
- 4) Which storage device is commonly used in smartphones, cameras, and tablets?
a) HDD b) SSD c) Flash Drive d) Memory Card
- 5) Which storage device is best suited for transferring files between computers quickly?
a) HDD b) SSD c) Flash Drive d) Memory Card
- 6) Which storage device is most prone to mechanical failure?
a) HDD b) SSD c) Flash Drive d) Memory Card

a. Hard Disk Drive (HDD)

A hard Disk Drive (HDD) is traditional storage device that uses spinning magnetic platters and a moving read/write arm.

Data is stored magnetically on the platters. The arm moves to read or write information. HDDs are known for large storage capacity at a lower cost, but they are slower and more prone to mechanical failure compared to SSDs.



Picture 2. 41. Hard disk

b. Solid State Drive (SSD)

A Solid State Drive (SSD) is modern storage device that uses flash memory chips instead of moving parts.

SSDs store data electronically, which allows much faster access speeds. They are more durable, quieter, and energy efficient than HDDs. However, they are more expensive per gigabyte.



Picture 2. 42. Solid State Drive

c. Flash Drive (USB Drive)

USB Drive is small, portable device that plugs into a USB port and uses flash memory. Flash drives are used for transferring files between computers and for portable storage. They are easy to carry, and reliable for small to medium amounts of data, though they usually have less capacity than HDDs or SSDs.



Picture 2. 43. USB drive

d. Memory card

Picture 2. 44: Memory card

A memory card is a small, removable storage device used to store data such as photos, videos, music, and files in electronic devices.



Picture 2. 45. Memory card

Application activity 2.3.3

- 1) Which storage device uses spinning magnetic platters?
- 2) SSDs have moving parts. (True or False)
- 3) Fill in the blank: Adrive is small, portable, and plugs into a USB port.
- 4) Which storage device is most commonly used in smartphones and cameras?
- 5) Which device is best for transferring files between computers quickly?
- 6) Explain why SSDs are faster than HDDs.

2.3.4. Computer hardware issues

Activity 2.3.4

Match the hardware problem with its correct cause.

Hardware issue	Cause
a) No display on monitor	i) Damaged USB port
b) USB not detected	ii) RAM not properly connected
c) Overheating	iii) Dust blocking air vents
7) Beeping at startup	iv) Loose display cable

A. Hardware common issues

i. No display on monitor

This happens when the monitor shows a black screen even though the computer is turned on. Common causes include loose or damaged cables (HDMI, VGA, display port), the monitor is off or has no power, the computer is not fully starting (booting), faulty graphics card or motherboard.

ii. Device not detected

This occurs when you plug in a USB flash disk, printer, or other hardware but the computer does not detect it.

Possible causes are damaged USB port, faulty device (the USB or printer might be damaged), outdated drivers and loose connection

iii. Overheating

Overheating happens when the computer becomes too hot. Signs include fan making loud noise, computer slowing down and automatic shutdown. Automatic shutdown causes dust blocking air vents, heavy programs running for long hours, faulty cooling system (fan not working properly).

iv. Beeping sounds at startup

When a computer makes beeping sounds as it starts, it is giving error codes. Each pattern of beeps means something different, such as RAM not properly connected, hardware failure, motherboard error. This helps technicians identify internal problems.

B. Basic troubleshooting steps

- **Check power connections:** Make sure the computer, monitor, and other devices are plugged into working power sockets.
- **Ensure cables are firmly connected:** Loose cables can cause display issues, network problems, or devices not working.
- **Restart the computer:** A simple restart fixes many temporary errors by refreshing the system.
- **Allow the computer to cool:** If overheating occurs, stop using the computer and let it cool down.
- **Test with another device:** Try another monitor if the screen is blank, use another keyboard or mouse to see if the device has a problem.
- **Remove dust from vents:** Dust blocks airflow and causes overheating. Cleaning helps air circulate.

C. Safety rules when handling hardware

- **Always turn off and unplug before opening a computer:** This prevents electric shock and protects components from damage.
- **Do not touch internal components with wet hands:** Water can cause short circuits and damage hardware.
- **Avoid static electricity:** Static electricity can destroy computer parts. Using an anti-static wrist strap.
- **Handle components gently and by the edges:** Touching chips or circuits directly can damage them or leave oil from your fingers

Application activity 2.3.4

- 1) Answer by True or False for each statement.
 - a) A black screen on a monitor can be caused by loose display cables.
 - b) Overheating can cause a computer to automatically shut down.
 - c) Beeping sounds at startup always mean the computer is working normally.
 - d) Dust blocking air vents can lead to overheating.
 - e) Restarting a computer can fix some temporary errors.
 - f) USB devices are detected even if the USB port is damaged.
- 2) Choose the letter corresponding to the right answer:
 - a) What is the common cause of no display on a monitor?
 - i. Loudspeakers
 - ii. Loose HDMI or VGA cable
 - iii. Full hard disk
 - iv. Virus infection
 - b) When a USB device is not recognized, one possible cause is:
 - i. The monitor is off
 - ii. Faulty USB device
 - iii. Dust in the CPU fan
 - iv. Low screen brightness
 - c) Which of the following is a sign of overheating?
 - i. Bright screen
 - ii. Fast internet
 - iii. Loud fan noise
 - iv. Clear display
 - d) What does beep at startup usually indicate?
 - i. Internet connection problem
 - ii. Internal hardware error
 - iii. Full memory card
 - iv. Printer problem

2.4. Computer software

Computer software refers to a set of instructions that tell the computer what to do. Software cannot be physically touched. It either enables the computer hardware to perform its assigned tasks or helps the user to accomplish specific tasks. Software is also referred to as programs.

Computer software plays the role of controlling hardware, running application and allowing users to interact with the computer.

2.4.1. Main types of computer software

Activity 2.4.1

Starts your computer (with Windows installed) then use antivirus software to scan viruses and type notes using Microsoft Word.

- 1) Identify the operating system used.
- 2) Name the utility software which is in that computer.
- 3) Identify the application software that are in that computer.
- 4) State the role of each software identified.

Computer software is categorized into two main types, namely system software and application software.

A. System software

They are software designed to manage computer resources and provide these resources to the user. There are five types of system software namely operating system, utility software, firmware, programming languages and Networking software.

- **Operating System (OS):** This is a group of programs that provide a platform for loading other programs known as application programs. The OS acts as an interface between the hardware and the application programs. Examples of operating systems include Linux, Ms DOS, MS Windows, OS X, ios, Android, and UNIX.
- **Utility software:** They are system software that are used to perform routine functions aimed at optimizing, analysing, and maintaining the operations of a computer. They manage computer files, diagnose, repair computer problems and help the computer to run more efficiently. Examples of utility software include anti-virus, Disk cleaners, Disk Checkers, Disk Compressors, Disk Defragmenter, Disk partitioners, File managers, language translators, graphic editors, and text editors.
- **Firmware:** They are sometimes referred to as **stored logic**. These are software programs that are stored in the computer's Read Only Memory (ROM) where they are available for immediate use. They are not affected by loss of power, hence the term **firm**. An example of firmware is Basic Input Output System (BIOS). Firmware may hold utility programs, language processors, or even operating systems.
- **Programming language software:** These are formal constructed languages designed to communicate instructions to a computer. They are used to create programs that control the operations of a machine. They have language translators integrated in them.

B. Application software

They are programs that are designed to enable the user to accomplish a given task. There are two groups of application software, namely customised software and off-the-shelf software.

- **Customized software:** They are also known as in house developed software. They are tailor-made to solve a specific problem at hand, or to meet a specific need of the user. They can be modified any time at the user's request. They only contain features that are required by the user.
- **Off-the-shelf software:** They are also known as **standard** software. They are software written by a software engineer or a programmer, packaged, and then made available through a vendor, a distributor, or a developer. Examples are Microsoft Word,

Open Office Writer and Ms Access.

Application activity 2.4.1

- 1) State one role of computer software.
- 2) Differentiate between system software and application software.
- 3) Give one example of firmware.
- 4) A hospital needs software designed only for managing patient records.
 - a) Identify the type of application software needed.
 - b) Give one reason for your choice.

2.4.2. Software installation

Activity 2.4.2

- 1) Where do we get software from?
- 2) What happens when we click “Install”?
- 3) Why do we sometimes click “I Agree” before installation continues?
- 4) Why is it dangerous to download software from unknown websites?

A. Installation of application program

Installation steps:

- **Obtain the program:** This may be from a website, a CD/DVD or a flash drive. Only obtain software from trusted or official sources.
- **Run the installer file:** Double-click the file (e.g., *setup.exe*) to start installation.
- **Accept license agreement:** This shows the rules for using the software. You must accept to continue.
- **Choose installation location:** You can install in default folder (recommended) or Custom folder
- **Wait for installation to complete:** The installer copies files and configures the program.
- **Open the software to confirm installation:** Check if it runs properly and appears in start menu, desktop and applications list.

B. Basic features of installed software

- **Icons:** Small pictures that represent programs; clicking them opens the application.
- **Toolbars and menus:** Provide quick access to commands (File, Edit, View, etc.).
- **Buttons and commands:** These perform actions such as Save, Print, Delete.
- **Help and support options:** Provide guidelines or troubleshooting assistance.

C. Safety and best practices for software installation

Download from trusted sources to avoid malware, scan files for viruses before installation, avoid installing unknown programs that may harm the computer and ensure compatibility (check if your system meets requirements).

D. Common software issues

- **Installation error:** This happens when the software cannot install properly. The reasons are wrong installer (e.g., 64-bit installer on a 32-bit system), missing installation files and corrupted download
- **Software freezing:** The program becomes slow or stops responding. This can be caused by low RAM, bugs in the software or too many programs running at the same time.
- **Compatibility issues:** Some software cannot run on certain systems. Examples: old software on a new operating system and software requiring higher RAM or more storage

E. Basic solutions

Some basic solutions are to restart the program to refresh its memory, update the software to fix bugs, reinstall the software if files are corrupted and check system requirements to ensure compatibility.

Application activity 2.4.2

- 1) Install an application program of your choice by going through the following steps:
 - a) Obtain the installer file from a trusted source.
 - b) Run the installer.
 - c) Accept the license agreement.
 - d) Choose installation location.
 - e) Wait for installation to complete.
 - f) Confirm installation by opening the program.
- 2) A program fails to install. What can be the cause?

End unit assessment 1

- 1) Choose the correct answer.
 - a) Which function of a computer converts data into information?
i) Input ii) Processing iii) Output iv) Storage
 - b) Which device is used to display information on a computer screen?
i) Keyboard ii) Mouse iii) Monitor iv) Scanner
 - c) The first generation of computers used:
i) Transistors ii) Integrated circuits iii) Vacuum tubes iv) Microprocessors
 - d) Which of the following is a microcomputer?
i) Supercomputer ii) Mainframe iii) Laptop iv) Minicomputer
 - e) Which storage device uses spinning magnetic platters?
i) SSD ii) Flash drive iii) Memory card iv) HDD
 - f) Which component controls and processes instructions?
i) RAM ii) CPU iii) PSU iv) Motherboard
 - g) Which port is commonly used to connect a modern monitor?
i) PS/2 ii) HDMI iii) RJ11 iv) Serial port
 - h) Which of the following is application software?
i) Windows OS ii) Antivirus iii) Microsoft Word iv) BIOS

i) Which generation of computers uses Artificial Intelligence?

i) Third

ii) Fourth

iii) Fifth

iv) Second

2) Short answer questions

- a) State two functions of a computer system.
 - b) What is the role of an operating system?
 - c) State two characteristics of third generation computers.
 - d) Identify two types of computer software and state one purpose of each.
- 3) Describe how the IPOS cycle works when a person types, saves, and prints a document.
- 4) A computer shows no display when switched on.
- (a) State two possible causes.
 - (b) Suggest two troubleshooting steps.
- 5) Differentiate between internal and external hardware using two examples each.
- 6) The computer is very slow and makes loud fan noise.
- a) State two possible causes.
 - b) Suggest two solutions.
- 7) Describe the importance of following safety rules when installing or repairing computers.
- 8) Explain why using legal and licensed software is important.

UNIT 3: COMPUTER NAVIGATION



Key unit competence

Manage files and folders in Windows Explorer efficiently.

Learning objectives

At the end of this unit, you should be able to:

Knowledge and understanding:

- Explain how to customize the desktop, taskbar, and start menu.
- Describe the purpose of files, folders, Recycle Bin, and Control Panel
- Explain the importance of organizing files and folders.

Skills:

- Customize desktop settings and taskbar features.
- Create, rename, copy, cut, paste, delete, and restore files and folders.
- Navigate Windows Explorer and manage files and folders.

Attitudes and values

- Appreciate the importance of correctly managing files and folders
- Show care and responsibility when customizing or managing computer settings.
- Demonstrate organization in arranging desktop items and folders.

3.0. Introduction

This unit introduces learners to basic computer navigation using the Windows operating system. You will explore the desktop environment, taskbar, start menu, and common accessories. The unit also emphasizes proper file and folder management, which is essential for organizing information, saving work correctly, and maintaining digital responsibility.

Introductory activity

Start the computer and answer the following questions:

- 1) What items do you see on the desktop?
- 2) Where do you click to open programs?
- 3) What happens when a file is deleted?
- 4) Why is it important to save files in folders?

3.1.Desktop

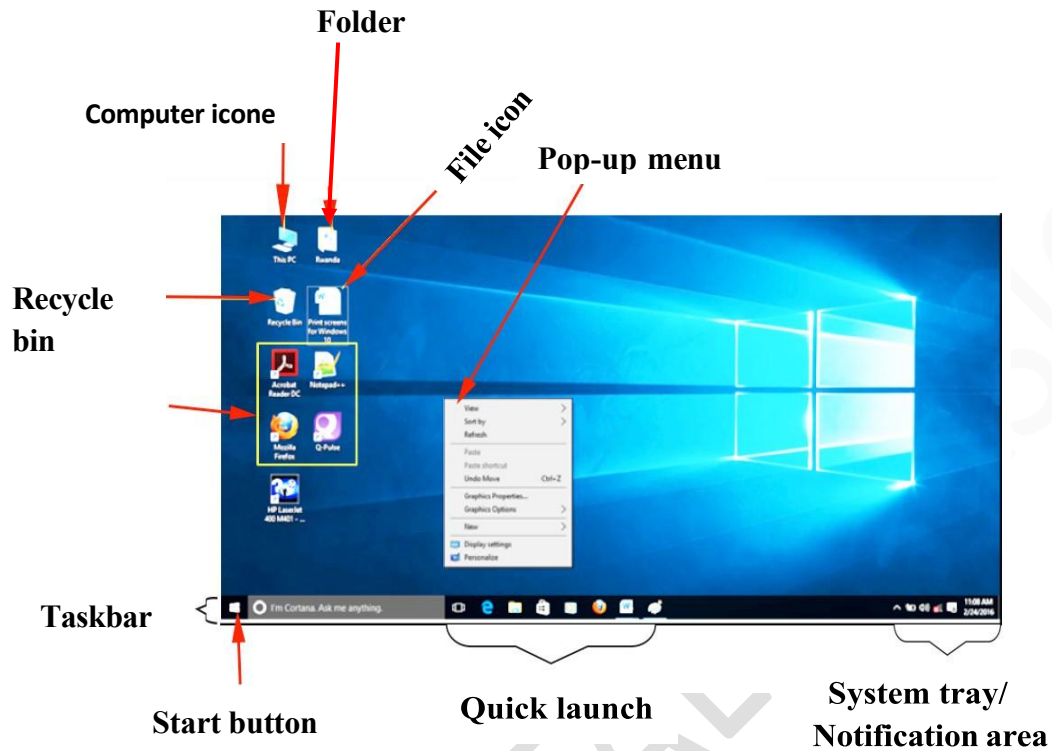
Activity 3.1

- 1) Answer by Yes or No:
 - a) Is the desktop the first screen displayed when a computer starts?
 - b) Does the desktop contain icons?
 - c) Is the taskbar found at the top of the screen by default?
 - d) Are deleted files stored in the Recycle Bin?
- 2) What is the desktop?
- 3) Mention two things found on the desktop.
- 4) Why is the desktop important to a computer user?

3.1.1. Introduction to desktop

A. Definition

- **Desktop** is the first screen that is displayed when a computer boots up. To boot a computer is to start it so that it can load the operating system and make it ready for use. The desktop consists of icons, the task bar, and other features.
- **Desktop icons:** Icons are images that represent application programs, folders, and files. They are usually manipulated using pointing devices. Icons on the desktop include Computer, Documents and Recycle Bin among others.
- **Task bar:** It is a bar usually at the bottom of a computer screen showing running programs. These programs can be clicked to start software applications or switch between open applications or active Windows.
- **Start button:** It is the button in the far left bottom corner of taskbar. The use of the start button is to provide a central launching point for computer programs and performing other tasks.
- **Task manager:** It is used to provide information about the programs that are running on a computer, as well as the general status of the computer.
- **System tray (or Notification area):** This is a part of the taskbar in the desktop interface that displays small icons on the opposite side of the Start Menu such as the clock, date and time, antivirus, and the icons of certain programs so that a user is reminded that they are running in the background.
- **Quick launch:** It is a part of the taskbar near the Start menu where shortcuts to Programs are added. This toolbar stays visible for shortcuts to be easily accessed, even when a window is open.



Picture 3. 1 Desktop window

3.1.2. Customizing the desktop and taskbar

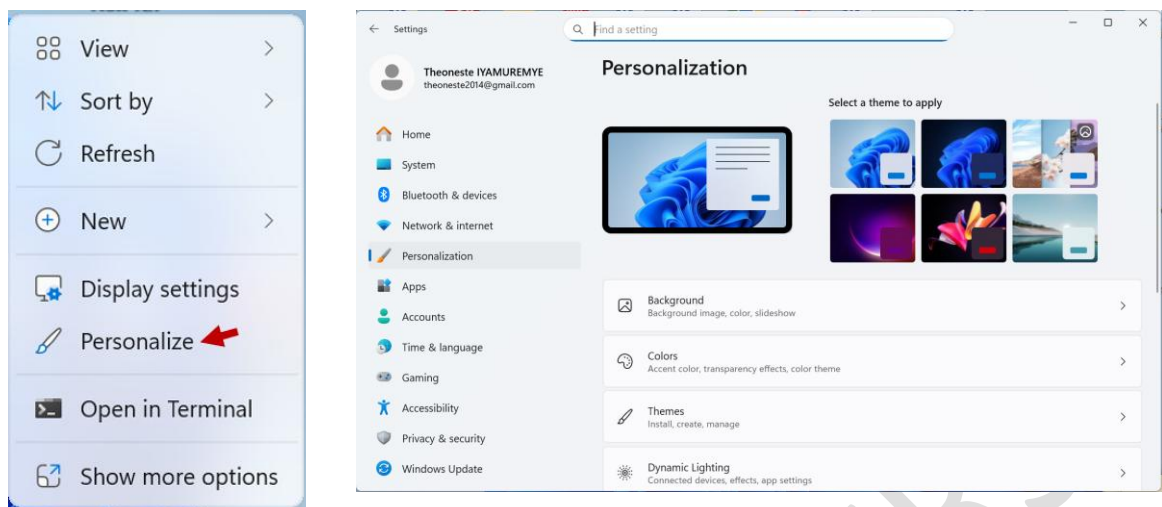
Customizing refers to personalizing the desktop to suit the user's needs and tastes.

A. Customizing the desktop

The changes that the user can make to customise include new themes, mouse pointers, backgrounds, user images, and gadgets among others.

To customize:

- Right-click on the desktop. The following dialog box appears:
- Click Personalize.
- Select the desired option from the left pane of the Personalization window as shown in Picture behind.
- Click on the desired option on the right pane.

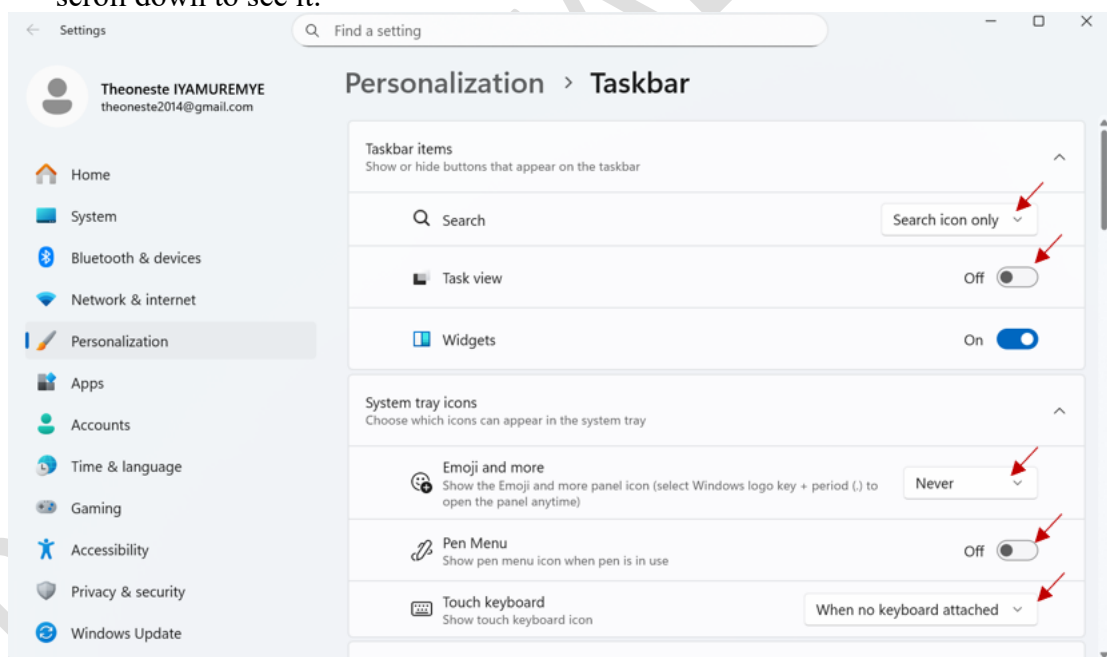


Picture 3. 2: The dialog box for Customising Theme and different themes.

B. Customizing the taskbar

To customize the taskbar in Windows 11, do the following:

- Right click on the taskbar and choose **Taskbar settings**
- In the Settings windows that opens click on any setting that you want to customize so that it gets selected or becomes **ON**. If the setting you want to customize is not visible, scroll down to see it.



Picture 3. 3. The Settings window for customizing the taskbar in Window 11

The following are some of the taskbar customisation options:

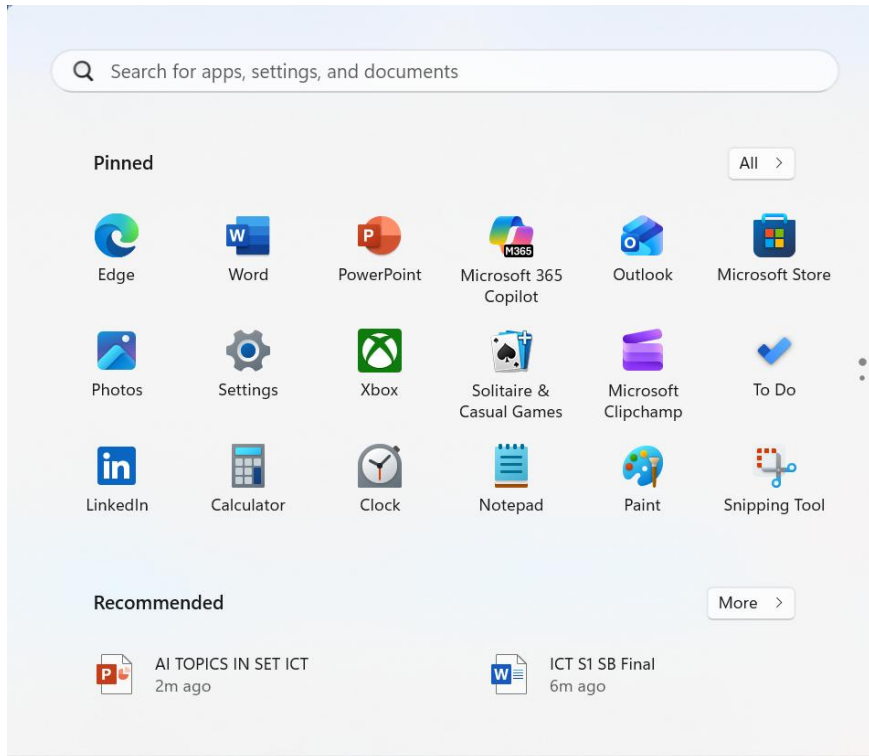
- **Search icon:** You can hide it, have the search icon only, search icon and label, search box.
- **Task view:** You can make it appear on the taskbar or not. The task view shows all

the opened programs and is used to navigate among them

- Etc.

3.1.3. Explore and customize the start menu

In Windows 11, when the start button is clicked, all the programs that are pinned are shown in the window that appears immediately. When you click on All the other programs that were invisible will appear.

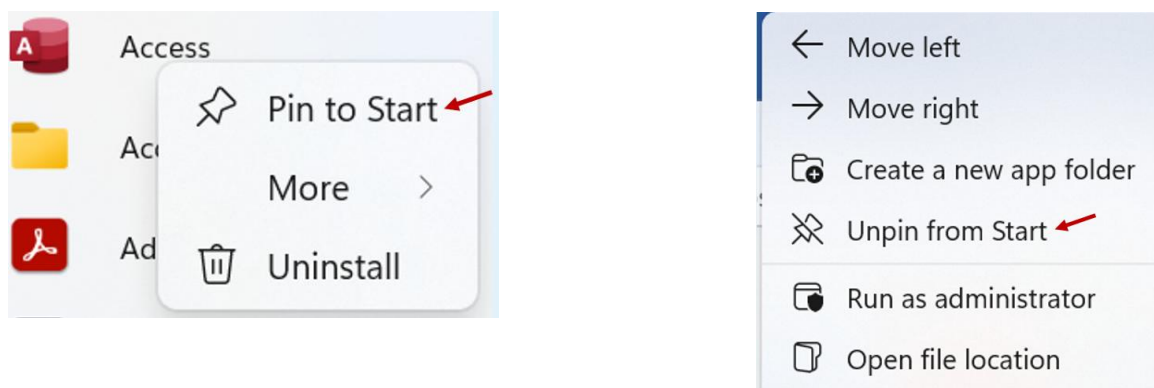


Picture 3. 4. Programs displayed after clicking the Start menu

To add or remove a program icon to the **Taskbar** do the following:

- To add it right-click the **Program** icon then select **Pin to Start**.
- To remove it right click the program icon then select **Unpin from Start**.

Note that right clicking on a program displayed after clicking the start menu shows different options depending on whether a program is already pinned to the start menu or not.



Picture 3. 5. From left to right: Pin to Start and Unpin from Start options

3.1.4. Recycle bin

The Recycle Bin is a container that temporarily stores deleted files and folders before they are permanently deleted. It allows restoring files and folders that have been deleted while also allowing the user to restore them if they were deleted by mistake or permanently delete them when no longer needed.

a. Deleting and restoring files from the recycle bin

Deleting is the act of erasing files or folders. Deleted files are moved to the Recycle bin. To delete a file permanently, you go to the Recycle bin and delete it again while restoring is the act of retrieving the deleted files and folders from the recycle bin.

The deleted files remain in the **recycle bin** until it is emptied. If files have been deleted by mistake, they can be recovered by clicking the **Restore** button in the Recycle Bin. However, pressing **Shift + Delete** permanently erases the file without sending it to the Recycle Bin.

To delete a file or a folder, do one of the following:

- Click on the file or folder to be deleted. Press the **Delete** key on the keyboard; or
- Right-click on the file or folder and then select **Delete** option from the pop-up menu.

The operating system will prompt the confirmation of the deletion by displaying a dialogue box as shown below.



Picture 3. 6. Prompt message when deleting a file

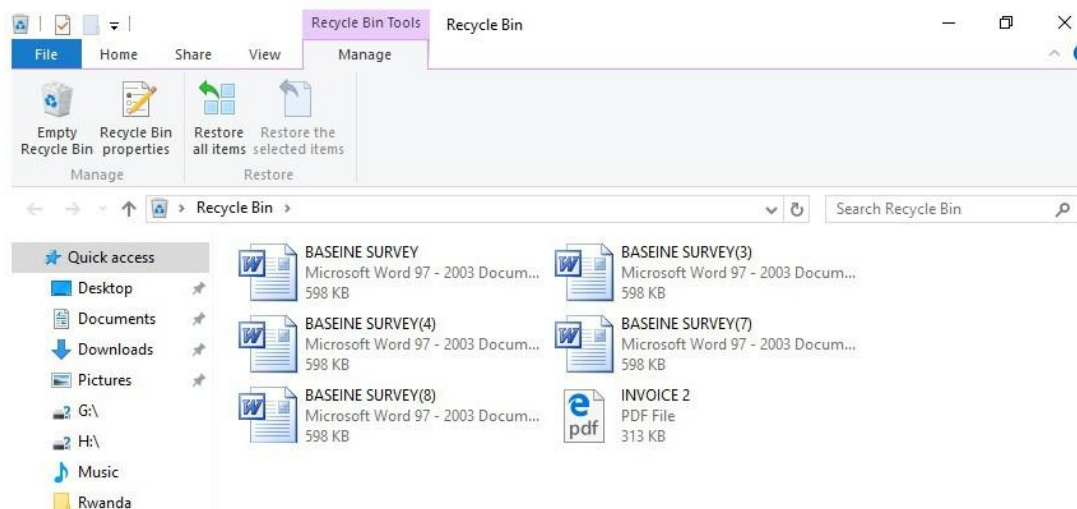
- Click **Yes** to confirm deletion and **No** to cancel.

To restore a file or folder from the Recycle Bin, do the following:

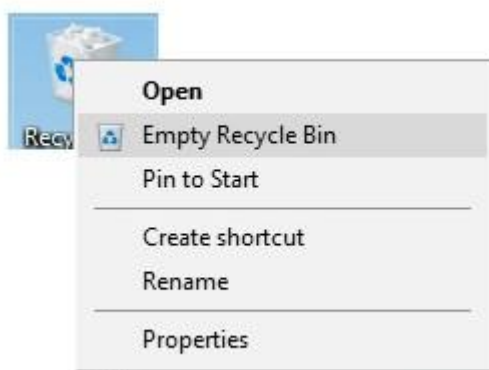
- Right-click on the **Recycle bin** icon on the **Desktop** and select **Open** from the pop-up menu or double-click the icon.
- Right-click on the file(s) or folder(s) to be restored then select **Restore** from the pop-up menu. The file or the folder will be taken back to its original location.

To empty the recycle bin, do the following:

- Right-click on the **Recycle bin** icon on the **Desktop** and select **Open** from the pop-up menu; or double-click on the icon.
- Click on **Manage** tab under **recycle bin tools** in the menu bar.
- Click on **Empty Recycle Bin** icon.
- A dialog box appears prompting you to confirm the deletion. If you are sure of the action, click **Yes** to conform the deletion otherwise, click **No**.



Picture 3. 7: The Recycle bin window



Picture 3. 8: Emptying the Recycle Bin.

Alternatively, right-click on the **Recycle Bin** icon then select the **Empty Recycle Bin** Command from the pop-up menu.

Note that once items have been deleted from the recycle bin, they cannot be restored. Always keep a backup of your work. Do not attempt to delete program files. If you are unsure about an action, seek guidance from your teacher.

Application activity 3.1

- 1) Answer with Yes or No:
 - a) Does the desktop show icons for programs and files?
 - b) Is the Recycle Bin an example of a desktop icon?
 - c) Is the Recycle Bin used to open programs?
 - d) Can icons represent programs?
- 2) Draw a simple desktop and label three components.
- 3) Explain how desktop icons help users work faster.
- 4) What is the Recycle Bin and what is its role?

3.2.The Control Panel

Activity 3.2

MUGISHA notices that the computer screen is too bright and the sound is too loud. His teacher explains that these settings can be adjusted using the Control Panel.

- 1) Answer with Yes or No
 - a) Is the Control Panel found in the Start Menu?
 - b) Is the Control Panel a hardware component?
 - c) Does changing the pointer affect stored files?
 - d) Does keyboard layout affect typing?
- 2) Mention three settings that can be changed using the Control Panel.
- 3) Explain what gadgets are used for.?
- 4) What is a desktop background?
- 5) What is screen resolution?

A. Desktop background and theme

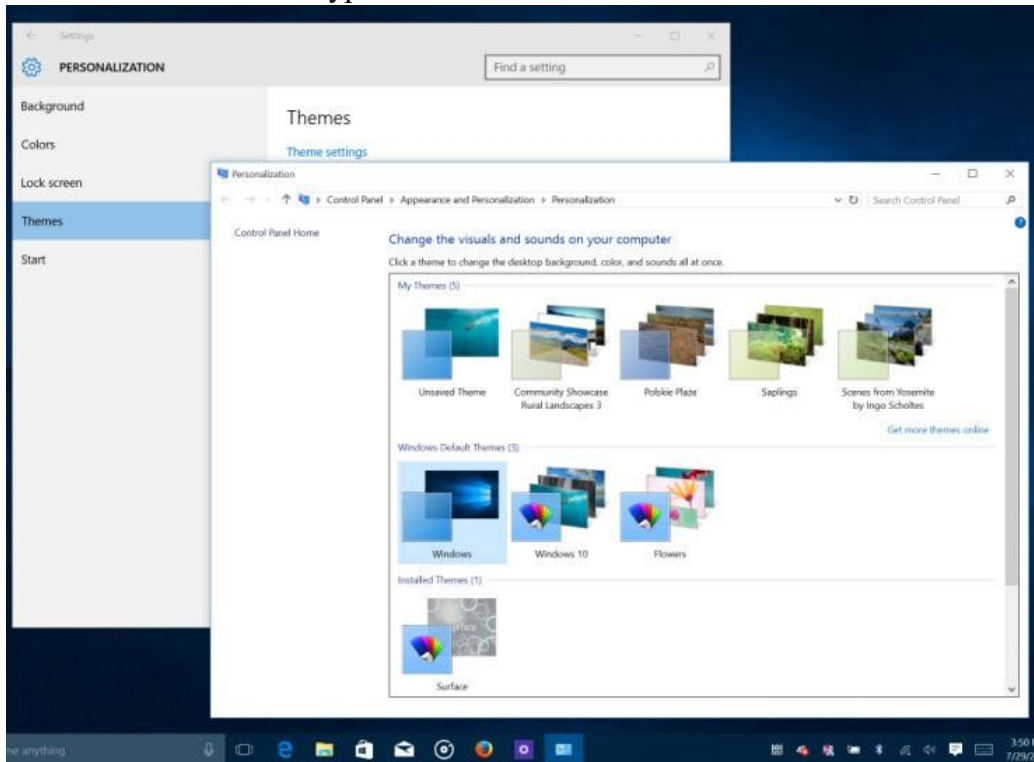
A background also known as **wallpaper** is an image used to decorate the graphical user interface on the screen.

A theme is a preset package used to customise the look and feel of graphical appearance details. It replaces a computer's ordinary icons, sounds, pointers, background, or screen saver with a designed pattern to reflect the interests of the user.

To customise the desktop background and theme, do as follows:

- Click the **Start** menu, select **All Apps** then click on **Setting**. The setting dialog box appears.
- Click on **Personalization**.
- To change the background, click on **Background**.

- To change theme, click on **Themes** in the left pane then click **Theme Settings** in the right pane.
- Click on the desired background or theme.
- Click **Save Theme**. Type the name of the theme then click **Save**.

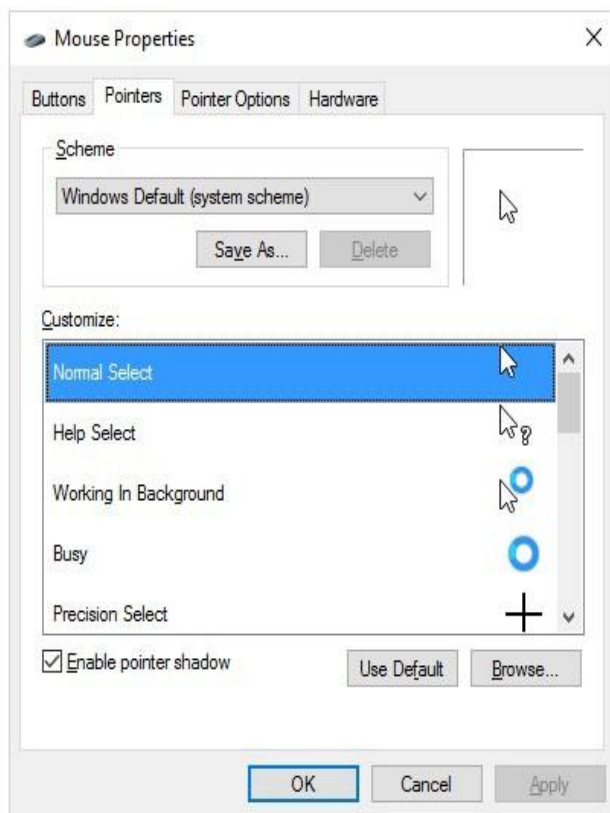


Picture 3. 9: Theme settings in Windows 10

B. Mouse pointer

To customise the mouse pointer, do the following:

- Click **Start** button, select **All apps** then click Setting.
- Select **Themes** in the left pane.
- Click on **Mouse Pointer settings** hyperlink in the **Related Settings** section.
- Click **Pointers** tab.
- Select the desired pointer option under **Customize**.
- Click **Apply** then **OK**.



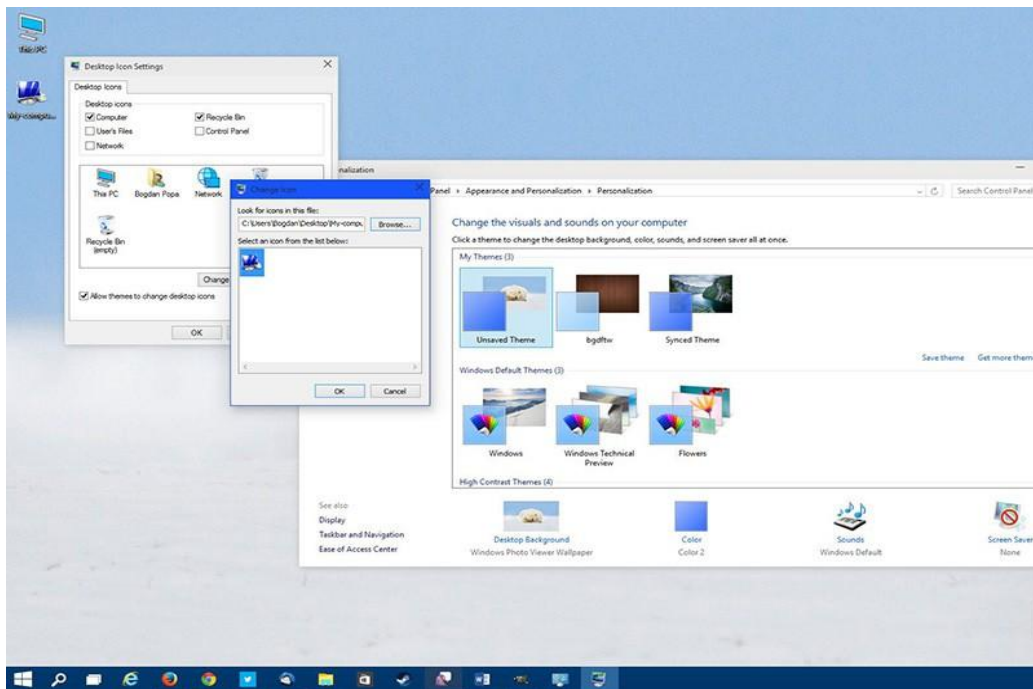
Picture 3. 11. Window for changing the mouse properties

C. Desktop icons

Icons are small images used to represent a program, file, folder or commands. The icons are accompanied with a brief description of what it does.

The following procedure is used to customize desktop icons:

- Click **Start** button, select **All Apps** then click **Setting** and select **Personalization**.
- Select **Themes** in the left pane.
- Click on **Desktop Icon settings** hyperlink in the **Related Settings** section.
- Select the icons to appear on the desktop, for example, **Computer**, **Network** and **Control Panel**.
- To change the way an icon appears, click on **Change Icon** button, select a desired option from the window displayed and click the **OK** button.
- Clicking on **Restore Default** button reverts the setting to the original factory settings.
- Click **Apply** then **OK** to close the window.

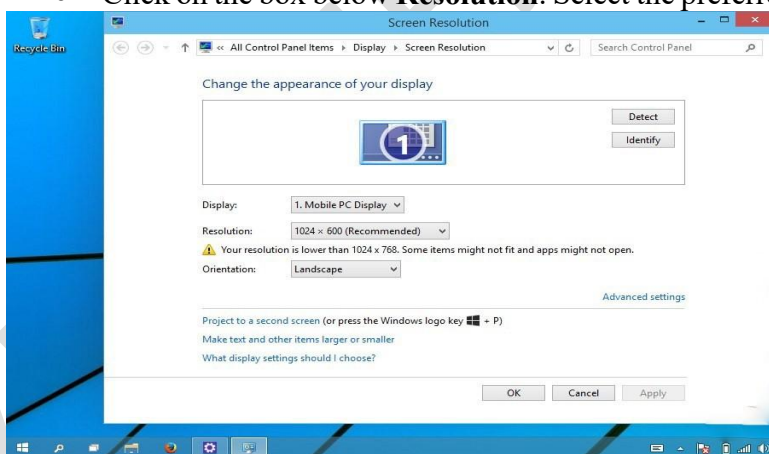


Picture 3. 12: Desktop icons settings

D. Screen resolution

Screen resolution determines the clarity of the text and images displayed on the screen. To change the screen resolution:

- Click **Start** button, select **All Apps** then click **Setting** and select **System**.
- Select **Advanced Display Settings** link in the right pane.
- Click on the box below **Resolution**. Select the preferred resolution using the slide bar.



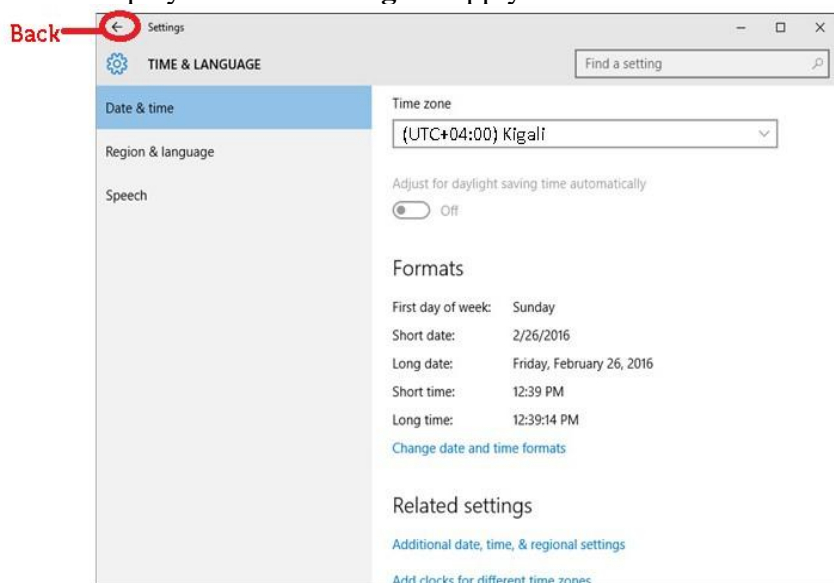
Picture 3. 13: Screen Resolution dialog box

E. Changing date and time

To change date and time, do either of the following:

- Click **Start** button, select **All Apps** then click **Setting** and select **Time & Language**.

- Click on **Date & Time** link in the left pane.
- Click on **Change Date and Time** formats. Select the preferred date and time on the respective list boxes then click on the **Back** button.
- Click **Change** button then select new date and time format on the list boxes displayed. Click **Change** to apply.

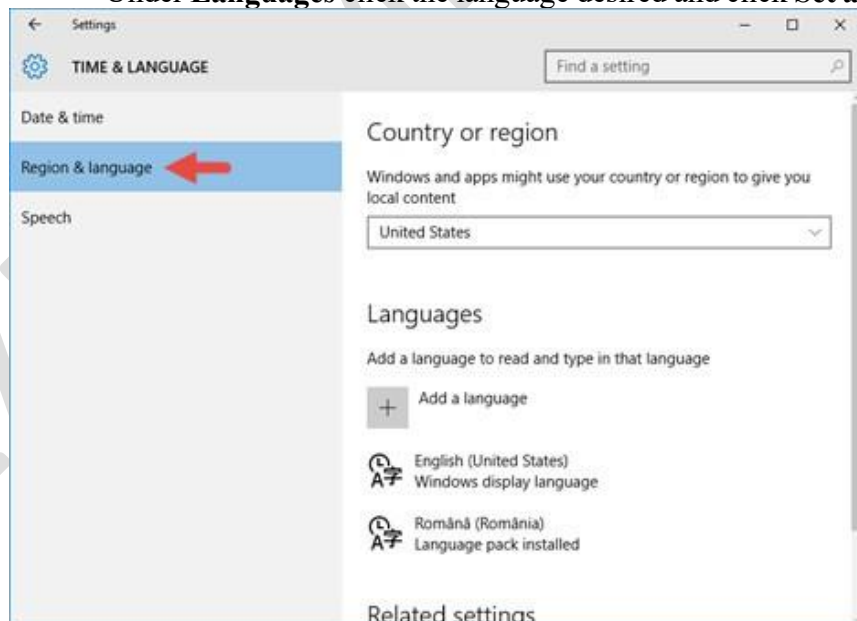


Picture 3. 14: Date and Time dialog box

F. Keyboard layout

This refers to the arrangement of keys. To change keyboard layout:

- Click the **Start** button and select **Settings**.
- Select **Time & Language**, click on **Region & Language** on the left pane.
- Under **Languages** click the language desired and click **Set as default**.



Picture 3. 15: Region and Language dialog box

Application activity 3.2

MUGISHA wants to make the computer comfortable for use.

- 1) Can you change the display settings using the Control Panel? (Yes/No)
- 2) Can date and time be adjusted from the Control Panel? (Yes/No)
- 3) Open the Control Panel and identify three settings you can change.
- 4) Explain one problem that can be solved using the Control Panel.

3.3. Windows

Activity 3.3

- 1) When MUTONI opens Microsoft Word, a large rectangular area appears on the screen. Inside it, she can type her work. Answer with Yes or No:
 - a) Is Windows an area on the screen?
 - b) Does Windows display information for a program?
 - c) Can more than one window be open at the same time?
 - d) Is Microsoft Word an application software?
 - e) Is the typing page inside Word a document window?
- 2) What is a Windows?
- 3) Give one example of a program that opens in a window.
- 4) Why are Windows important when using a computer?
- 5) Open two programs and describe what you see on the screen.
- 6) Explain one advantage of working with more than one window.

A window is an area on the screen that displays information for a specific program. This often includes the user interface GUI as well as the program content.

A. Types of windows

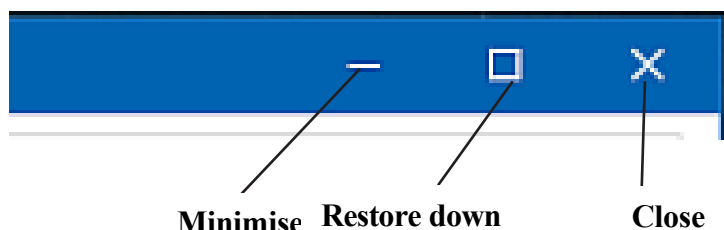
There are two types of windows, namely **application** and document **windows**.

- **Application window:** It is the larger window that is displayed when an application is opened, for example, MS Word and MS Excel among others. It helps the user to communicate with the application program.
- **Document window:** It is the smaller window inside the application window. It is used for typing, editing, drawing and formatting the text and graphics.

B. Window control buttons

There are four window control buttons, namely Minimize, Restore Down, Maximize and Close.

- **Minimize:** It removes a window from a view to an icon on the taskbar.
- **Restore Down:** It reduces the size of a window.
- **Maximize:** It enlarges the window to occupy the entire screen.
- **Close:** It exits the active program.

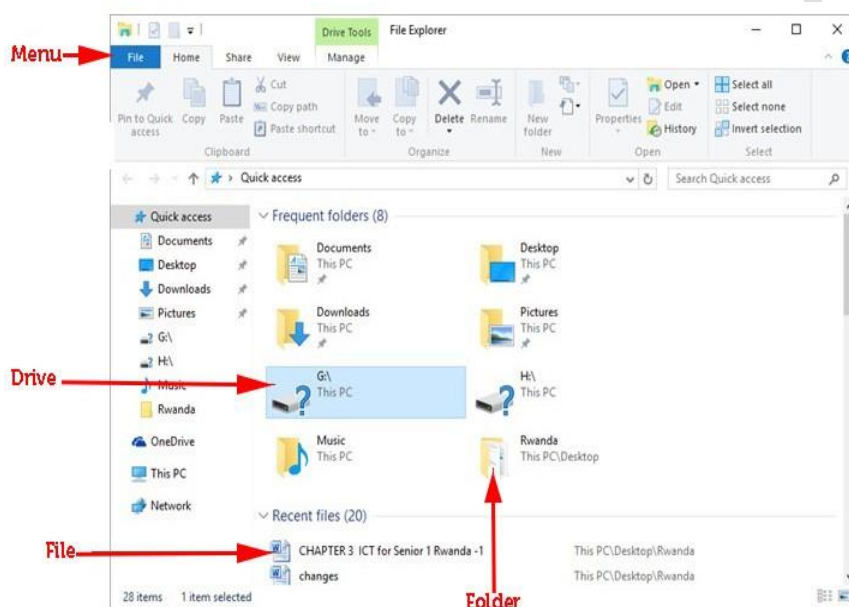


Picture 3. 16: Window control button

C. Common windows accessories

The following are some of the common windows accessories:

- (i) **File Explorer:** It is also known as windows explorer. It refers to the graphical file management utility that displays the drives and contents of folders and files in the computer.



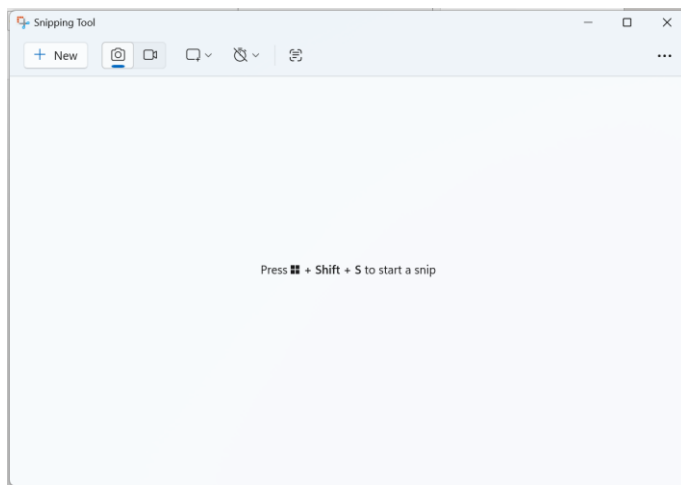
Picture 3. 17: Windows File Explorer

- (ii) **Sticky Notes:** It is an application program that enables the user to write small notes that are kept on the screen as a reminder on special tasks. They are like a small agenda to the computer owner.

To open Sticky notes search for it from the Search box located at the task bar then Write “Sticky Notes” and click on it.

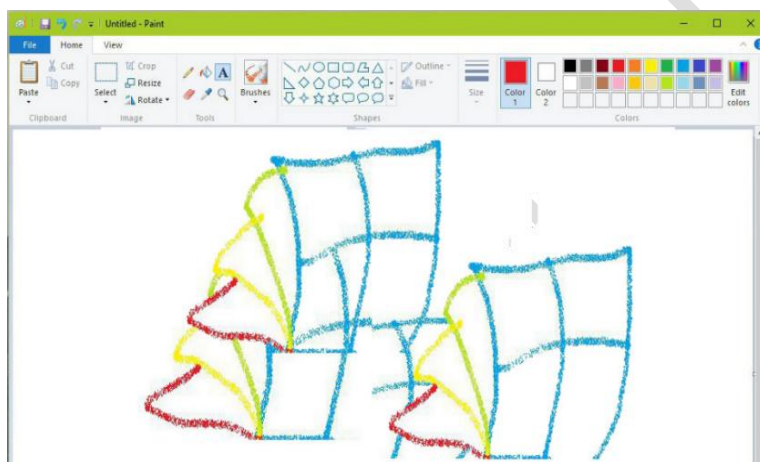
- (iii) **Snipping tool:** This is a tool which is used to take an image of the screen as it is. The image taken is of high resolution compared to an image of the screen taken with a camera.

Open the Snipping tool by using the search box, type “Snipping tool” and click on it.



Picture 3. 18. The Snipping tool window

- (iv) **Paint:** It is an application program designed for drawing images on the screen using a pointing device. The images are generated as bitmaps (see picture: windows paint).



Picture 3. 20: Windows Paint

Application activity 3.3

- 1) Answer with Yes or No:
 - a) Can two windows be opened at once?
 - b) Does each window belong to a program?
 - c) Can Windows accessories be opened from the Start Menu?
 - d) Is File Explorer a Windows accessory?
- 2) Explain one advantage of working with more than one window.
- 3) State one difference between an application window and a document window.
- 4) Name the four window control buttons.
- 5) Practice minimizing, maximizing, and closing a window.
- 6) Explain when you would use the restore down button.

3.4. File and folder management

Activity 3.4

1) Answer with Yes or No:

- | | |
|---|--|
| a) Is a file a collection of related data? | b) Must a file have a name? |
| c) Can files be saved on the computer?
saving? | d) Can a file be opened later after |
| e) Is a folder used to store files? | f) Can a folder contain other folders? |
- 2) What is a file?
- 3) Give two examples of files stored on a computer.
- 4) What is a folder?
- 5) Why are folders important in organizing data?

A. Definitions

- A **file** is a group of related data given a unique name for ease of access, manipulation and storage. A file is created by opening a program and entering data in the program. A file may be designed to store a picture, a written message, a video, a computer program or a wide variety of other kinds of data.
- A folder is a specific storage location in a storage media. Folders are used to organize and store related files.

B. Creating a file and a folder

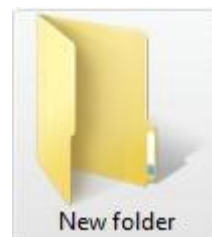
i. Creating a file

To create a new file, click the **File** menu in the active program and then click **New**. A dialog box appears. For the steps up to when the new file will be created.

ii. Creating a folder

A folder may also contain other folders within it which are called **subfolders**.

It is important to understand the organisational plan of the folders in the computer system when creating folders.



Picture 3. 21: New Folder Icon

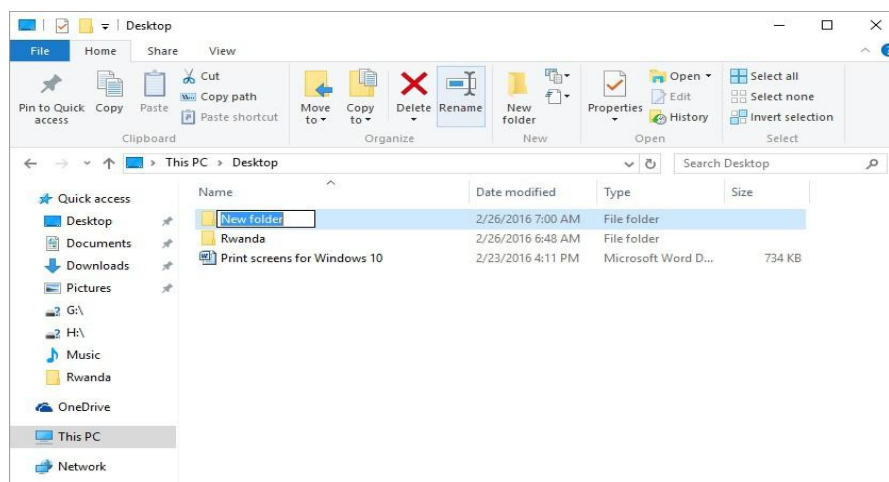
To create a new folder, use one of the following options:

(a) Using the menu

- Double click **This PC** icon on the desktop.
- Select the folder/drive where the folder was to be stored. For example, **Documents**.
- Click on the **Home** tab in the menu bar select the **New folder** command in the

New group.

- (iv) A folder will be automatically created with a default name “**New folder**” highlighted and surrounded with a box. Type a name for the folder and press **Enter**.



Picture 3. 22: Typing the name of a New Folder

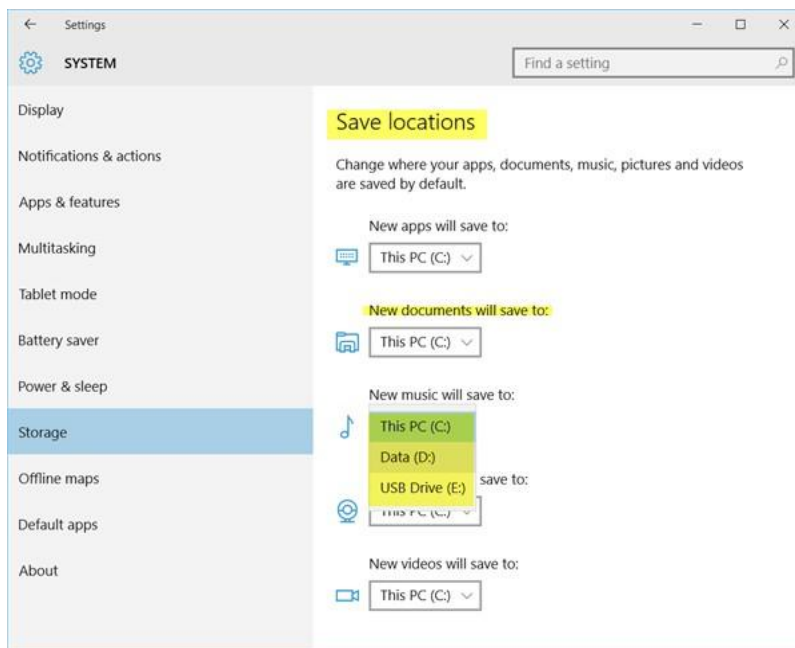
(b) Using right click

- (i) Double click **This PC** icon on the desktop.
- (ii) Select the folder/drive where the folder was to be stored. For example, **Documents**.
- (iii) Right-click on an empty space.
- (iv) Select **New** from the drop down menu that appears then click on **Folder**.
- (v) Type the desired name to replace the name **New Folder** then press **Enter**.

C. Saving

Saving is the process of storing a document in a storage medium or device for future reference or later use. There are two ways of saving a document:

- (i) **Save:** It updates an active or existing document in a given location. If it is a new document, clicking on **Save** displays the **Save As** dialog box.
- (ii) **Save As:** It allows one to enter a new file name and specify the location where to store the document. If there is an existing file with the same name, the file is overwritten.



Picture 3. 23: Saving a file

When saving a file for the first time, do the following:

- (i) Click on **File** tab and select **Save As** command. A dialogue box appears.
- (ii) Select the location to save the file from the left pane, for example, in a folder in **Documents**, or in **This PC**.
- (iii) Type the file name in the **File name** box.
- (iv) Click the **Save** button. The document is saved in the selected location.

D. Opening a file

To open a file, do either of the following:

Double-click on the file or Right-click and then click **Open** from the pop-up menu.

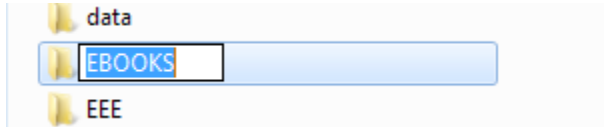
E. Renaming a file and a folder

Renaming refers to changing the name of a file or a folder. To rename a folder, do the following:

- (i) Double click **This PC** icon on the desktop.
- (ii) Select the folder/drive where the folder was initially stored. For example, **Desktop**.
- (iii) Click on the folder to be renamed.
- (iv) Click on the **Home** tab in the menu bar, select the **Rename** command in the **Organize** Group. The original name will be highlighted and surrounded with a box.
- (v) Type the new name and press the **Enter** key

To rename a file, do the following:

- (i) Right-click on the file to be renamed.
- (ii) Select and click **Rename** option from the pop-up menu.
- (iii) Type the new name over the old name then press the **Enter** key.



Picture 3. 24: Renaming a Folder



Picture 3. 25:
Renaming a file

F. Cutting and pasting folders and files

Cutting is also known as moving. It is the transfer of the selected file or folder to a new location without leaving a copy at the source location.

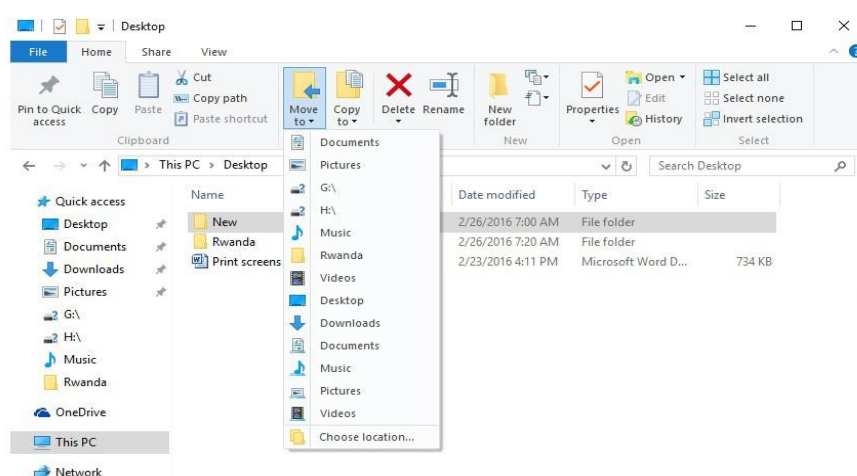
It is done using the **Cut command**. Once the file(s) or folder(s) to be moved are selected, they can be cut and pasted at any location by use of **menu**, **keyboard shortcuts** or by **dragging and dropping**.

The operating system automatically stores the cut data in a clipboard from where it will be pasted to the new location. Pasting refers to placing cut file or folder to a new location.

(a) Using the menu

To move the selected file(s) or folder(s) to another location using menus, do the following:

- (i) Double click **This PC** icon on the desktop.
- (ii) Select the folder/drive where the folder/file was initially stored, for example, **Desktop**.
- (iii) Click on the folder to be moved.
- (iv) Click on the **Home** tab in the menu bar, select the **Move to** command in the Organize group.
- (v) Click on a location from the drop down menu that appears.



Picture 3. 26: Moving files to another location

(b) Using keyboard

To move selected file(s) or folder(s) using the keyboard commands, do the following;

- Select the folder to be moved. Press **Ctrl + X** to cut.
- Click on the new location for the folder. Press **Ctrl + V** to paste.

G. Copying and pasting folders

Copying is duplication of text which means that a copy remains at the source location.

a) Using Menu

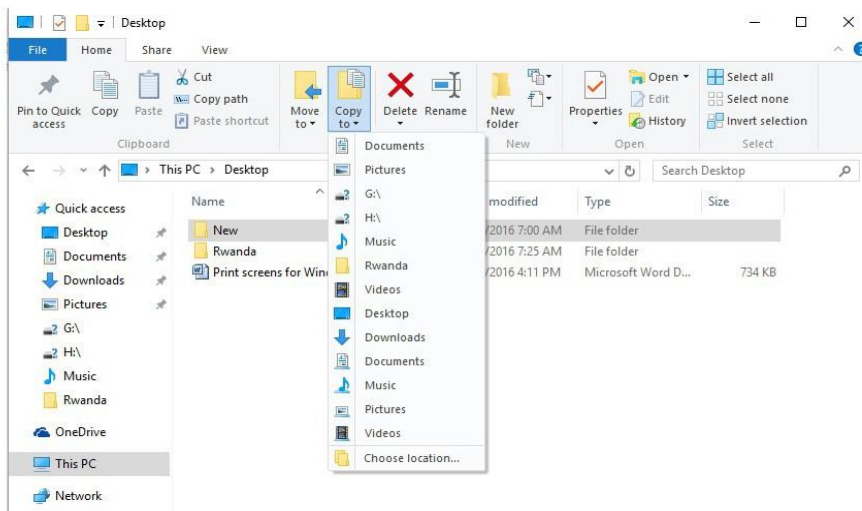
To copy selected file (s) or folder (s) to another location using menus, do the following:

- Double-click **This PC** icon on the desktop.
- Select the folder/drive where the folder was initially stored, for example, **Desktop**.
- Click on the folder to be copied.
- Click on the **Home** tab in the menu bar, select the **Copy to** command in the **Organize** group (see picture below: Copying files to another location).
- Click on a location from the drop down menu that appears.

b) Using the keyboard

To copy selected file (s) or folder (s) using the keyboard commands, do the following;

- Select the folder to be moved. Press **Ctrl + C** to copy.
- Click on the new location for the folder. Press **Ctrl + V** to paste.



Picture 3. 27: Copying files to another location

c) Cutting and pasting using drag and drop.

This option can be used to move or copy file(s) or folder(s). To use drag and drop do the following:

- (i) Click the folder(s) or file(s) to be copied or moved.
- (ii) Hold down the left mouse button.
- (iii) Drag the selected folder to the preferred destination.
- (iv) Release the mouse button.

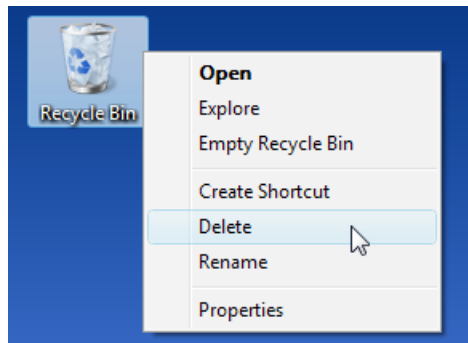
Note: Dragging and dropping file or folder in the same drive moves the file or folder to the new location. Dragging and dropping the file or the folder on different drives creates a duplicate.

H. Deleting files or folders

Most Windows Operating Systems usually protect files and folders deleted from the hard disk by placing them in a **Recycle Bin**.

The deleted files and folders remain in the **Recycle Bin** until it is emptied.

Files or folders deleted accidentally or prematurely can be retrieved from the recycle bin.



Picture 3. 28: The Recycle Bin holds the deleted files and folders

Since the Recycle Bin has been emptied, its contents are permanently deleted from the computer and cannot be recovered. To delete a file, do either of the following:

Click on the file or folder to be deleted. Press **Delete** or Right-click on the folder to be deleted and then select **Delete** option from the pop-up menu.

Application activity 3.4

1. Explain the difference between cut and copy.
2. Describe what happens when a file is pasted.
3. Create a text file and save it with a meaningful name.
4. Create a folder and give it a suitable name.
5. Explain how folders help you find files easily.
6. Move one file and copy another to a new folder.
7. Explain why file organization is important.

End Unit assessment

- 1) Answer with Yes or No.
 - a) Is the desktop the first screen displayed when a computer starts?
 - b) Is the taskbar found at the bottom of the screen by default?
 - c) Is the Control Panel used to change system settings?
- 2) State two functions of the Start button.
- 3) Mention three settings that can be changed using the Control Panel.
- 4) State the two types of windows.
- 5) A learner accidentally deletes an important file from the Desktop.
 - a) Where is the deleted file first stored?
 - b) Explain how the learner can restore the file.
 - c) What will happen if the Recycle Bin is emptied?
- 6) MUGISHA finds that the computer screen is too bright and the keyboard is typing wrong characters.
 - a) Which tool should he use to solve these problems?

- b) Name two settings he can change to solve the problem.
- c) Does changing these settings affect stored files? Explain.
- 7) MUTONI opens Microsoft Word and Paint at the same time.
 - a) What type of windows are Microsoft Word and Paint?
 - b) Name the button used to make a window occupy the whole screen.
- 8) Match the item in Column A with the correct function in Column B.

Column A	Column B
a) Recycle Bin	i) Stores deleted files temporarily
b) Taskbar	ii) Stores related files
c) Control Panel	iii) Used to change system settings
d) Minimize button	iv) Hides a window on the taskbar
e) Folder	v) Shows running programs

- 9) Open two programs and demonstrate how to switch between them.
- 10) Create a folder named ICT_S1 and save a text file inside it. Demonstrate how to:
 - a) Rename a file b) Copy a file c) Move a file
 - d) Delete a file and restore it from the Recycle Bin.
- 11) How does adjusting screen brightness and sound responsibly help in protecting health and equipment?

UNIT 4: WORD PROCESSING



Key unit competence

Create documents using Word Processing features

Learning objectives

At the end of this unit, you should be able to:

Knowledge and understanding

- Identify the parts of the Word Processing environment and their uses
- Explain steps to carry out different activities in a Word Processing environment
- Explain the purpose of paragraph tools

Skills

- Manipulate documents by using different tools and features available in Word.
- Move in sections of text from one place to another in a Word document
- Navigate in different locations of a Word document
- Use Track Changes and Comments to review documents.
- Create and apply styles and headings for document organization.

Attitudes and values

- Show interest in formatting and manipulating a document.
- Appreciate advanced formatting and proofing features of Word Processing for editing a document
- Demonstrate responsible use of technology and proper saving of work.

4.0. Introduction

This unit helps you understand Word Processing and how it is used to create, edit, and manage text documents using a computer. You will explore Word Processing applications, how to start them, and become familiar with its environment. The unit also covers working with documents including creating, saving, opening, and renaming files. You will develop skills in selecting and formatting text, using paragraph tools, moving and editing text, navigating through documents, setting page layout options, and using document reviewing tools such as track changes, comments, styles, headers, and footers.

Introductory activity

Kalisa is asked to prepare a two-page school report using a Word Processing application. He opens the application and creates a new document. He types the title of the report and formats it using a larger font size, bold style, and center alignment. While typing, he realizes that one paragraph should appear before another, so he cuts and pastes the text to the correct position. Later, he saves the document with a new name, changes the page orientation to landscape for a table, and adds a footer with page numbers. Before submitting the report, he uses the find and replace tool to correct repeated spelling mistakes and checks comments added by his teacher using track changes.

- 1) Which Word Processing activity did Kalisa perform when he created a new file?
- 2) Name two text formatting tools Kalisa used for the title.
- 3) Which command is used to move a paragraph from one place to another?
- 4) What page layout feature did Kalisa change for the table?
- 5) Which tool helped Kalisa correct repeated words quickly?
- 6) Why is saving a document with a new name important?

4.1. Introduction to Word Processing

Activity 4.1

Here are software: Microsoft Word, Paint, Excel, WordPerfect, PowerPoint, Notepad.

- 1) a) Tick the Word Processing applications.
b) State the use of each selected application.
- 2) Identify three roles of a Word Processing application.
- 3) Give two examples of documents that can be created using Word Processing software.

4.1.1. Introduction to Word Processing application

A Word Processing application is a software that enables the user to create, save, edit, format, and print text documents. It is used to create documents such as letters, memos, reports, and minutes among others. Examples of Word Processing applications are: Word Perfect, Word Star, Lotus Word Pro, AmiPro and Microsoft Word among others.

4.1.2. Starting Word Processing applications

Microsoft Word is an example of a Word Processing application. It is available in various versions incorporated in an integrated package called **Microsoft Office Suite**.

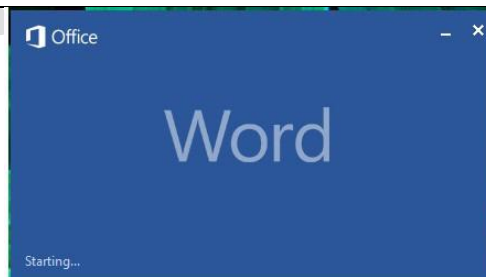
Examples of Ms Word versions include Ms Word 2000, Ms Word 2003, Ms Word 2007, Ms Word 2010, Ms Word 2013, Ms Word 2016 and Ms Word 2019 among others. Ms Word 2013 has been used in this unit to illustrate the use of a Word Processing application.

To launch Ms Word 2013, do one of the following:

- (i) Click on **Start button**, go to **All Programs** and select **Microsoft Office**; then click on **Microsoft Word 2013**.
- (ii) If the Microsoft Word icon is on the Desktop, do one of the following:
 - Right-click on the icon then select **Open** from the pop-up menu that appears.
 - Double-click on the icon.
 - Click on the icon then press **Enter** key.
- (iii) Click on **Start** button then select **Microsoft Word 2013** if it was pinned to the **Start Menu**.
- (iv) Click **Microsoft Word 2013** if it is pinned to the Task bar.
- (v) Click on the **Start** button go to **Search programs and files**, type **Run** then press **Enter** key. On the dialog box that appears, type the word **WINWORD**. Click **OK**.



Picture 4. 1: Microsoft 2013 Word icon



Picture 4. 2: Starting Microsoft Word screen

Application activity 4.1

- 1) State four functions of a Word processing application.
- 2) List four examples of Word processing applications.
- 3) Explain two ways of starting Microsoft Word and do it on your computer.

4.2. Word Processing environment

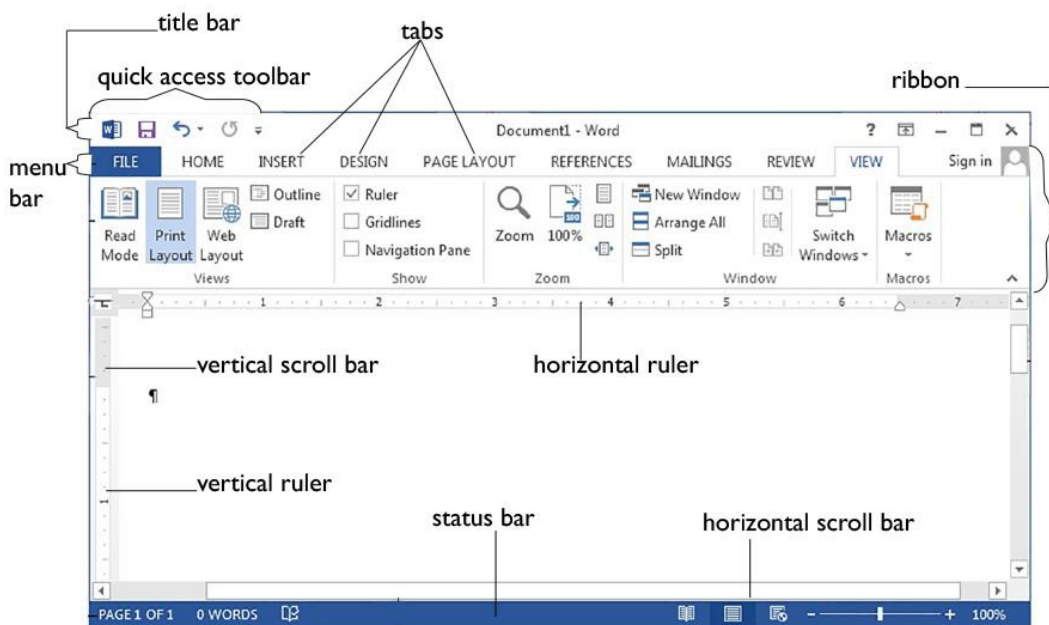
Activity 4.2

Jeanne opens Microsoft Word to type a school report. She uses different parts of the Word window to complete her work.

- 1) What is the top bar that shows the program and document name called?
- 2) Which tab is used to save a document?
- 3) Which part helps move up and down the document?
- 4) What is the name of the large blank area in the middle where text is typed?
- 5) Which part shows the page number and word count at the bottom?

In the environment of a Word Processor, the following basic tools or features can be used: title bar, tabs, menus, toolbars, scroll bar, ruler, and status bar.

- **Title bar:** This is the topmost bar on the window. It contains minimize, restore/maximize, and close buttons as well as the name of the program and document. Any time a new word document is opened, Ms-Word gives it a temporary or a default name which could be Document1, Document2, and so on. When the document is saved, the default name is automatically changed to the new name.
- **Tabs:** This term refers to the part of a window or a dialog box which when clicked, displays different parts of a menu, program window, or webpage.



Picture 4. 3: The Word processing environment

- **Menu Bar:** This bar displays a series of tabs namely File, Home, Insert, Page Layout, References, Mailings, Review and View. Other tabs appear if an image is inserted. Clicking on each tab displays a different ribbon.
- **File tab:** It is the first item on the menu bar. When clicked it produces a pull-down menu which contains a list of options to choose from such as New, Open, Save, Save As, Print, and Close.
- **Home tab:** It contains icons used for text formatting, paragraph formatting, and editing among others.

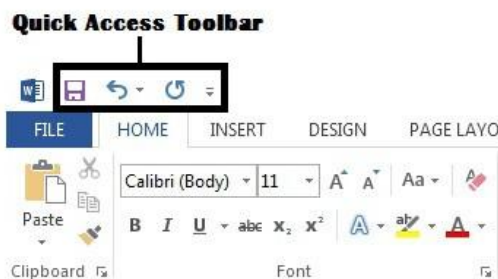
- **Insert tab:** It contains icons that allow the user to add graphics and images in a document. It can be used to add pictures, tables and word art among others.
- **Design tab:** It consists of document formatting features, themes and features for formatting page background such as watermark.
- **Page Layout tab:** It contains icons that are used for formatting an entire page and paragraph formatting icons among others.
- **References tab:** It contains icons that are used for creating and manipulating Tables of Contents, Footnotes, Captions, Citation, and Bibliography among others.
- **Mailings tab:** It contains icons that are used during mail merging.
- **Review tab:** It contains icons used for proofreading, adding comments, tracking, and protection among others.
- **View tab:** It contains icons that are used to show the document views, add macros, zoom, and change the window appearance.
- **Ribbon:** It consists of icons of commands which are organized into groups and given a group name. For example, clicking on the **Home** tab displays a ribbon with five groups namely Clipboard, Font, Paragraph, Styles, and Editing.
- **The Quick Access Toolbar:** This is a small selection of the most commonly used commands. In most cases, it is located at the top left corner of the Title Bar but it can be placed above or below the Ribbon.

It can be customized by adding more commands on it or removing commands from it.

By default, this bar contains Save, Undo and Redo commands.

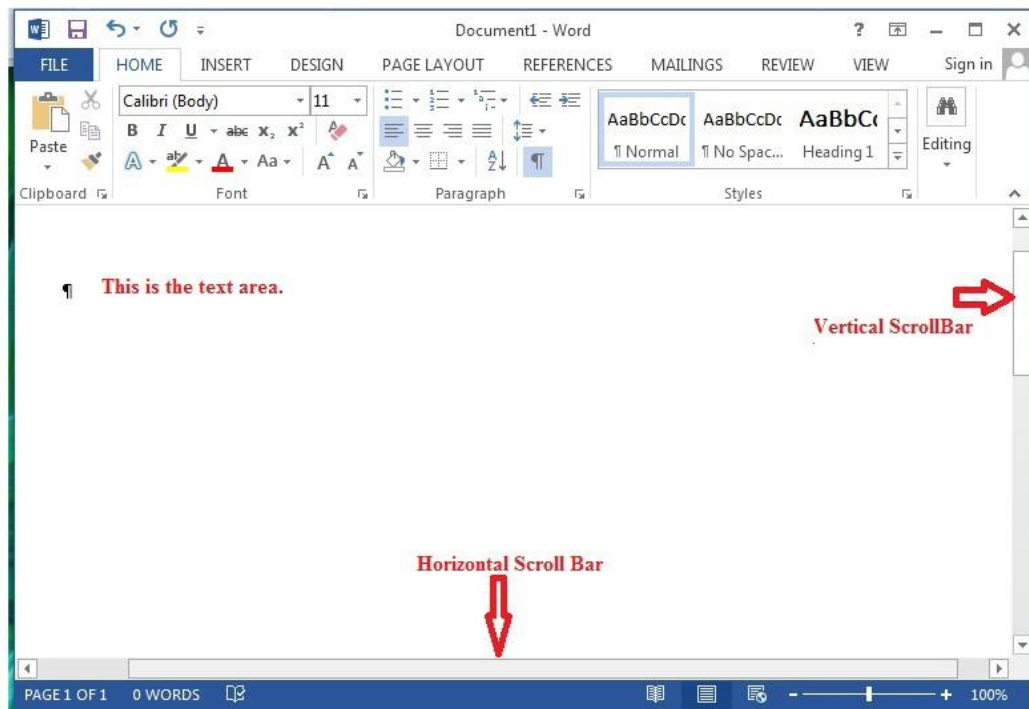
To customize it:

- Right click within the quick access toolbar, select **Customize Quick Access Toolbar** option. A dialog box is displayed.
 - Select the commands to add from the left pane then click **Add>>** button.
 - To remove the commands from the quick access toolbar, click on the command from the right pane of the window then select **<<Remove** button.
- **Toolbars:** These are stripes that contain icons used for formatting, editing or inserting data. Examples of toolbars are drawing toolbar, formatting toolbar, standard toolbar, and picture toolbar among others.
 - **The Scroll bars:** They allow one to move or navigate through a large document. There are two types of scroll bars. The **vertical scroll bar** allows the content of a document to



Picture 4. 4: The Quick Access toolbar

be moved from top to bottom. The **horizontal scroll bar** allows the content of the document to be moved from left to right and vice versa.



Picture 4. 5: The vertical and horizontal scroll bars

- **Rulers:** The ruler is used to set the layout of the text across the page such as margins, tab positions, and indents. There are two rulers namely the **horizontal ruler** on top of the document and the **vertical ruler** on the left side of the document. Refer to picture below. It shows the horizontal and vertical rulers.
- **Status bar:** This bar is usually located at the bottom of the window. It displays information about the activities being carried out in the current document and certain general information about the document such as the current page, word count, view buttons and zoom level tools.
- **Working area/zone:** Also called text area it is the largest part of a Word Processing window. It is the area where text and objects are displayed. It is enclosed with the top, bottom, left, and right margin.
- **View tabs:** These are used to set the working mode in a Word document. That mode can be Read mode , Print Layout  and Web Layout . They are located at the bottom right of a Word document.

Application Activity 4.2

1) Answer with True or False

- A Word Processing application is used to create and edit text documents.
- Word processing software cannot save documents.
- Letters and reports can be written using a Word processing application.

- d) Microsoft Word is an example of a word processing application.
 e) Microsoft Word is part of the Microsoft Office Suite.
- 2) List the names of the visible parts of the Word window when it is open.
- 3) Write the function of each part using the table below:

Part of Word Window	Function
Title bar	
Ribbon	
File tab	
Scroll bar	
Status bar	
Working area	

4.3. Working with documents

Activity 4.3

- 1) What is a document in a computer?
- 2) When you open Microsoft Word, what appears on the screen?
- 3) Why is it important to save your work?
- 4) What is the difference between Save and Save As?
- 5) How can you open a document that you saved before?
- 6) Complete the table below:

Document Operation	Purpose
Create document	
Save document	
Save As	
Open document	
Rename document	

A document can be created, saved, opened, and renamed within the work area.

A. Creating a document

It is the process of starting a new blank page from scratch or from an existing document. When the Microsoft Word program is launched, a new document is automatically created. However, to create another new document do one of the following:

- Click the **File** tab. Select **New** on the window that appears; then select **Blank document**; or
- Press **Ctrl + N** on the keyboard; or

- Click the **New** icon if it is on the **Quick Access Toolbar**.

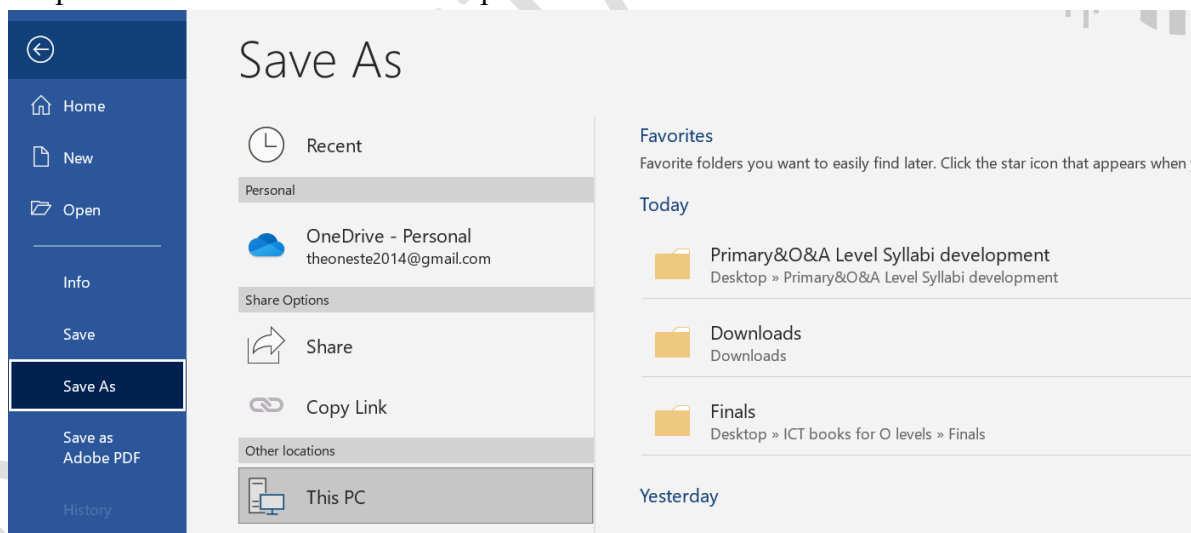
B. Saving a document

Saving is the process of copying a document from primary memory to a secondary memory for future reference or later use. There are **two** main commands used when saving a document which are: **Save** and **Save As**.

- Save:** It is used when updating an existing document. Clicking on **Save** if it is a new document results in the display of the **Save As** dialog box. To update a document do one of the following:
 - Click **File** tab then select **Save** command.
 - Press **Ctrl + S** on the keyboard.
 - Click **Save** icon on the **Quick Access Toolbar**.
- Save As:** It is used when:
 - Saving a document for the first time in the computer.
 - Saving the document using a new file name. Changing the storage location of a document.

To use **Save as** command do the following:

- Click **File** tab then select **Save As** command.
- Specify the location where the file is to be stored by clicking **My Computer** in the first pane then **Documents** in the last pane.



Picture 4. 7. The Save As option dialog box

- In the **File name** box, type the name of the file.
- Click the **Save** button or press the **Enter** key.

C. Open an existing document

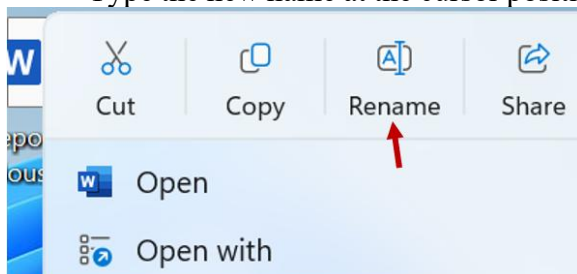
It is also known as **retrieving** a document. This is the process of viewing the content of an existing document. To open a document, do one of the following:

- Click on the **File** tab on the menu bar. A pull-down menu appears. Select **Open** command. Click **computer** from the left pane then select **Browse** if the document was not among the recent ones. A dialog box is displayed. Specify the location where the document was previously stored and the file name. Click the **Open** command; or
- Click on the **File** tab on the menu bar. A pull-down menu appears. Select **Recent Documents** command. All recent documents are displayed on the right pane. Click on the desired file name to open it; or
- Open the location where the document was stored then open the document.

D. Renaming a document

To rename a document:

- Right-click on the document name then select **Rename**.
- Type the new name at the cursor position then press the **Enter** key.



Picture 4. 8. From left to right: The Rename option and a file ready for renaming

Application activity 4.3

Opens Microsoft Word and perform the following tasks step by step:

- 1) Create a new blank document using any method.
- 2) Type a short paragraph of 3 to 5 lines.
- 3) Save the document using Save As with the name “My First Document”.
- 4) Close Microsoft Word.
- 5) Reopen Microsoft Word and open the saved document.
- 6) Rename the document to “My Edited Document”.

4.4. Text selection

Selecting a character, word, sentence, paragraph and the whole document.

Activity 4.4

- 1) What does it mean to select text in a document?
- 2) Why do we need to select text before formatting or editing it?
- 3) How can you quickly select a whole document?
- 4) How do you remove selection to a selected text?

A. Selecting a text

The selecting feature allows text or another object to be selected with the mouse or keyboard. When text is selected for formatting or editing, it becomes highlighted. The highlight goes off when the action of formatting or editing is completed, or when one clicks on another area on the screen.

To select any block of text, take the cursor to the beginning of the block of text then do one of the following:

- Click and press the left button of the mouse then drag the mouse over the text. Or
- Press **Shift + arrow keys**, depending on the direction of the desired text. However, to quickly select the specified text, the following options can be chosen:

To select:	Action to be taken
One character	• Place your cursor before or after it and use Shift+Right Arrow (to select forward) or Shift+Left Arrow(to select backward) • Or use your mouse to click and drag over just that single character
A word	Double-click anywhere within the word.
A paragraph	Triple-click anywhere within the paragraph.
A sentence	Press the Ctrl key then click anywhere within the sentence.
An entire document	Press CTRL +A
A line of text	Take the pointer to the left of the line until it changes to a right pointing arrow and then click.
Multiple sequential items	• Click the first item • Hold down the shift key and click the last item then release the shift key.

Table 4. 1: Options for selecting text

B. Deselecting text

To deselect text, click anywhere on or away from the selected text.

Application activity 4.4

- 1) Answer with True or False
 - a) Can text be selected using only the arrow key?
 - b) Does selected text appear highlighted on the screen?
 - c) Is Ctrl + A used to select the entire document?
 - d) Is a Word selected by double-clicking on it?
 - e) Is a paragraph selected by triple-clicking inside it?
 - f) Is Ctrl + Click used to select a sentence?
 - g) Does selected text remain highlighted after deselecting it?
- 2) Explain how you can select a single character using the keyboard.
- 3) Describe two ways of selecting a block of text.

- 4) How do you select an entire document?
- 5) What steps do you follow to select a sentence?
- 6) How can you deselect selected text?

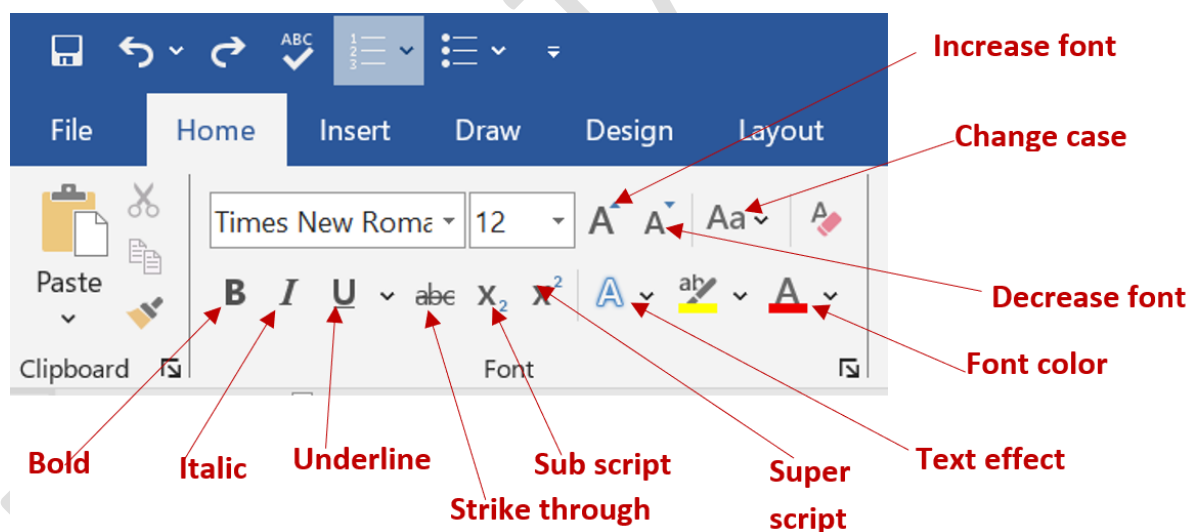
4.5. Text font

Activity 4.5

- 1) Answer with True or False:
 - a) Text formatting deal with the appearance of text.
 - b) A font is a set of printable characters in a specific style and size.
 - c) Bold formatting makes text appear thicker or darker.
 - d) Italic formatting make text get underlined.
 - e) Superscript is used to place text slightly below the typing line.
- 2) What is a font? Give two examples of fonts.
- 3) Explain how font size affects the appearance of text.

Formatting is the process of enhancing the appearance of a document, making it attractive or stand out from the rest. Formatting features are divided into two broad categories namely: text and paragraph formatting.

In the image below are some tools used in text formatting:



- A. **Font**: A font is a set of printable text characters in specific style and size. Font is a combination of typeface and other qualities such as size and space. Examples of font include Calibri, Times New Roman, Calibri, Arial Black and Arial Narrow among others.
- B. **Font style**: There are four types of font styles namely; bold, italic, regular and bold italics.
 - **Bold**: Makes the selected text appear thicker or darker than the rest of the text in a

document.

- **Italic:** Makes the selected text slant forward.

C. Font size: It refers to how small or large the characters are set. The standard size is 12.

D. Font colour: The default font colour is black. The user can, however, change this colour to the desired one, by selecting options from **Font Color** box

E. Underline: It is a line placed under the letter, word, or sentence. There are various underline styles that the user can choose from. Examples include words only, double underline and dotted lines.

F. Font Effects: There are various font effects that can be applied on a document to enhance the text appearance. These include superscript, subscript, strikethrough, double strikethrough, all caps, small caps, and hidden.

Superscript: It is where text is formatted to appear slightly above the normal typing line. For example, in cm^3 , text 3 is a superscript.

Subscript: It is where text is formatted to appear slightly below the normal typing line. For example, in “ H_2O ”, the text 2 is a subscript.

Strike through: It is used to cancel already typed text. It draws a line over the text. To use any of the text formatting features, do the following:

- (i) Select the desired text.
 - (ii) From the **Home tab**, either select the desired text formatting icon on the **ribbon** under the **Font** group or click on the dialog box launcher (arrow) besides the **Font** group label. From the dialog box that appears, select the **font** tab then select the desired formats and click **OK**.
- G. Change case:** Refers to the process of switching between the available cases. The following are the various types of cases available:
- **Sentence case:** Only the first character of a sentence is capitalized. In this sentence the first letter of the first word in the sentence is in capital letters.
 - **Upper case:** All the characters in a word or sentence are capitalized.
 - **Lower case:** All the characters in a word or sentence are displayed in small letters.
 - **Capitalize Each Word:** The first character of each word in a sentence is in capital letters. It is also known as title case in the lower versions of MS Word.
 - **Toggle CASE:** All characters in upper case are changed to lower case and vice versa.

The table below illustrates changing case. The original sentence reads as follows:

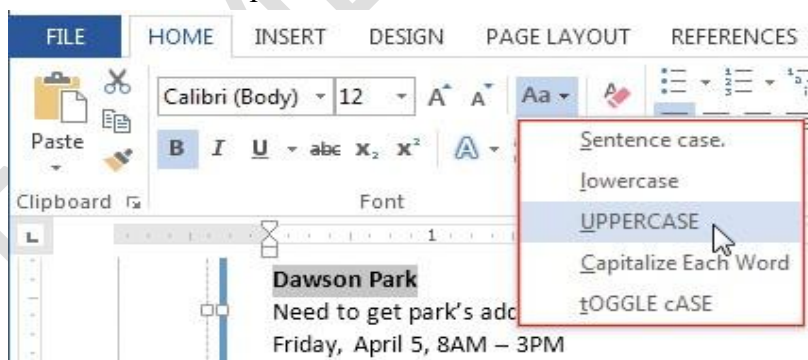
Example of sentence	Type of case
Abstaining from premarital sex is the most assured way of avoiding contracting the hiv and aids and other sexually Transmitted infections.	Sentence case

ABSTAINING FROM PREMARITAL SEX IS THE MOST ASSURED WAY OF AVOIDING CONTRACTING THE HIV AND AIDS AND OTHER SEXUALLY TRANSMITTED INFECTIONS.	UPPER CASE
Abstaining from premarital sex is the most assured way of avoiding contracting the hiv and aids and other sexually transmitted Infections.	Lower case
Abstaining From Premarital Sex Is The Most Assured Way Of Avoiding Contracting The Hiv And Aids And Other Sexually Transmitted Infections.	Capitalise Each Word
Abstaining FROM PREMARITAL SEX IS THE MOST ASSURED WAY OF AVOIDING CONTRACTING THE hiv AND aids AND OTHER SEXUALLY TRANSMITTED INFECTIONS.	Toggle case

Table 4. 2: Changing case

To change case, do the following:

- Select the text to be formatted.
- From the **Home** tab, click on the arrow in the **Change case** icon under the **Font** Group.
- Select the desired option.



Picture 4. 9: Changing case in Microsoft Word 2013

Application activity 4.5

- Answer with True or False:
 - Strikethrough used to delete text so that it is no longer visible on the screen.

- b) Is Ctrl + A used to change case in a document?
- c) Does UPPER CASE convert all selected letters to capital letters?
- 2. With examples explain the difference between superscript and subscript.
- 3. Describe the steps used to apply text formatting in a document.

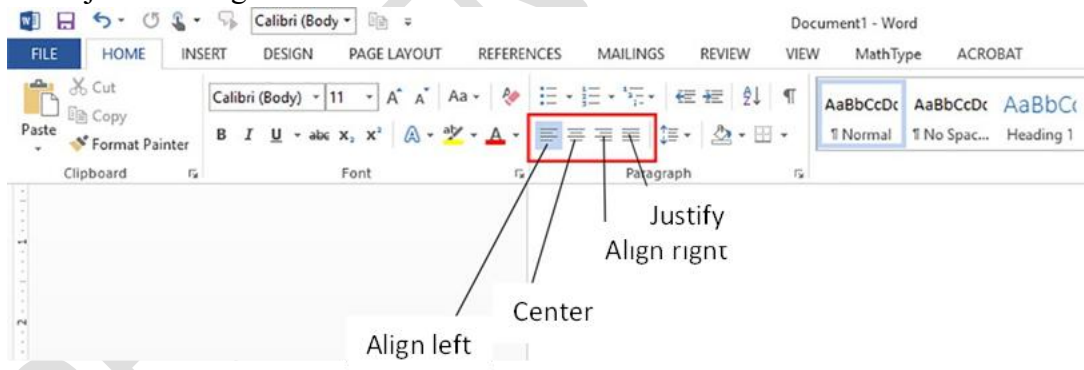
4.6. Paragraph tools

Activity 4.6

- 1) Which type of alignment that makes the text evenly arranged at both the left and right margins?
 - a) Left Alignment
 - b) Center Alignment
 - c) Justified Alignment
 - d) Right Alignment
- 2) Answer with True or False:
 - a) In a Hanging Indent, the first line is pushed away from the margin while the rest of the lines stay in the same place.
 - b) The default line spacing in most Word Processors is "Double."
 - c) Sorting data in "Descending order" means arranging it from Z to A or largest to smallest.

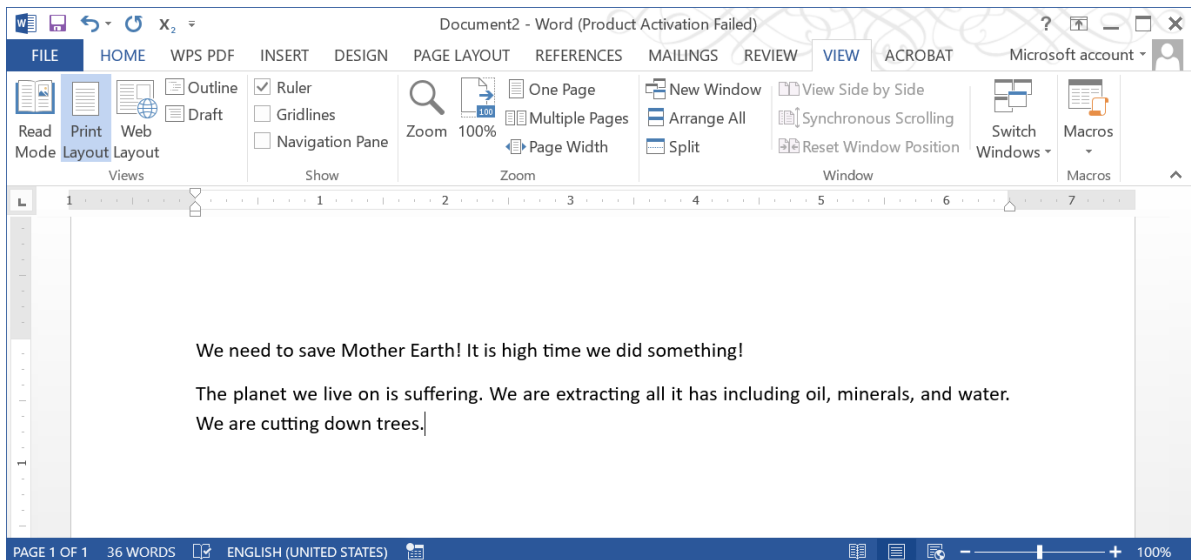
Paragraph tools are features used to improve the appearance of an entire paragraph. They include alignment, indentation, spacing, tabs, Bullets and Numbering.

A. Alignment: It refers to the arrangement of text relative to the left or right margin of a page. There are four types of alignments namely: left, right, centre, and justified alignments.



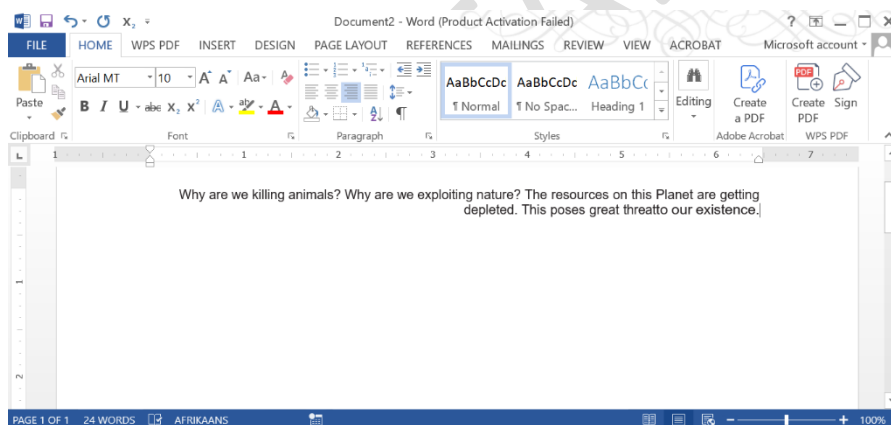
Picture 4. 10: Alignments

- i. **Left alignment:** This is where text is evenly positioned along the left margin but uneven at the right margin. Typing begins at the left margin and does not have to end at the right margin.



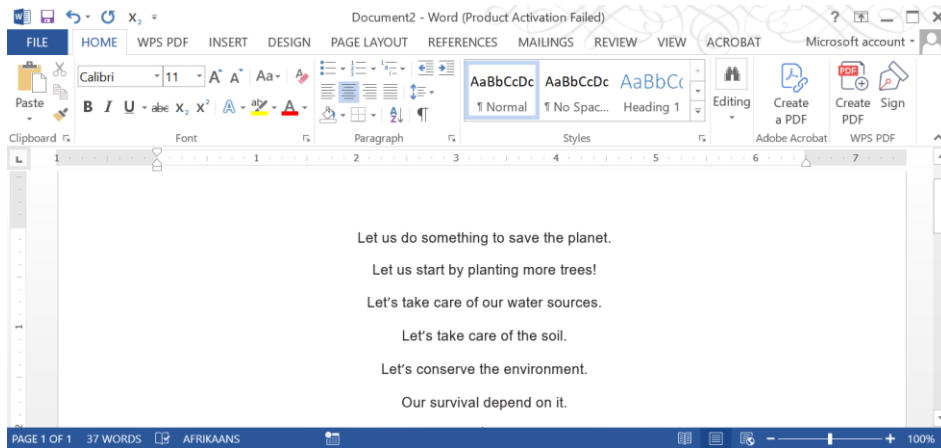
Picture 4. 11: Left alignment

- ii. **Right alignment:** This is where text is evenly positioned along the right margin but uneven at the left margin. Typing begins at the right margin and does not have to end at the left margin.



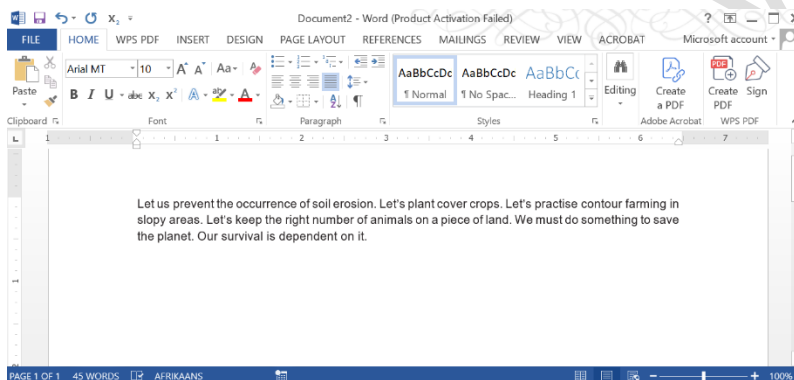
Picture 4. 12: Right alignment

- iii. **Center alignment:** This is where text is evenly arranged at the center of the page but uneven both at the left and right margins. Typing begins at the centre of the page.



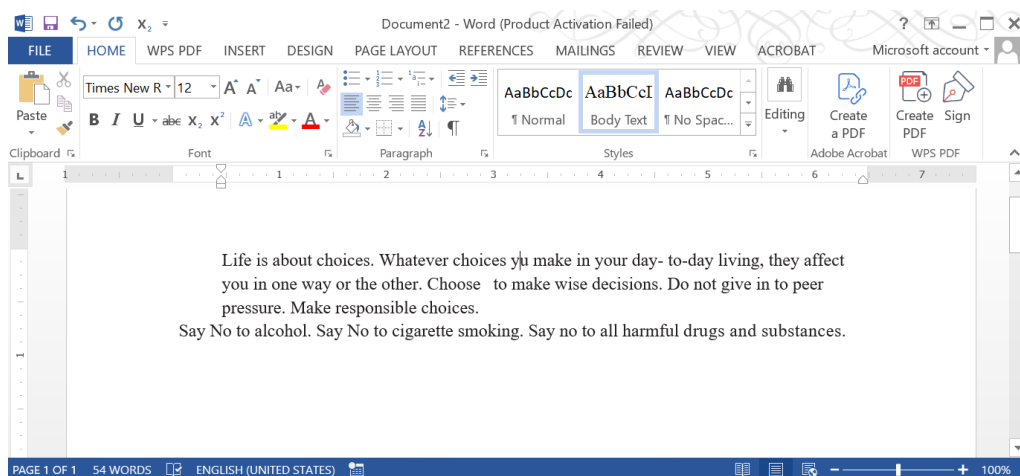
Picture 4. 13: Center Alignment

- iv. **Justification:** This is where text is evenly arranged both at the right and left margins.



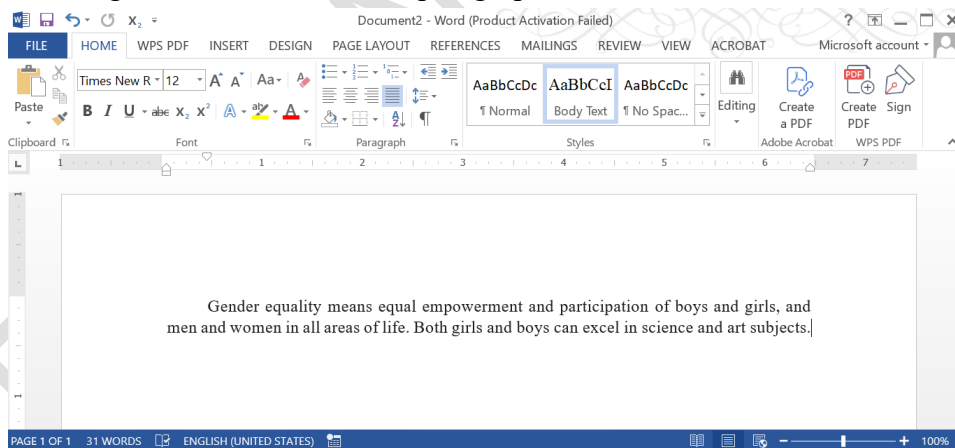
Picture 4. 16: Jusify alignment

- B. Indentation:** It is the process of moving text away from the right or left margins by a given interval. They include right, left, and special indents.
 - i. **Left indent:** The lines of text in an entire paragraph are pushed away from the left margin by several units of measurements. The first paragraph in the following text is left indented.



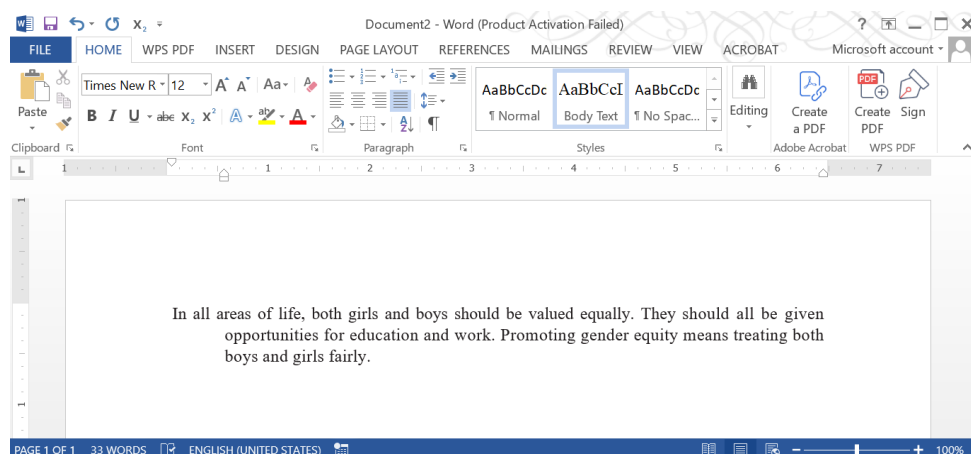
Picture 4. 17: Left indent

- ii. **Right indent:** The lines of text in an entire paragraph are pushed away from the right margin by several units of measurements. The first paragraph in the following text is right indented.
- iii. **Special indents:** There are two types of special indents, namely first line and hanging indents.
 - **First line indent:** The first line in a paragraph is pushed away from the left margin by several units of measurements. The other lines in the paragraph remain on the left margin. The first line in the paragraph is indented.



Picture 4. 18: First indent

- **Hanging indent:** The first line of a paragraph remains at the left margin. The rest of the lines in the paragraph are pushed several units of measurements away from the left margin. Hanging indent is applied in this paragraph.

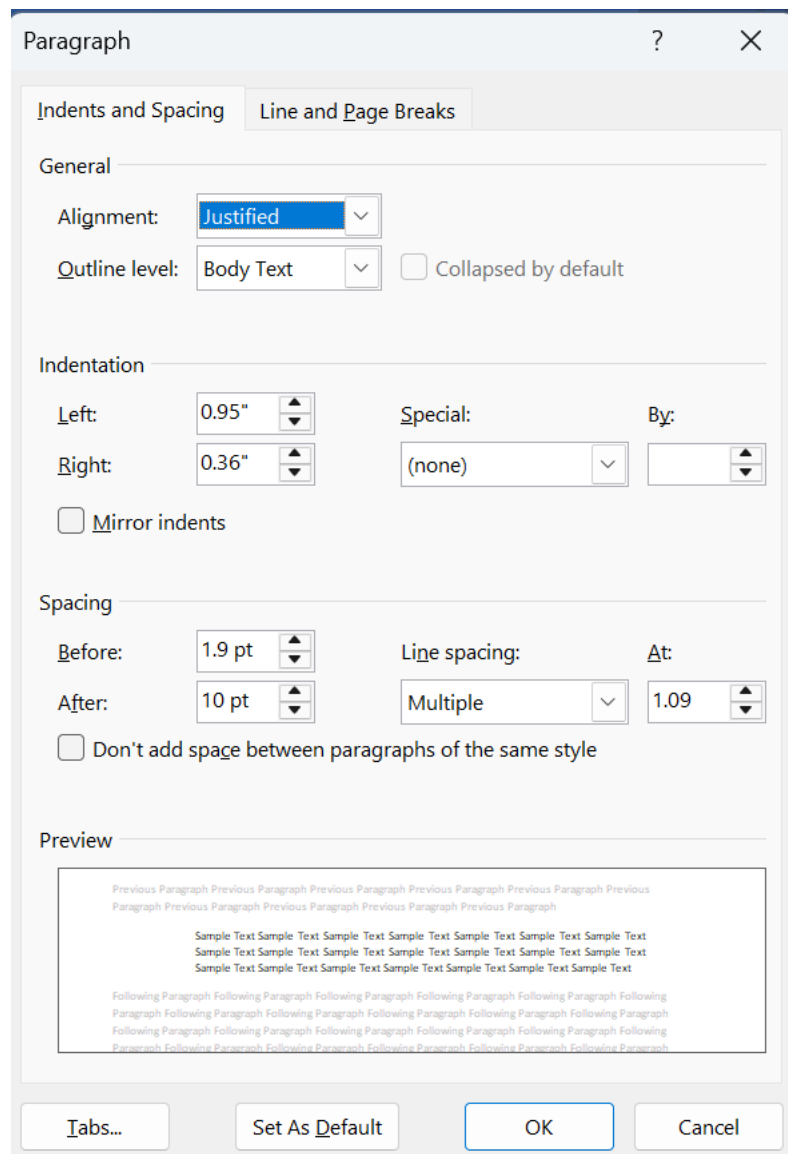


Picture 4. 19. Hanging indent

C. Line spacing: It refers to the distance between lines of the text in a paragraph or a document. The default line spacing is Single. Other options are 1.5 lines, Double, At least, Exactly, and Multiple line spacing.

To apply line spacing, indentation, bullets and numbering, and alignment do the following:

- Highlight the text. From the **Home tab**, either select the desired paragraph formatting icon on the **ribbon** under the **Paragraph** group; or click on the dialog box launcher besides the **Paragraph** group label. A dialog box appears as shown in Picture below.
- From the dialog box that appears, select the Indents and Spacing tab then select the desired formats and click OK.

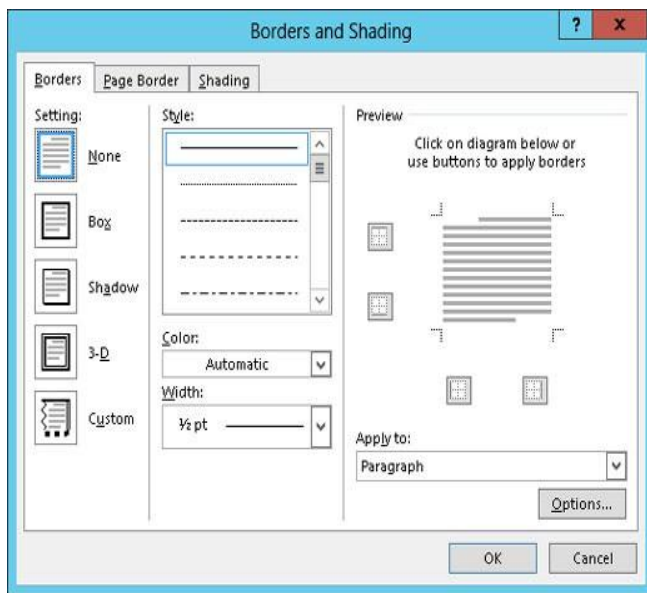


Picture 4. 20: Formatting paragraph

- D. Borders and shading:** The term **border** refers to a line separating two parts in a document. For example, to separate two cells in a table, a border can be used. **Shading** refers to the process of making an area in a document to have a relatively dark tone that produces a shade.

Shading can be applied to a drawing, a table, or a picture. To create borders and shading on a table, do the following:

- Click anywhere within the text.
- Click the **Page Layout** Tab, in the **Page setup** Group, click on the dialog box launcher.
- A dialog box is displayed. Select the **Layout** tab from the **Page Setup** dialog box.
- Click on the **Borders** command. The **Borders and Shading** dialog box is displayed as shown in below:.



Picture 4. 21: Creating borders and shading

- Select the **Borders** tab and specify the settings, style, colour, and width.
- In the **Apply to** section, specify the area where the border will be effected.
- Click **OK** to close the dialog box and effect the changes.

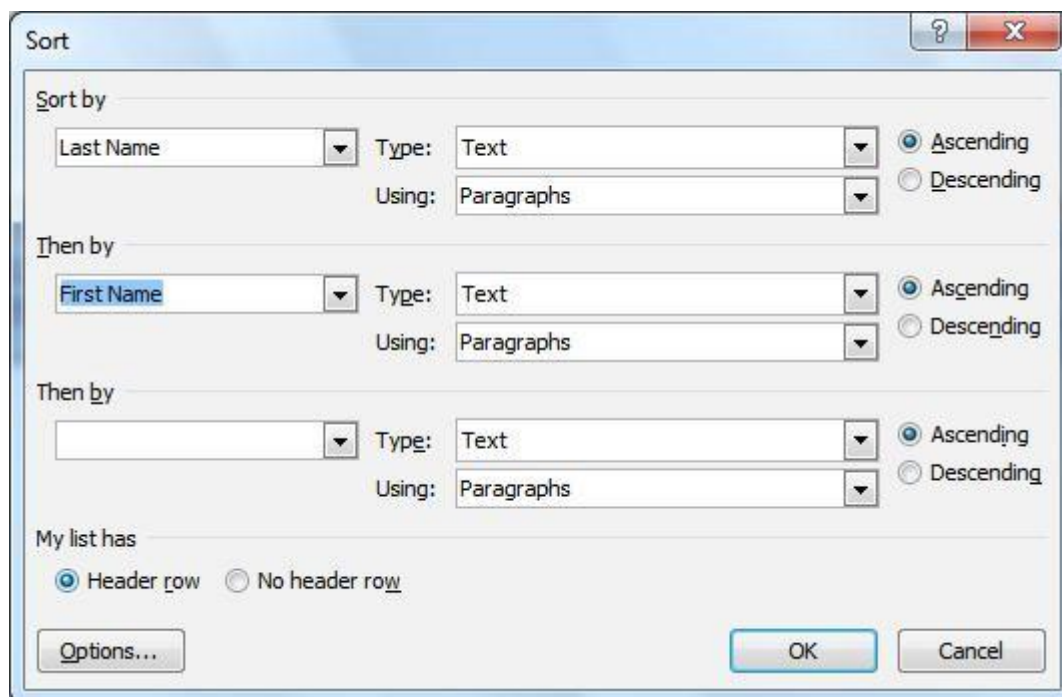
E. Sorting: This is the process of arranging data in a particular order; that is, either in ascending or descending order. Data can be sorted when it is in a paragraph.

Sorting a paragraph of data: A paragraph can be sorted using the first letter in each line. For example, when the following data is sorted in ascending order it will appear as follows:

Unsorted	Sorted
I am a patriot.	I am a law abiding citizen.
I am a law abiding citizen.	I am a patriot.
I will always defend my country. Let us work as a team.	I love my country.
We must protect our country.	I will always defend my country. Let us work as a team.
I love my country.	We must protect our country.

To sort data in a paragraph do the following:

- Select the paragraph. Click on **Sort** from the **Paragraph** group in the **Home** tab. A dialog box appears as shown in Picture below. Ensure that the words to be sorted are separated by a comma or by tab stops.
- Under **Sort by** select **Paragraphs**. Under **Type** select **Text**. Choose either ascending or descending sort orders. Click **OK** to apply.



Picture 4. 22: Sorting text alphabetically

F. Lists and nested lists

Bullets and numbers are used for the purpose of easier identification of items in the list. Nested lists refer to lists at different levels (sub-lists) that is a list within another list.

To create a list and a nested list, do the following:

Create the list with each item on its own line as shown below:

Vegetables

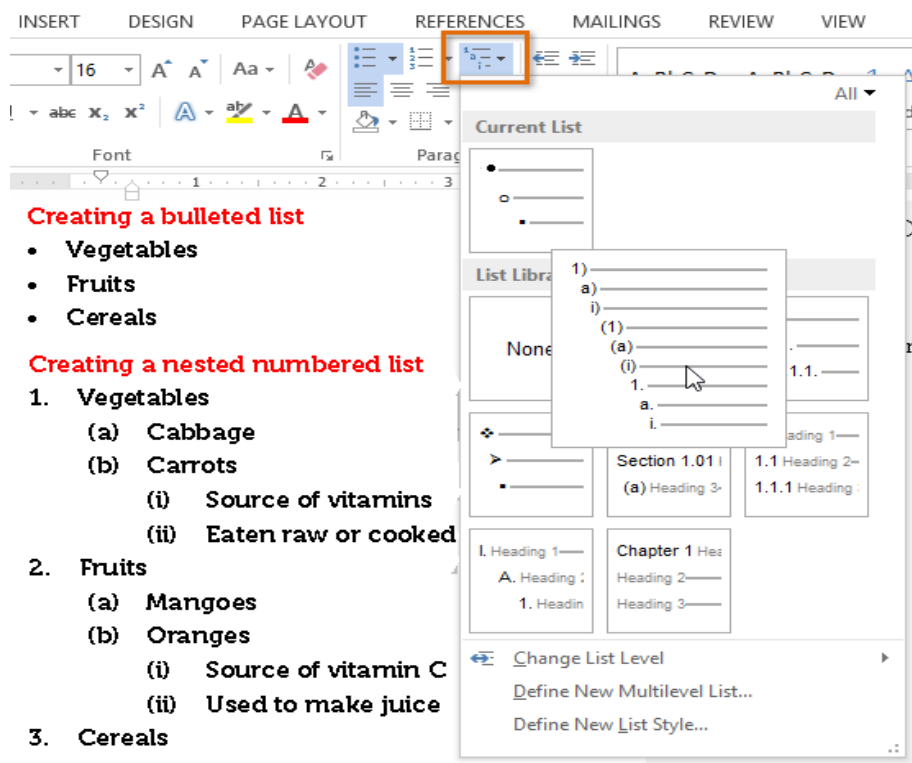
Fruits

Cereals

- (i) Select the list then, click on the **Numbering** or **Bullets** icons from the **Home** tab in the **Paragraph** group.
- (ii) From the resulting menu, select the desired type of bullet or numbering list to automatically apply it. For example:
 - Vegetables
 - Fruits
 - Cereals

To create a nested numbering list under vegetables, do the following:

- (iii) Click at the end of the word vegetables.



Picture 4. 23: Creating a nested list

- (iv) Press the **Enter** key and then press on the **Tab** key. A number is automatically inserted. Type the lists and subsists.
 - (a) Vegetables
 - Cabbage
 - Carrots
 - (b) Fruits
 - (c) Cereals

To close the current list level, either press the **Enter** key twice or press shift + tab

Application activity 4.6

- 1) Fill in the blanks

To create a nested list (a list within a list), you should press the enter key to start a new line and then press the _____ key to move to the next level.

- 2) Under which tab on the Microsoft Word ribbon can you find the sort command and alignment icons?
- 3) Compare and contrast a First Line Indent and a Hanging Indent.
- 4) What is purpose of using borders in a document? Explain how applying shading to a specific paragraph can improve the readability of a report.

4.7. Moving a section of text

Activity 4.7

- 1) Answer with True or False.
 - a) Ms Word allow text to be moved and copied within a document
 - b) Cut is used to move text from one location to another but text remains to its original location.
 - c) The clipboard is a temporary storage area.
 - d) Ctrl + C is the keyboard shortcut for Cut.
- 2) What is meant by moving a section of text in Ms Word?
- 3) Explain the difference between cutting and copying text.

MS Word contains commands that can be used for moving and copying text from one part of a document to another. These commands are cut, copy, and paste.

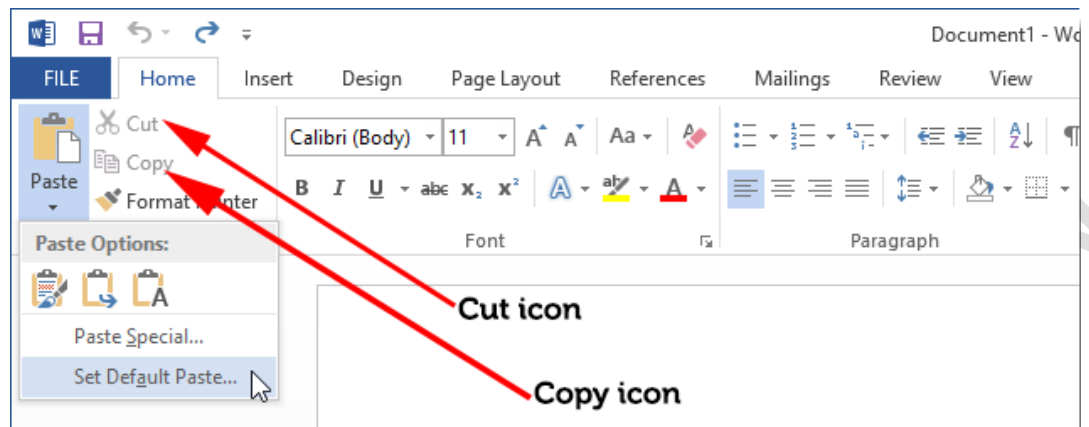
A. Moving a text (Cut)

Moving refers to changing the position of text to a new location. The original text is moved to a clipboard awaiting to be pasted to the new location.

A clipboard is a temporary space where all items to be moved or copied are stored before they are pasted to the desired location.

To move text, do the following:

- (i) Select or highlight the text. Use one of the following options:
 - From **Home tab**, click **Cut** icon.
 - Right-click on the selected text and click **Cut** command from the resulting pop-up menu.
 - Press **Ctrl + X** on the keyboard.
- (ii) Position the cursor where the text is to be moved.
- (iii) To paste, use one of the following options:
 - Click **Home tab**, under the **Clipboard** group, click **Paste** icon.
 - Right-click and select **Paste** Command from the pop-up menu.
 - Press **Ctrl + V** on the keyboard



Picture 4. 24: Picture that shows copy and cut icons

B. Copying text

Copying refers to creating a duplicate of text. To copy, do the following:

- (i) Select or highlight the text or document to be copied.
- (ii) Use one of the following options:
 - From **Home** tab, click **Copy** icon under the **Clipboard** group.
 - Right-click on the selected text and click **Copy** command from the pop-up menu.
 - Press **Ctrl + C** on the keyboard.
- (iii) Position the cursor where the item is to be copied.

C. Paste

This is the action of inserting copied or cut text, images, or other content from the clipboard into a document at the current cursor position.

Use one of the following options:

- From **Home** tab, click **Paste** icon under the **Clipboard** group.
- Right-click and click **Paste** command.
- Press **Ctrl + V** on the keyboard.

Another copy is created in the new location while the original remains in the previous location.

D. Deleting Text or image in a Document

To delete text or image refers to remove or erase content such as words and pictures from a document. Deleting can be done using the **Backspace** key and the **Delete** key.

- **Backspace key:** Erases character by character from the left side of the cursor position to the right.
- **Delete key:** Erases character by character from the right side of the cursor position to the left.

E. Undo commands

The **Undo** command is used to cancel or negate a series of the last commands executed starting with the most recent in order to restore the status of the document.

Assuming that the following command have been executed in the following order: Delete, copy, and paste. When the undo command is executed, the “paste” command will be negated first.

The next execution of undo command will negate “copy” and another execution of the same command will negate “delete”. The undo command is executed using one of the following options:

- Press **CTRL + Z** on the keyboard.
- Click on the **Undo** command icon on the **Quick Access Toolbar**.

F. Redo command

The **Redo** command is activated only when the undo command had been used earlier. It nullifies a series of the negated actions. The redo command is executed using one of the following options:

- Press **CTRL + Y** on the keyboard.
- Click on the **Redo** command icon on the **Quick Access Toolbar**.

Application activity 4.7

- 1) Answer with True or False.
 - a) Ctrl + C used to copy text.
 - b) Ctrl + V used to paste text.
 - c) Text can be pasted after cutting it.
 - d) Moving text create a duplicate in the original location?
- 2) Explain the difference between cutting and copying text.
- 3) Describe the steps used to copy text from one part of a document to another.
- 4) Why is the Paste command important after using Cut or Copy?
- 5) In which situations would you use Copy instead of Cut?

4.8. Moving through the document

Activity 4.8

- 1) Answer with True or False:
 - a) Keyboard shortcuts can be used to move through a document?
 - b) Home key move the cursor to the end of the current line?
 - c) Is the vertical scroll bar used to move up and down a document?
- 2) What is the Find command used for?
- 3) Explain how Find and Replace can save time when editing a document.
- 4) Why is it important to know document navigation and editing commands?

The keyboard shortcuts can be used for moving through a document. When they are used,

the position of the cursor is automatically changed.
The following are some keyboard shortcuts that can be used:

Press Keys	To move
Page down (Pgdn)	Down one screen
Page up (Pgup)	Up one screen
End	To the end of the current line
Home	To the beginning of the current line
Ctrl + Home	To the beginning of the document
Ctrl + End	To the end of the document

Table 4. 3: Keyboard shortcuts used in moving around a document

A. Scrolling through the text

Scrolling is the process of moving around a document without changing the position of the cursor. The **vertical scroll bar** is used to move up or down the pages. The **horizontal scroll bar** is used to scroll from side to side.

To Scroll	In the vertical scroll bar
Up one line at a time	Click on the up scroll arrow.
Down one line at a time	Click on the down scroll arrow.
Multiple pages up or down the document	Drag the scroll box to the top, bottom or middle of the scroll bar.
To the previous page	Click the double-up arrow.
To the next page	Click the double-down arrow.

Table 4. 4: Using the vertical scroll bar

B. Find and replace

The **Find** command is used to locate a word or phrase within a document. This makes it easy to identify a word or words to be formatted or changed with another.

If a document has many occurrences of the article “the” and needs to be replaced by the word “an”, the **find** and **replace** command facilitates this change. The **Replace** command is used to automatically substitute the located word with a new word.

h) The Find command

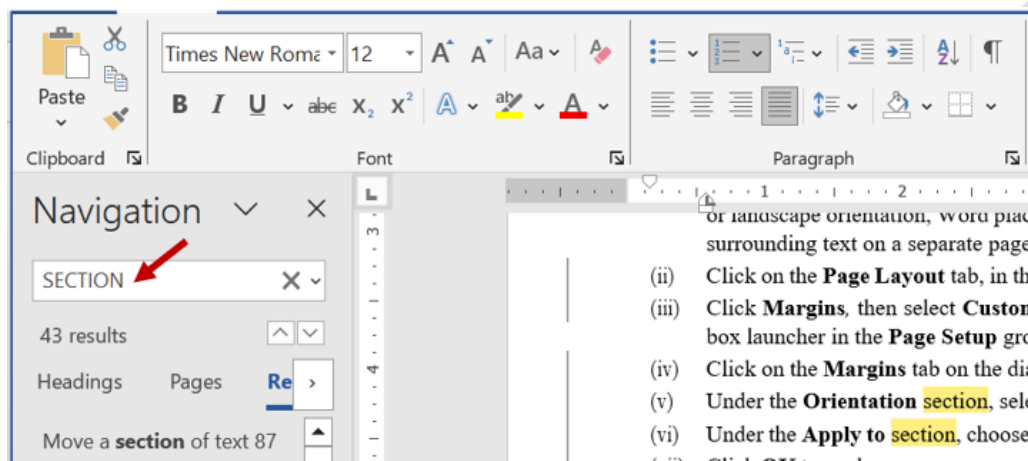
The following are the steps followed to locate words:

- Click on **Home tab** on the menu bar.
- In the **Editing** group click on the **Find** command. The **Navigation task pane** appears as shown in Picture.

- Type the word to search in the **Search Document** box. Press the **Enter** key.
- All the occurrence of the word will be automatically highlighted on the document and a summary displayed as shown in Picture below.

Note that this task pane can also be shown by pressing **CTRL + F** on the keyboard.

- Click on the **Close button** in the navigation pane to exit the search.

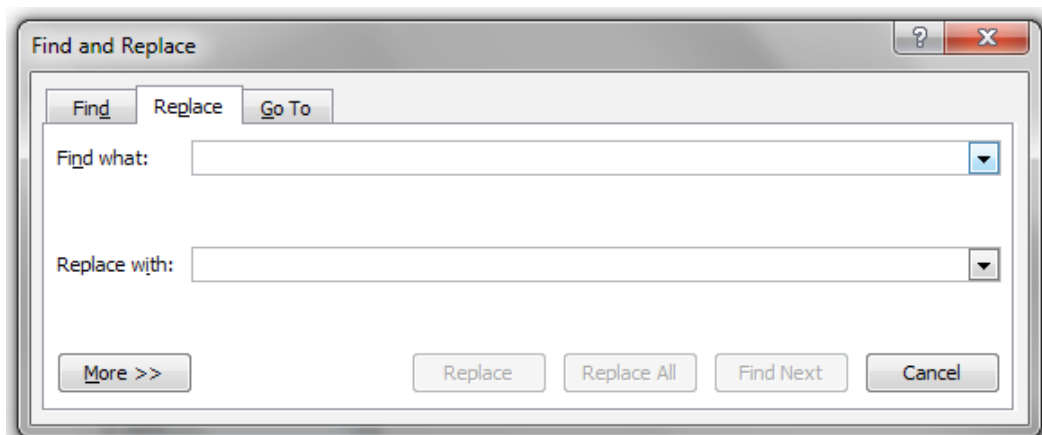


Picture 4. 25. Using the Find command

i) The Replace command

To replace an existing word or phrase with a new one do the following.

- Click on **Home tab** on the menu bar, in the **Editing** group, click on the **Replace** command. A dialog box is displayed as shown in below. The box can also be displayed automatically by pressing **CTRL + H** on the keyboard.
- Type the word to be replaced on the Find what box, and the word to substitute it in the Replace with box
- Click on the **Find Next** button to locate the word in the document.
- Either click on **Replace** button to substitute one occurrence of the word at a time or click on the **Replace All** button to substitute all the occurrence of the word in the document at once.
- Click **Cancel** button to stop the procedure and close the dialog box.



Picture 4. 26. The Find and Replace dialog box with replace tab activated

Application activity 4.8

- 1) Answer with True or False.
 - a) All occurrences of a searched word are automatically highlighted in the document.
 - b) The Replace command can be accessed using Ctrl + H?
- 2) What is the difference between Replace and Replace All options?
- 3) How can the Find task pane be closed after searching?
- 4) Give an example of a situation where using Find and Replace would save time

4.9. Page layout

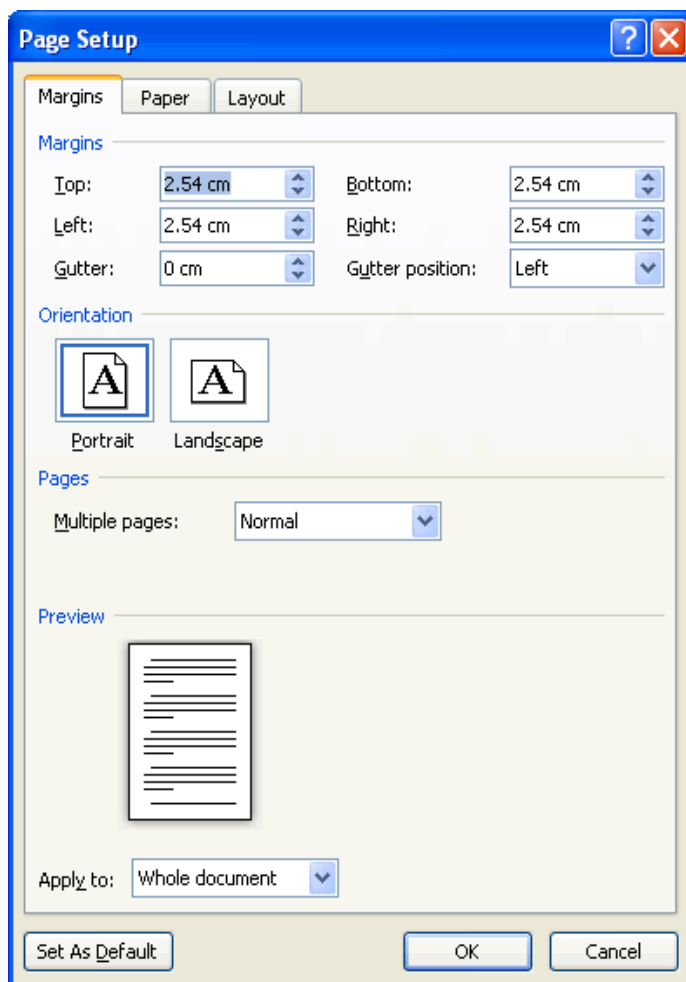
Activity 4.9

- 1) Answer with True or False:
 - a) Page margins are blank spaces around the edges of a page.
 - b) In a Portrait orientation the longest side is vertical.
 - c) In a landscape orientation, the longest side is horizontal.
- 2) Differentiate Portrait and Landscape page orientations.
- 3) How can you apply both portrait and landscape orientation in the same document?

A. Setting margins

Page Layout is used to specify the margins. Page margins are the blank spaces around the edges of a page. To set the margins do the following:

- Click **Page Layout** tab, from the **Page Setup** group and click **Margins** command.
- Click the margin type desired from the pull-down menu that appears. The margins are automatically adjusted.
- To set a customized margin, either click **Margins**, then select **Custom Margins** command or click on the dialog box launcher at the bottom of **Page Setup** group. A dialog box is displayed.
- Click on **Margins** tab on the dialog box that appears.
- Specify all the desired options and click **OK** to apply.



Picture 4. 27: Setting margins and orientation

B. Page orientation

This refers to the positioning of the page in relation to the text or graphics. There are two types of page orientation.

- **Portrait:** This is where text and graphics are printed with the longest side placed vertically and the shortest placed horizontally.
- **Landscape:** This is where text and graphics are printed with the longest side placed horizontally and the shortest placed vertically.

To change the page orientation of an entire document, proceed as follows:

- Click on the **Page Layout (or Layout)** tab, then in the **Page Setup** group, click **Orientation**.
- Select either **Portrait** or **Landscape**.

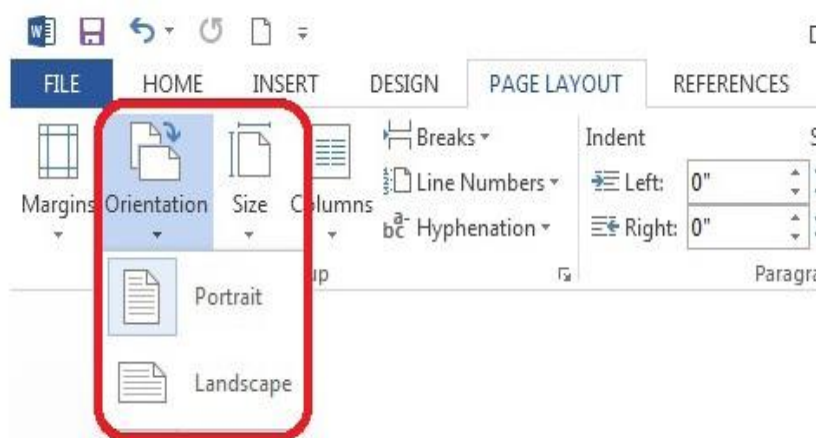
Using both portrait and landscape orientation in the same document:

- Select the content on the pages or paragraph(s) whose orientation is to be changed to portrait

or landscape.

Note: If some but not all of the text on a page is selected to change to portrait or landscape orientation, Word places the selected text on its own page, and the surrounding text on a separate page.

- Click on the **Page Layout** tab, in the **Page Setup** group.
- Click **Margins**, then select **Custom Margins** command or click on the dialog box launcher in the **Page Setup** group.
- Click on the **Margins** tab on the dialog box that appears.
- Under the **Orientation** section, select the desired orientation.
- Under the **Apply to** section, choose **Selected text** option and click **OK** to apply.



Picture 4. 28. Page orientation can either be portrait or landscape

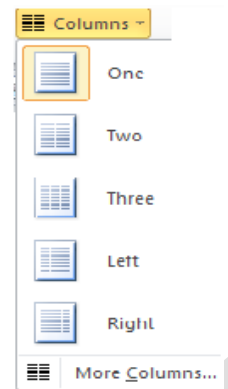
C. Columns

Columns are created to enhance the look of a document. The flow of the content from the bottom of one column to the top of the next one in the same page should be consistent. Columns can either be created using predefined or user-defined options.

i. Predefined columns

To create a column, do the following:

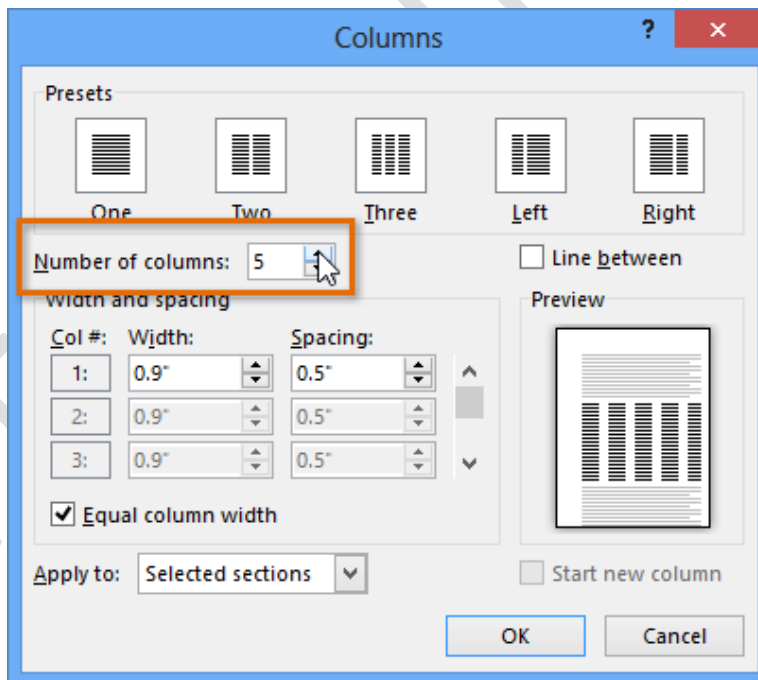
- Select the text then click **Page Layout** tab from the menu bar; then click on the **Columns** command under **Page Setup** group.
- Click on the number of columns desired from the drop-down menu displayed.



Picture 4. 29: The Columns menu

ii. Defined columns

- Select the text then click **Page Layout** tab from the menu bar then click on the **Columns** command under **Page Setup** group.
- Select **More Column** option from the resulting pull down menu.
- Specify the column width and spacing, line between columns, number of columns, and where the column is to be applied. Click **OK** to apply once all the desired options have been selected.



Picture 4. 30: Column dialog box

D. Breaks

A break is inserted in order to demarcate the end of a part and beginning of another. There are four types of breaks, namely the page break, column break, section break, and text wrapping break.

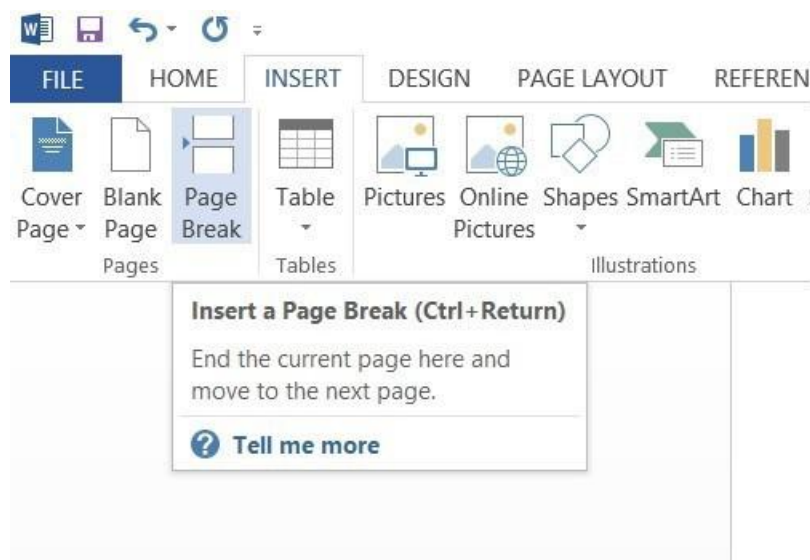
- i. **Page Break:** It demarcates the end of one page and the beginning of the next. It is automatically inserted when the bottom margin is reached and a new page created.

To insert a page break, place the cursor where the page break is to be inserted then click **Insert** tab, then click the **Page Break** under **Pages** group.

ii. Section break:

The section breaks include the Next page, Continuous, Even, and the Odd page section break.

Use sections breaks to make other formatting changes on different pages in a document, including page size and orientation, header and footer, page numbering, line numbering, footnote and endnote numbering.



Picture 4. 31: Inserting a Page break

To insert a section break click **Page Layout** tab, in the **Page Setup** group, click **Breaks**. Select the desired type of section break. There are different breaks namely the **Column break**, **Text wrapping break**. There are also different section breaks such as **Next Page**, **Continuous**, **Even Page**, **Odd page**.

Application activity 4.9

1) Answer with True or False:

- a) Columns are used to change the margins in a Word document?
- b) Columns can be predefined or user-defined.

- c) Page break indicate the end of one page and the beginning of another.
- d) Section breaks are used to apply different formatting on different pages.
- 2) What is a section break, and what types are available?
- 3) Describe the use of a continuous section break.
- 4) When would you use an even or odd page section break?
- 5) Explain how to insert a column break.
- 6) What is a text wrapping break and when is it useful?

4.10. Document reviewing tools

Activity 4.10

- 1) Answer with True or False:
 - a) Activating Track Changes highlight all edits.
 - b) Track Changes can be turned on from the status bar.
 - c) Comments are displayed in balloons in the margin.
- 2) Explain the purpose of the Track Changes feature in Ms Word.
- 3) Describe the steps to add a comment to a document.

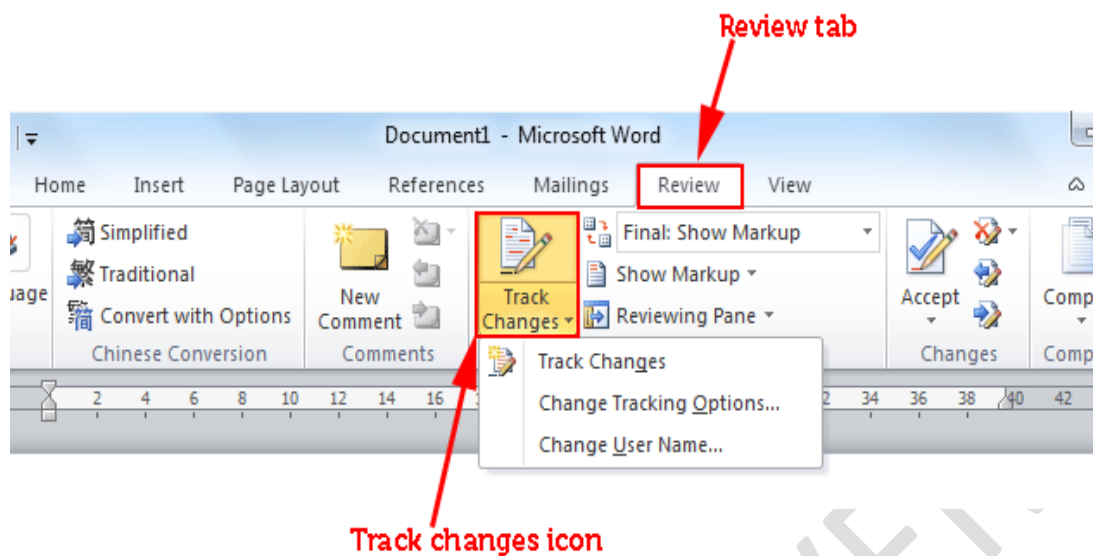
Document reviewing tools are features in a Word Processing that help people check, correct, and improve a document before it is finalized or shared..

A. Track changes

Track Changes is a document reviewing tool that shows all edits made in a document. When it is turned on, the Word processor records every change such as Inserted text, deleted text, formatting changes, etc.

To turn on track changes do the following:

- Click on the **Review** tab in the menu bar, in the **Tracking** group, click the **Track Changes icon**. Click on Track Changes again. The **Track Changes** option will automatically be turned on. Any changes done to the document will always be highlighted in red or any color specified in the settings.
- To add a track changes indicator to the status bar, right-click the status bar and click **Track Changes**. Then click the **Track Changes** indicator on the status bar to turn Track Changes on or off instead of using the menu option.



Picture 4. 32: Tracking changes

B. Comments

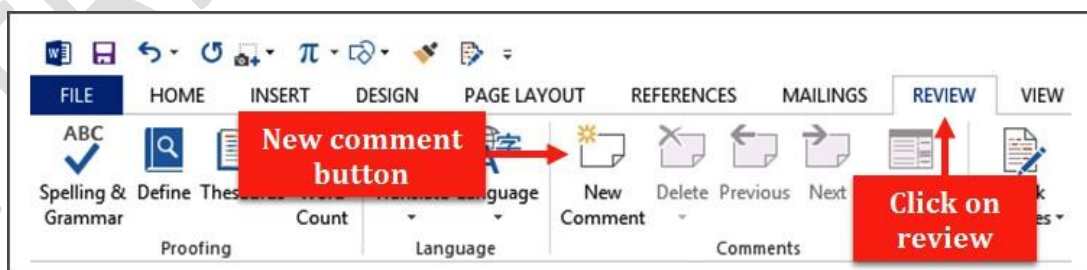
A comment is a note that an author or reviewer adds to a document.

Microsoft Word displays the comment in a balloon in the margin of the document or in the Reviewing Pane. Comments can also be cleared or hidden to ensure that they do not appear in the document.

i. Adding a comment

To add a comment do the following:

- Select the text or item or click at the end of the text.
- Click on the **Review** tab, in the **Comments** group, click **New Comment**. A comment line is added with a comment balloon as shown in below. Type the comment text in the comment balloon.



Picture 4. 33: Inserting comments in a document

Comments group, click **New Comment**. A comment

Comment [LKM1]:

Picture 4. 34: Type the desired comment in the balloon at the cursor position.

Note: To respond to a comment, click its balloon, and then click **New Comment** in the **Comments** group on the **Review** tab. Type the desired response in the new comment balloon.

ii. Deleting a comment

- To delete a single comment, right-click the comment, and then click **Delete Comment** or click on it then select **Delete** command in the **Comments Group** from the **Review** tab.
- To quickly delete all comments in a document, click a comment in the document. On the **Review** tab, in the **Comments** group, click the arrow below **Delete icon**, and then click **Delete All Comments in Document**.

Application activity 4.10

- 1) Why is it important to see the changes made in a document?
- 2) What will you click first to insert track changes in a document?
 - a) Insert tab
 - b) Home tab
 - c) Review tab
 - d) View tab
- 3) Which of the following is recorded by Track Changes?
 - a) Inserted text
 - b) Deleted text
 - c) Formatting changes
 - d) All of the options
- 4) By using a search engine such as Google do a research on the water cycle and make a document having at least 2 pages. After turning on Track Changes edit your document so that it does not contain errors and most of the sentences are written in your own words.
 - a) What is the color of the inserted words
 - b) Explain how you can change the appearance of those inserted words
 - c) What will happen if you click on “Accept all Changes and Stop Tracking”

4.11. Document structuring

Activity 4.11

- 1) Explain why it is good to have different heading styles in a document
- 2) In a document where are found headers and footers?
- 3) What are their importance in a document?
- 4) Why are page numbers important in a document

A. Headings

These are styles that have been defined in Word processing application. Every heading comes along with its font size, colour, font and indentation among others. There are four main heading styles that the user can choose from namely: Heading 1, Heading 2, Heading 3 and Heading 4.

The heading styles are necessary when the user wants to create a table of contents. They are defined in a heading/subheading. Main heading is given heading 1 style, sub heading, heading 2 style and so on. For example:

ICT	Heading 1
Senior	Heading 2
Course Book	Heading 3
Students' Book	Heading 4

Picture 4. 35. : Heading styles

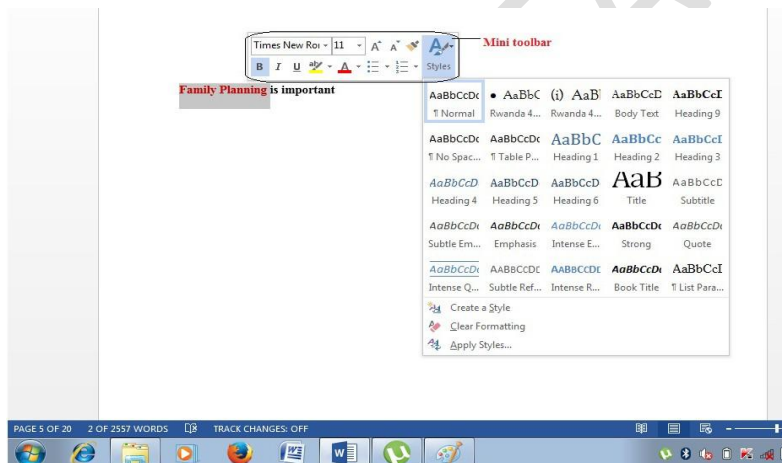
i. Apply a style

Click on the **Home** tab, in the **Styles** group, select a set of styles desired. Apply the styles from the convenient **Quick Styles** gallery as the document is being created.

ii. Create a style

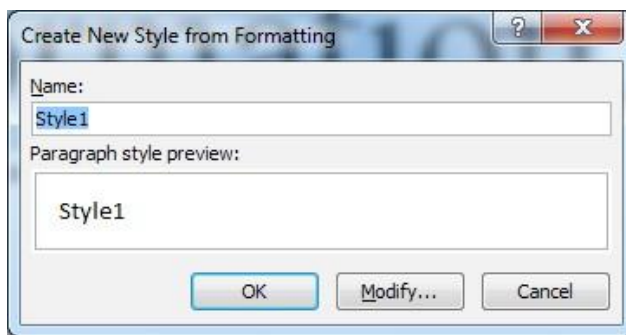
It creates a style based on the formatting of the selected text. To create a new style and add it to the **Quick Styles** gallery do the following:

- Select the text to be used to create the new style. For example, if the words “Business Information” is to be formatted so that it always appear as a red Heading 1 in a document. On the Mini toolbar that appears above the selection, select **Styles** then **Heading 1** and finally click on **Red color** to format the text as shown in below.



Picture 4. 36: Applying style

- Right-click the selection, click on **Styles** then click **Create a style** command. A dialog box appears as shown in picture below.



Picture 4. 37: Creating a new style

- Give the style a name, for example, Business Info, and then click **OK** or **Modify** to make other changes. The style created appears in the Quick Styles gallery with the given name, ready for application whenever the user desires to apply it.



Style added

Picture 4. 38: Adding a new style to the quick style gallery

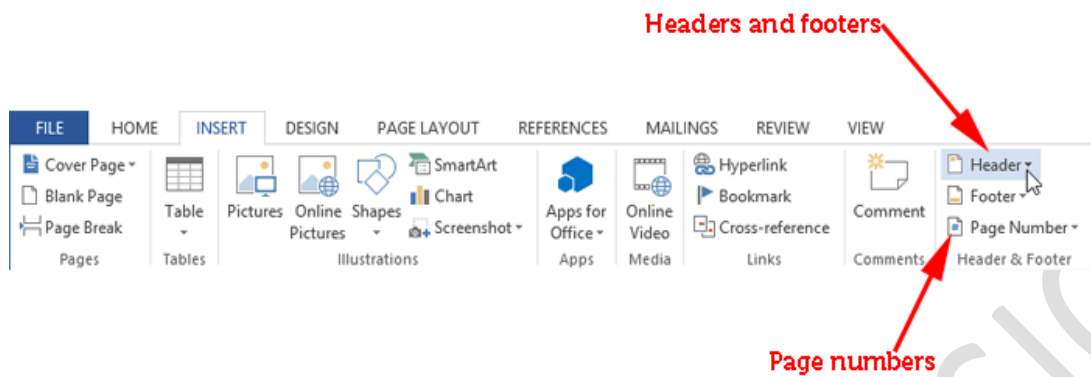
B. Clear Formatting

- Select the text whose style is to be erased.
- Right-click the selection, click on **Styles**, and select **Clear Formatting** command. The formatting style will be changed back to the default style automatically.

C. Headers and footers

Headers are lines of text or graphics that appear above the top margin of a page or selected pages. Footers are lines of text or graphics that appear below the bottom margin of a page or selected pages. To insert a header and footer do the following:

- Click **Insert tab** from the menu bar; then click on **Header or footer** command from the ribbon.
- Under **Header & Footer** group, type the desired content to insert a header or footer respectively. Then press **Enter**.
- A drop down menu appears. Click on the desired option.

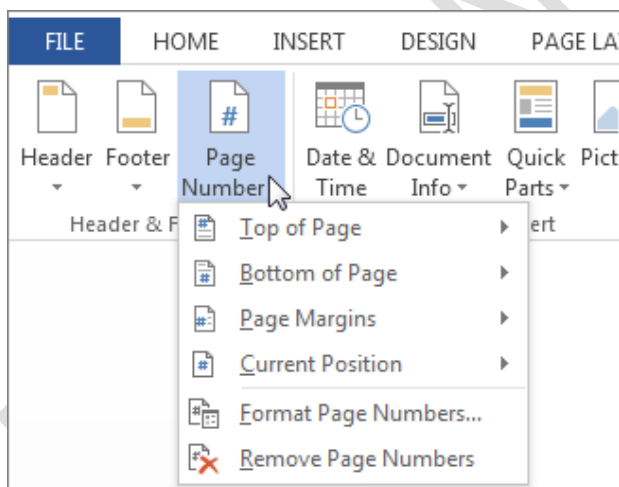


Picture 4. 39: Inserting header, footers and page number

D. Page number

This is a formatting feature that organizes a large document for ease of reference and makes it easy to track by inserting numbers beginning with the first page. To insert page numbers, do the following:

- Click **Insert tab** on the ribbon that appears.
- Click on **Page Number** from **Header and Footer** group
- Click on the **Page Number** icon. The options shown in the picture below will be displayed.
- Choose the preferred option.



Picture 4. 40. Page numbering options

Application activity 4.11

- 1) Answer with True or False.
 - a) A user can create a new style and add it to the Quick Styles gallery.
 - b) Clear Formatting revert text to the default style.

- c) Headers are placed above the top margin of a page.
- d) Page numbers can be inserted from the Insert tab.
- 2) Describe the process of creating a new style in MS Word.
- 3) How does the Clear Formatting command work?
- 4) What is the difference between a header and a footer?
- 5) How do you insert page numbers in a document?
- 6) Give an example of when it is useful to use headers, footers, and page numbers

4.12. End unit assessment

4.12.1. Project Title: Designing a school's history document

- Write a document related to your school history
- Use different styles and size.
- Insert your school picture
- Insert page numbers and create a table of content

Project questions

- 1) Which software used in your project?
- 2) How do you choose the styles and font sizes for different parts of the document?
- 3) What steps do you follow to format the text and make it look professional?
- 4) How do you ensure the document is clear and easy to read?
- 5) What skills do you develop while completing this project?

4.12.2. Assessment questions

- 1) Answer with True or False:
 - a) A Word Processing application is used to create and edit text documents.
 - b) Ctrl + S is used to open a document.
 - c) Bold formatting makes text slanted.
 - d) Page orientation can be portrait or landscape.
- 2) Which tab is used to format text?

a) Insert	b) Home	c) View	d) Review
-----------	---------	---------	-----------
- 3) Which feature allows tracking changes in a document?

a) Find	b) Replace	c) Track Changes	d) Sort
---------	------------	------------------	---------
- 4) State two examples of Word processing applications.
- 5) What is the purpose of Save As command?
- 6) Name two paragraph formatting tools.
- 7) Create a document with a heading, paragraphs, bullets and numbering, page number, header or footer
- 8) Explain why you will be saving your work regularly.

UNIT 5: NETWORK AND INTERNET FUNDAMENTALS



Key Unit Competence

Explain network concepts and the use of networks in real life.

Learning objectives

At the end of this unit, you should be able to:

Knowledge and Understanding:

- Explain network and computer network terms.
- Explain the role, advantages and disadvantages of a computer network.
- Identify the types of computer networks according to different criteria
- Explain the importance of using the internet in daily life.
- Explain the different types of search engines.

Skills:

- Classify a computer network according to different criteria.
- Apply different methods to connect to the internet
- Explore different search engines to perform searches on the internet.
- Create and manage an email account.

Attitudes and values

- Appreciate the importance of communication and connectivity through the internet.
- Show respect for privacy and confidentiality when using email and online resources.
- Develop curiosity and lifelong learning attitudes toward new internet tools.
- Demonstrate discipline and time management when using the internet.

5.0. Introduction

This unit helps learners understand how computers and digital devices communicate through computer networks and the Internet. It covers the meaning, characteristics, roles, advantages, disadvantages, and types of networks such as PAN, LAN, MAN, and WAN. The unit also introduces the internet, including its history, components, how people access it, search for information, and communicate using e-mail.

Introductory activity

- 1) How do computers in a school computer lab share a printer?
- 2) Name one way people connect to the internet.
- 3) Give one example of a search engine and explain how it can be used.
- 4) What is an email used for?
- 5) Mention one way the internet helps people in daily life.

5.1. Computer network basics

Activity 5.1

- 1) What is a computer network?
- 2) State three characteristics of a computer network.
- 3) List three roles of a computer network.
- 4) Mention four advantages and three disadvantages of a computer network.

A. Definition of network

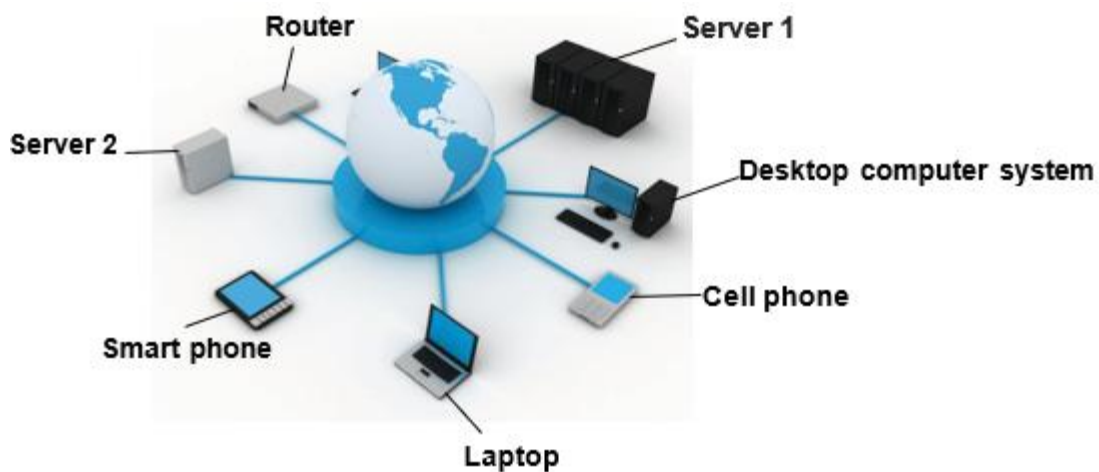
A network is generally described as a system or group of interconnected objects or people.

A computer network is a set of interconnected computing nodes which communicate with one another through a transmission medium using a set of rules. The nodes share resources and services. A computer network must have sharable resources, transmission medium and set of rules (protocols).

B. Characteristics of a computer network

The following are some characteristics of a computer network:

- **Security:** A network must provide protection data from unauthorized access, manipulation and implement policies for recovery from data loss.
- **Reliability:** A computer network should work consistently to perform according to its requirement.
- **Transmission capacity:** It must have two or more computing nodes connected using a transmission medium.



Picture 5. 1: A computer network

C. Role of a computer network

A computer network can be used to do the following:

- Store data centrally for ease of access and backup.
- Communicate with other network users.
- Share resources.

D. Advantages and disadvantages of a computer network

i. Advantages of a computer network

- **Resource sharing:** A computer network allows sharing of resources such as printers among the users in a network thus eliminating the need to have individual printers for each computer. This reduces the cost of purchasing and maintaining the resources.
- **Speed:** A computer network provides a very fast means of sharing and transferring files thus saving time while maintaining file integrity.
- **Communication:** A computer network provides the hardware necessary for exchanging information between the users in the network, for example, use of e-mail.
- **Work group computing:** Many users in different geographical locations can work on a document or a project at the same time using workgroup software and hardware, thus saving time and traveling expenses.
- **Security:** A computer network allows administrators to manage critical data by regularly backing up and implementing security measures to control access to critical information.
- **Increased storage capacity:** Many computers in a network provide a lot of memory that can be shared among the computers to increase the overall storage capacity.
- **Entertainment:** A computer network supports multi-player computer games and

entertainment.

ii. Disadvantages of a computer network

- **Expensive to install and maintain:** High costs are involved in installing and maintaining the network, for example, the cost of cables, file servers, software and technical support services among others.
- **Failure of the server and other hardware:** The whole network breaks down if the server fails and, in some configurations, one broken cable or a malfunction of a node may terminate the functioning of the entire network.
- **Insecurity of information:** Shared information in a network is exposed to hacking, cracking and eavesdropping among others.
- **Spread of computer viruses:** Viruses can easily spread to other computers within a computer network.

E. Classification of computer network according to geographical areas

There are four main types of computer networks classified according to geographical areas. They are Personal Area Network (PAN), Local Area Network (LAN), Metropolitan Area Network (MAN) and Wide Area Network (WAN).

i. Personal Area network (PAN)

It is also known as **Wireless Personal Area Network (WPAN)**. It is the smallest network used for data transmission among devices within the environment of an individual user.

These devices may include bluetooth-enabled devices or infra-red enabled devices such as a wireless printer, a telephone, a TV remote control, a wireless computer keyboard and a mouse among others.

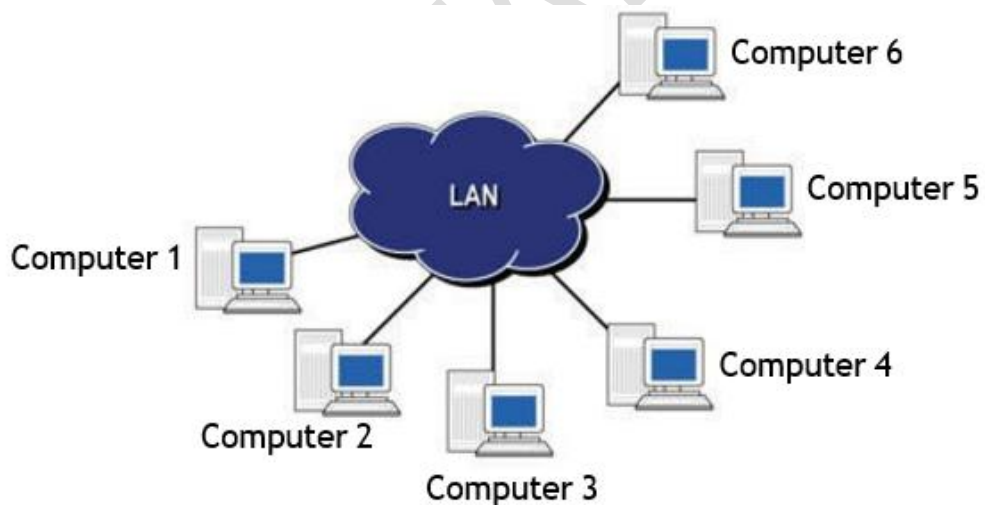
- **Bluetooth:** This is a wireless protocol that connects electronic devices over a short distance of up-to about 10 meters creating a PAN with a high level of security.
- **Infra-red:** This is a wireless electromagnetic energy technology with a wavelength longer than those of red visible light. It is used in conveying data from one device to another.



Picture 5. 2: The Personal Area Network (PAN)

ii. Local Area Network (LAN)

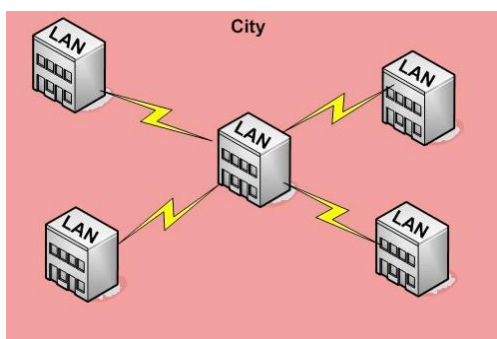
This is a network that covers a small geographical area for example, an office, a building, an institution, or an organization.



Picture 5. 3: The Local Area Network (LAN).

iii. Metropolitan Area Network (MAN)

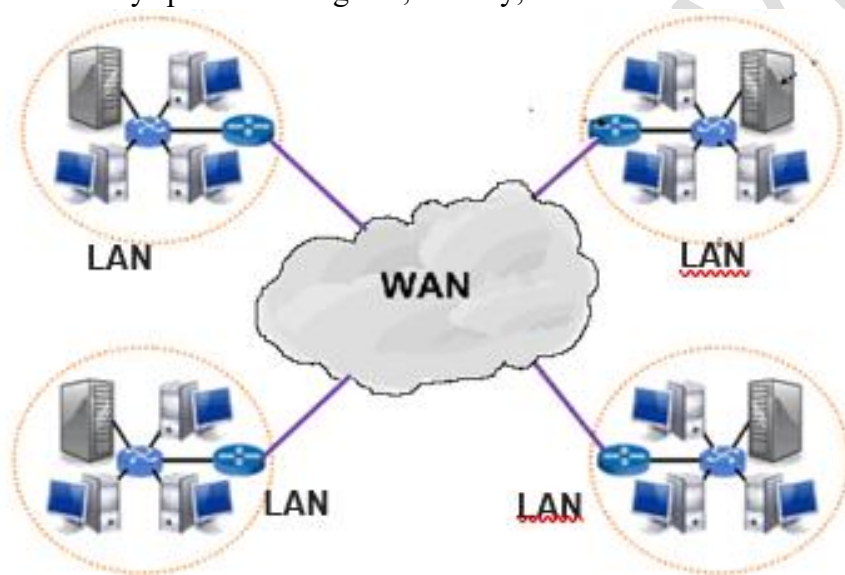
A metropolitan area network covers a city or town. It can be made up of many LANs and can provide internet connectivity in a Metropolitan region.



Picture 5. 4: The Metropolitan Area Network covers a city or a town

iv. Wide Area Network (WAN)

This is a network that interconnects LANs and MANs within a very large geographical area and may span across regions, country, continents or even the whole world.



Picture 5. 5: The Wide Area Network (WAN)

LAN and WAN are two types of computer networks used to connect devices for communication and sharing resources. They differ mainly in size, coverage area, and how they are used. The table below shows the differences between LAN and WAN.

Local Area Network	Wide Area Network
Covers a small geographical area.	Covers a very large geographical area usually world-wide.
Transmits data at high speeds (higher bandwidth) because of small distance covered.	Transmits data at lower speeds (lower bandwidth) because of very large distance covered.

Less transmission costs because it is managed within an organization.	High transmission costs because it is managed by other organizations such as satellite and telephone companies
---	--

Table 5. 1: Differences between LAN and WAN

Application activity 5.1

- 1) Write True or False.
 - a) A computer network allows sharing of resources.
 - b) Security is not important in a computer network.
 - c) Viruses can spread easily in a computer network.
 - d) A network must have a transmission medium.
 - e) Computer network exist only in a small geographical area.
- 2) Give two examples of resources shared in a network.
- 3) Explain why a computer network is expensive to maintain.

5.2. Internet

Activity 5.2

- 1) Define the internet.
- 2) What is World Wide Web?
- 3) Differentiate between a webpage and a website.
- 4) List four examples of web browsers.
- 5) What is an ISP? Give two examples.

5.2.1. Definition of terms

- Internet stands for **International Network**. It is a worldwide network of computers connecting millions of computer networks via communication media such as telephone lines, wireless connections, and fiber optic.
- World Wide Web (WWW): It is commonly referred to as the Web. WWW refers to a large virtual space in the Internet where information pages called web pages are installed. The web pages can be accessed using a web browser.
- **Webpage**: This is Hypertext and hypermedia documents containing information about an individual or organization.
- **Website**: A collection of related web pages stored as a single file in a web server.
- **Web browser**: It is commonly referred to as a browser. These are programs used for displaying and viewing web pages on the World Wide Web (WWW), for example, Mosaic, Netscape Navigator, Internet Explorer, Mozilla Firefox, Opera and Google Chrome among others.
- **URL**: It is an abbreviation for **Uniform Resource Locator**. A website address is

commonly known as **URL**. It is a unique address of each web page in a websites which identifies the location of a specific web page on the Internet. The URL consists of three parts namely; protocol, server name and resource ID. An example of URL is <http://www.techrwanda.com>

- **ISP**: This is the abbreviation for **Internet Service Provider (ISP)** is a licensed Organization or company that provides access to the Internet usually at a fee. Examples of isps include Rwandatel, Tigo, ISPA, and MTN Rwanda among others.

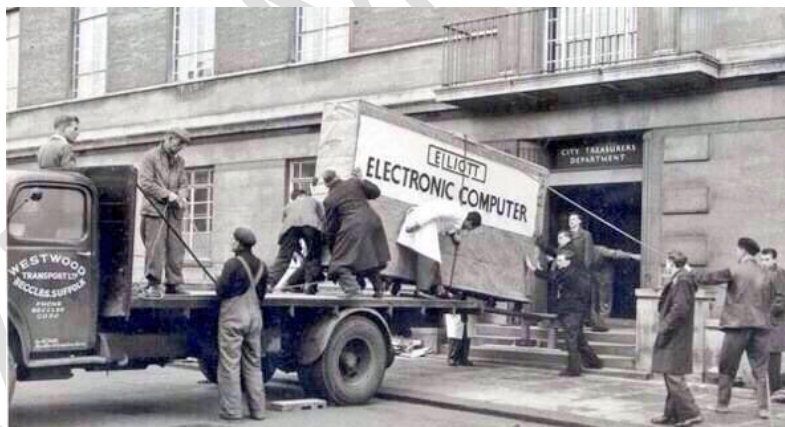


Picture 5. 6: The URL consists of the protocol, the server name and the resource

5.2.2. History of the internet

The root of today's internet came from visionary people in the 1960's for the purpose of information and research sharing. In 1962 packet switching was developed and it formed the basis of internet connection. The Advanced Research Projects Agency (ARPANET) which was a branch of the Department of Defense was the first known fully operational packet-switching network.

In 1969, ARPANET began working on an experiment to determine whether computers at different universities could communicate with each other without a central system.



Picture 5. 7: Evolution of computers: A large computer being delivered by 13 men

The corporation Bolt Baranet and Newman developed Interface Message Processor (IMP), the basis of the new communications system. IMPs were small machines which were dedicated to

forming the network between computers. They used a technology which split large sections of data into small parts called packets, each labelled with its destination address, and later re-assembled at the destination computer.

In 1970, Vinton Cerf and others, published a proposal for protocols and later ARPANET began operating Network Control Protocol (NCP) to enable computers to communicate. In 1974, Vinton Cerf together with Bob Kahn presented their protocol for packet network interconnection, which became the basis of the modern Internet. In 1978, TCP was split into two protocols TCP/IP (Internet Protocol).

In 1982, TCP/IP was established as the protocol for ARPANET and by January 1983, all of the ARPANET was switched over from NCP to TCP/IP.

By the end of the 1980's, a young British Scientist, Tim Berners-Lee assigned a common system of written addresses and hypertext links to all information. Hypertext is the organization of information units into connections that a user can make.

In 1990, Lee started working on a hypertext graphical user interface browser and editor. By 1991 the first World Wide Web (WWW) files were made available on the net for downloading using File Transfer Protocol (FTP). The mid 1990's saw the introduction of search engines and by late 1990's Independent Internet Service Providers (isps) such as Sprint and AT&T had so many subscribers.

In 1993, the world started waking up to the WWW and at least 200 known HTTP servers were connected. In 1994, Netscape Communications was born and Microsoft created a Web browser for Windows 95.

Later in 1998 Google search engine was born changing how users engaged with the Internet.

Today the Internet is everywhere, enabling people to communicate and share interests in many ways more faster using sites such as Facebook, Twitter, Linked-In, youtube, blogs, wikis, and many more.

5.2.3. Internet uses in daily life

Internet can be used in the following ways in daily life:

- **Communication:** Through electronic mails (e-mails), social sites, chat rooms and newsgroups, the public and employees in both government and private sector can easily exchange information within and outside the country.
- **Research:** People use the Internet to get information on various topics. For example, university students in Rwanda are able to use the internet to do research on different topics.
- **Business:** Customers can buy and sell goods and services through the internet. They can pay for the goods and services through Electronic Funds Transfer (EFT), cheques and credit cards among others.

In Rwanda, people are able to buy goods from other countries in the comfort of their homes. For example, through Visa card transaction systems, a Rwandan bank account holder in Bank de Kigali is able buy an electronic book from Amazon. Com.

- **Education:** The internet is used in various ways such as distant education or online

education. Internet or Web-based training can also be used to take up short term courses with the course materials available online, attend virtual classes, learn and take examinations.

- **Medicine:** Medical practitioners are using the internet to update themselves on specific areas of interest and even take refresher courses to enhance their medical knowledge for better services. This has enhanced delivery of services in Rwanda hospitals.
- **Internet banking:** Bank customers access their accounts and carry out transactions through the internet. People are able to transfer money online from one account to another without necessarily having to physically visit their respective banks.
- **Entertainment:** Internet provides various forms of entertainment. For example, one can play games, watch online movies, upload and download entertainment materials such as music.

5.2.4. Ways of connecting to the internet

Connecting to the Internet can be done using the following ways:

A. Dial up

It involves connecting a computer to an Internet Service Provider (ISP) by the use of a modem and telephone line such as a Subscriber Identity/ Identification Module (SIM) card, only when accessing the internet. The connection is terminated once the user logs out from the internet. Dial-up internet connection is mostly used in rural or remote areas where broadband internet access is not available. However, dial up internet is potentially approaching extinction because of the advancements in wireless technology.

B. Broadband

This is a high capacity transmission technique that uses a wide range of frequencies to enable a large number of messages to be communicated simultaneously. Some examples of broadband are Wi-Fi , 4G and satellite among others.

C. Wi-Fi

Wi-fi stands for Wireless Fidelity. This is a wireless networking technology that uses radio waves to provide high-speed internet and networking connections. Wi-Fi enabled devices includes laptops, cell phones, tablets, IPADs, and peripheral devices such as the mouse, printers and keyboards among others. They communicate with a single computer which is fitted with a Wi-Fi adapter.

Application Activity 5.2

1) Fill in the blanks

- WWW stands for _____.
- A website address is known as a _____.
- The protocol in a URL shows the _____ used.

d) An ISP provides _____ access.

2) Match left column with the right column

Item	Role
Web browser	Displays web pages
URL	Website address
ISP	Internet provider
Webpage	Hypertext document

5.3. Getting online

Activity 5.3

- 1) Define a web browser.
- 2) List five basic parts of a web browser window.
- 3) What is an address bar used for?
- 4) State two functions of control buttons.
- 5) Write one example of a URL.

A. Web browser

A web browser is a program that is used for displaying and viewing web pages on the World Wide Web (WWW).

Examples of web browsers are Mosaic, Netscape Navigator, Internet Explorer, Mozilla Firefox, Opera Mini, and Google Chrome among others.



Chrome Chrome



Opera mini



Mozilla Fire
Fox

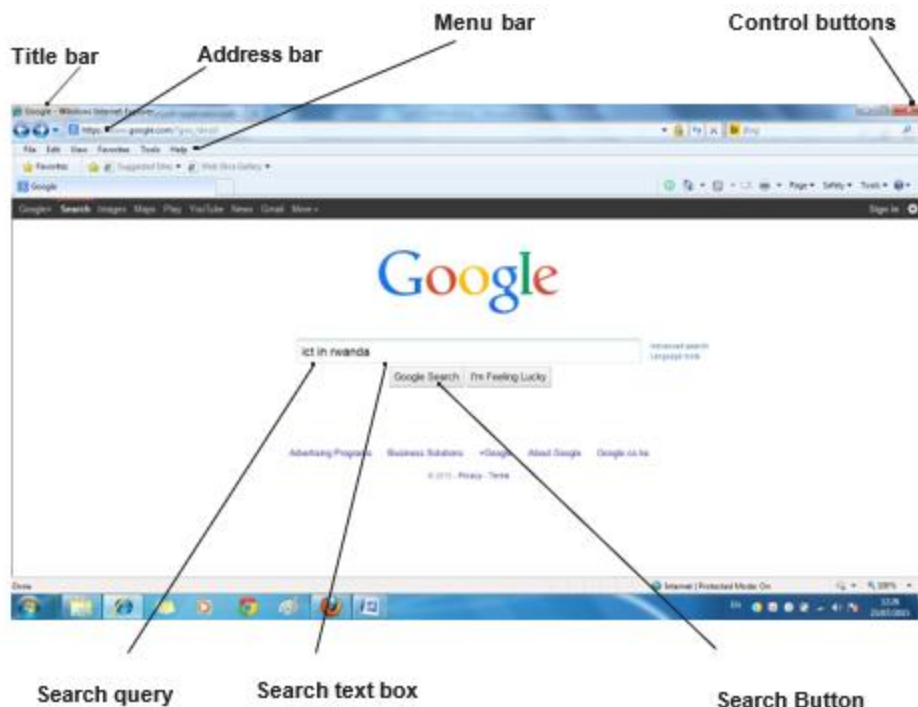


Internet
Explorer

Picture 5. 8: Web browsers

B. Basic parts of a web browser

The following are basic parts of a browser window:



Picture 5. 9. Parts of a browser window

(I) Menu bar: It consist of a list of menus to choose from, for example, File, Edit, View, Tools, and Help.

(i) **Control buttons:** These buttons are found on the far right-side of the title bar of the search engine and include the following:

- **Minimize:** It removes a web page window from a view to an icon on the taskbar.
- **Restore Down:** It reduces the size of a web page window.
- **Maximize:** It enlarges the web page window and fills up the screen.
- **Close:** It exits the active window.

(ii) **Address bar:** This is where an address (or name) is typed so as to go to a given web site page. An example of a web site address is **www.techrwanda.com**.

(iii) **Search Text box:** This is where keywords are typed to describe the information that one wants to find.

(iv) **Search button:** It is the command button clicked in order to initiate the search.

(v) **Search Query:** It is the information typed by the user to help the search engine decipher the required information.

C. Web address

A website address is commonly known as Uniform Resource Locator (URL). It is a unique address of each web page in a websites which identifies the location of a specific web page on the internet. The URL consists of three parts namely; protocol, server

name and resource ID.

An example of URL is <http://www.techrwanda.com>

Explanation of the address

- (i) **Protocol:** It is a set of rules that govern how data is transmitted in a network. There are many types of protocols used in a network.

In an URL address, the protocol is shown at the beginning of the address then followed with a colon and double slashes. In our example the protocol is **http (Hypertext Transfer Protocol)**.

- (ii) **Server name:** It identifies the computer on which the resource is located. This computer is known as a remote server.

Most web server names begin with the letters www, for example. www.techrwanda.com. The server name always ends with a dot and a three or two-letter extension called the domain name.

The domain is important because it usually identifies the company, agency or organization that is directly responsible for the information, or is providing the computer space where the information is kept. Sometimes it indicates the country where the server is located.

The **Domain Name System (DNS)** is a class or naming order system for computers, services, or any resources connected to the Internet or a private network.

A **top-level domain (TLD)** is a domain at the highest level in the Domain Name System of the Internet. For example, in the domain name **www.techrwanda. Com**, the top-level domain is com. Some common top level domain names are:

.com which identifies company or commercial sites.

.org for non-profit organization sites.

.edu for educational sites.

.go for government agencies sites.

.net for Internet service providers.

.co for company.

.ac for academic/educational institution.

.mil for military organization.

Examples of country domains include .rw for Rwanda, .ke for Kenya, .uk for United Kingdom, .jp for Japan, .tz for Tanzania, .us for United States, .au for Australia, and .ca for Canada.

- (iii) **Resource ID:** It identifies the file name and the folders or subfolders under which the webpage is stored on the remote server. An example of resource ID is /index. Php/ icons/services.

Application Activity 5.3

- 1) What is a search query?
- 2) State the three parts of a URL.
- 3) Using a web browser navigate to the elearning.reb.rw and identify different books which are there.

5.4. Search engines

Activity 5.4

- 1) Define a search engine.
- 2) Give four examples of search engines.
- 3) State three advantages of search engines.
- 4) State three disadvantages of search engines.

Search engines are programs that help a user to look for and identify items that correspond to keywords or phrases specified from the World Wide Web. Examples include Yahoo, Google, Bing and Ask.com among others.

5.4.1 Advantages and disadvantages of search engines

A. Advantages of search engines

- They enable users to describe the information being searched.
- They enable users to locate and access information in different websites.
- They reduce the time taken to locate desired information.
- They provide a wealth of information for professional and personal use.

B. Disadvantages of search engines

Tiresome and complex to search if the keyword typed is ambiguous. Many users get discouraged and frustrated because of the many links they have to search before getting the correct information. The disadvantages are:

- A lot of irrelevant information gets displayed after typing a keyword.
- A lot of distractions from advertisements that keep on popping up.
- A link to sites containing viruses could be displayed as a search result.

5.4.2 List of search engines

These are some examples of search engines: Google, Google scholar, Yahoo!, Bing Ask.com.

- **Google:** It is a general search engine which is used to find all types of information. It searches everything on the internet, not only academic or specific content.
- **Google Scholar:** It is a special search engine for education which finds books, research papers, journals. It focuses only on academic and school related information.
- **Yahoo!:** It is a web portal and search engine that offers email, news and search services. It combines search, news and email in one place, not just search.
- **Microsoft Bing:** It is a search engine created by Microsoft. It is an alternative to Google,

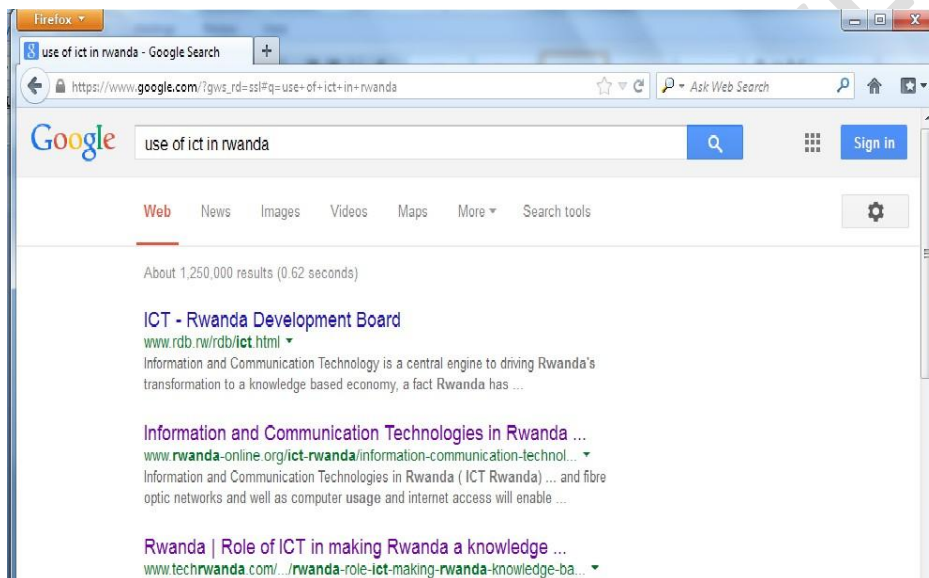
with a different design and results.

- **Ask.com:** It is a question-and-answer search engine with which users can ask questions in simple sentences. It focuses on direct questions and answers, not general browsing like Google.

5.4.3 Deciphering search results

A. Defining deciphering

Deciphering is the conversion of text written in code form into natural language. A search engine results page displays a lot of different information. In fact, one may not be aware of all the different types that appear. From vertical search engine results to related new stories, to suggested spellings, the results page offers all sorts of ways for one to dig deeper into a subject.



Picture 5. 10: Search results on a search engine

B. Procedure for deciphering search results

- (i) Type your search query in the Search Box.
- (ii) Click on the Search Button. The result is displayed with the following sections:
 - **Search Box:** The box where you type your search query. In this case it is “Use of ICT in Rwanda.”
 - **Search Verticals:** Links to the *vertical search engines*, the specialized ones that narrow your search into a specific type of result, such as news or images. Clicking one of these links takes you to a results page with only news or only images.
 - **Page Count:** The number of Web pages the search engine finds that match your search query in some way.

- **Time the Search took:** How long the search engine took to retrieve your results.
 - **Related Searches:** Other topics that contain your query or other searches might be relevant.
 - **Images:** Picture files that match your query. Clicking the link would take you to the vertical search results.
 - **News Results:** Any news results pertaining to your query or containing a keyword. These come from the vertical news engine. Clicking the link would take you to the news page.
 - **Sponsored Links:** These are paid advertisements (ads). Note how some of them relate to a specific geographical location near you. This is thanks to the *local* vertical search engine.
 - **Organic Results:** The listing results from a general search of Google's index, with algorithms applied to determine relevance.
 - **Pagination:** Links to the additional pages of results.
- (iii) Click on one of the options within the sections to access the web page.

Application activity 5.4

- 1) Answer with True or False and explain the reason:
 - a) Search engines help users locate information on the internet.
 - b) All information from search engines is always accurate.
 - c) Advertisements may appear in search results.
 - d) Google Scholar is not a search engine.
- 2) What does deciphering search results mean?
- 3) Mention three parts found on a search engine results page.

5.5. Electronic Mail (E-Mail)

Activity 5.5

- 1) Define e-mail.
- 2) What is the format of an e-mail address.
- 3) Explain the meaning of the @ symbol in an e-mail address.
- 4) State four advantages of e-mail.
- 5) State three disadvantages of e-mail.

5.5.1. Definition of terms

A. E-mail

E-Mail refers to a system of exchanging messages electronically over a computer network. The user must have an e-mail address to use e-mailing services.

B. E-mail address

An email address is a unique name used to send and receive electronic messages (emails) on the internet.

The following is a sample format of an e-mail address: Username@hostcomputer.domainname

It has the following parts:

- **Username:** It identifies the owner of the e-mail address.
- **@ (“at” sign):** Separates the username from the rest of the address parts.
- **Hostcomputer:** The name of the remote server on which the e-mail account is hosted. For example gmail, yahoo, and hotmail among others.
- **.(dot):** Separates the domain from the other part of the e-mail address.
- **Domain name:** It identifies the type of institution offering the services. This has been discussed earlier in this book.

For example, **nkeza@gmail.com**, **nkeza** is the username, **gmail** is the host computer and **.com** is the domain name for a commercial organization.

C. Advantages and disadvantages of email

i. Advantages of email

- It allows sending of one mail to many recipients at the same time.
- It is cheaper compared to traditional mails
- It is fast and efficient as it allows instant sending and receiving mails from anywhere in the world.
- E-mails can be saved for future retrieval on another computer.
- It is easier to reply and forward mail.
- Allows sending and receiving any form of information.

ii. Disadvantages of E-Mails

- Use of email requires computer literacy.
- E-mails are not accessible to everyone due to internet connectivity problems.
- There are security concerns. Hackers can gain unauthorised access to users’ accounts and, for example, solicit for money.
- Allows spread of viruses such as hoax.

D. Management of an email account

a) Creation of an email account

Follow these steps to create an email:

- Open e-mail application software like Gmail, Yahoo.com, etc. Then select **Check Mail** option. The **Sign in** dialog box is displayed.
- Click **Sign Up** and follow the procedure provided in the application.

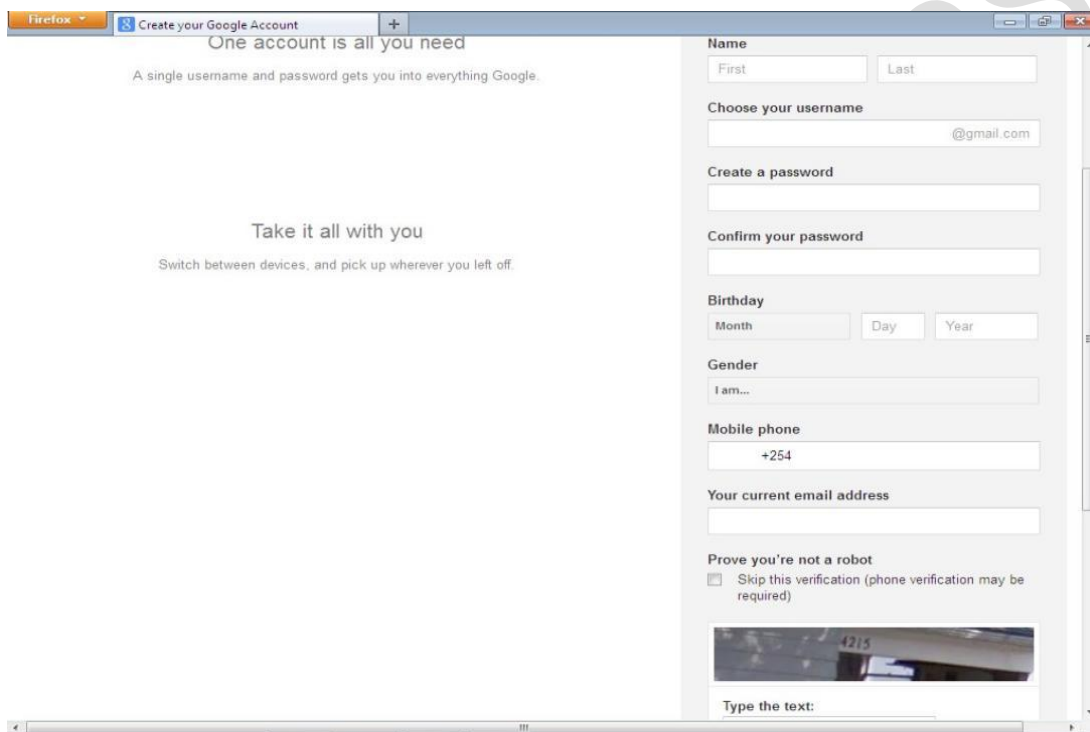
When creating a password for the account ensure that the password is strong enough. A strong password should have a minimum of 8 characters and should be a mixture of both letters, numbers, symbols and special characters.

Note that:

- Passwords are case sensitive.
- Always use a combination of characters that you can easily remember.
- Avoid using characters that can be guessed.
- Passwords are always encrypted for security purposes.

For example: To create a mail account in Gmail, the steps are as follows:

- Open Gmail application software then click **Create an account** link in the **Sign in** dialog box. This will open the **sign up** page as shown in picture below.



Picture 5. 11: A form to create an e-mail account in gmail

- Type the required information on the sign up page.
- Finish by clicking on the check box “I agree to the Google’s Terms of Service and Privacy Policy”.
- Click **Next Step** button.

b) Send and receive email messages

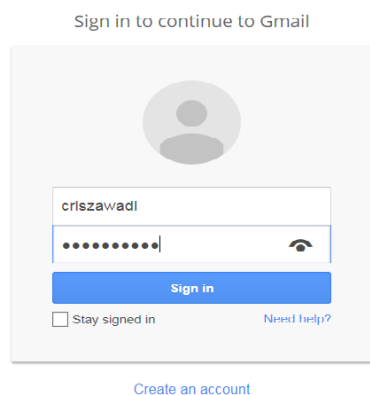
- **Sending message:** Before sending and reading a message, the user must open their account. The procedure of opening an account is referred to as **Signing in**.

To sign in to an existing account do the following:

- Open the web browser software installed in the computer.

- Open the e-mail application program to display the sign in window. A sample is shown in picture below.
- Type the e-mail address under **User Name** box and password under the **Password** Box.
- Click **Sign in** button to open the e-mail.

One account. All of Google.



Picture 5. 12: Sign in box of a Gmail software

The user can either compose a new message or send one that already exists in their inbox. Composing is the process of creating a new mail. To compose a message do the following:

- (i) Click on the **Compose** button in the e-mail account window. A dialog box appears.
- (ii) Click **To** text box and type the e-mail address of the recipient.

Note: To send the same message to a number of people the following options can be used.

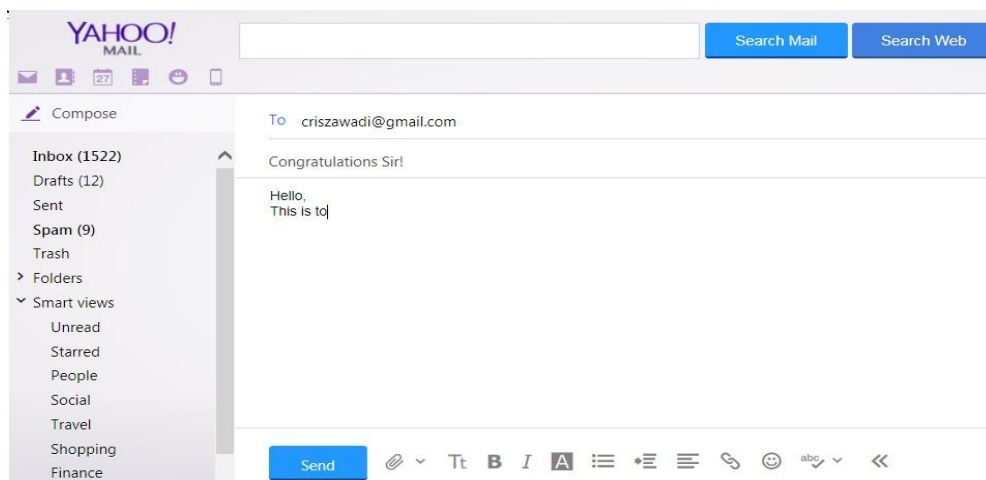
- **Cc (Carbon Copy):** When this option is used, the list of all the recipients appears in each recipient's email.
- **Bcc (Blind Carbon Copy):** When this option is used, the list of all the recipients is hidden from the other recipients of the mail.

(iii) Click at **Subject** text box to type the title that describes what the e-mail is about.

(iv) Type the message in the body section.

Note: E-mail text can be formatted the same way as in a Microsoft Word document using the formatting bar.

(v) Click **Send** button once the message is completed.



Picture 5. 13: Composing a new mail

c) Receiving a message

Upon signing in, the user's inbox is automatically opened. The messages are displayed in three columns showing the name of the sender, the subject and the Date/Time when the email was sent.

To read a message, click on the **Subject**. Once the message is read the user can decide to select one of the following options available on the message window: Reply, Reply to All, Forward, Move and Delete among others.

- **Reply:** This option is selected if the user desires to respond to the read mail.
- **Reply to All:** This option is selected if the user desires to respond to all the addresses listed in the **CC**, **BCC** and **To** box. Both CC and BCC forward a copy of the message to everyone listed as a recipient. The difference between CC or **Carbon Copy** and BCC (**Blind Carbon Copy**) is that, with the latter, the recipients do not get to know each other.
- **Forward:** This option is selected if the user desires to send the same message to other e-mail addresses.
- **Delete:** This option is selected if the user desires to erase the mail. Note that deleting the message only sends it to the **Trash** folder. The user can retrieve the message from this folder if so desired.
- **Move:** This option is selected if the user wishes to remove the message from the **Inbox** folder and transfer it to another folder.
- To permanently delete the message, empty the trash folder.

d) Organise email message

The following are some of the ways of organising an Email.

i. Move

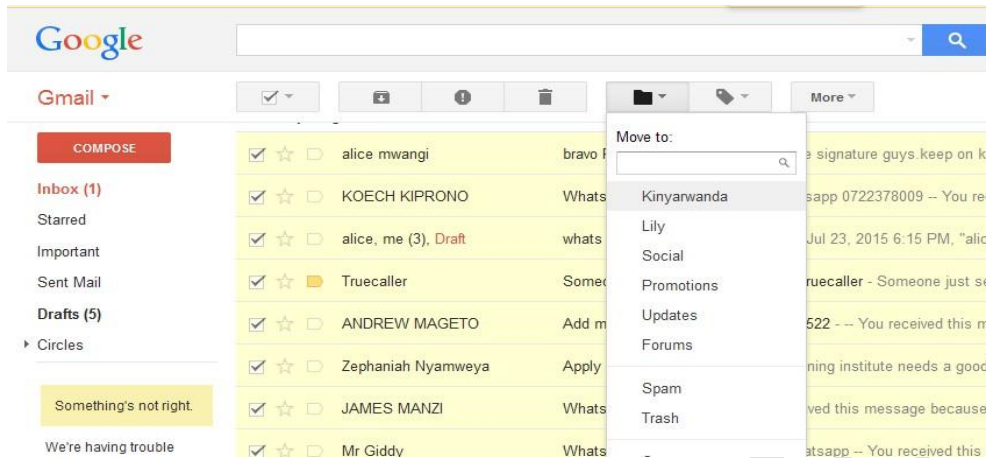
An email in the inbox can be transferred from the inbox to another folder. The following is the procedure for moving an email.

- Select the emails to be stored in a folder by checking the box to the left of the

email's name or thumbnail.

- Move the emails into the folder by clicking the **Move to** button at the top of the email list and select the name of the folder from the drop-down menu.

Note: If multiple emails have been selected, they will move together.

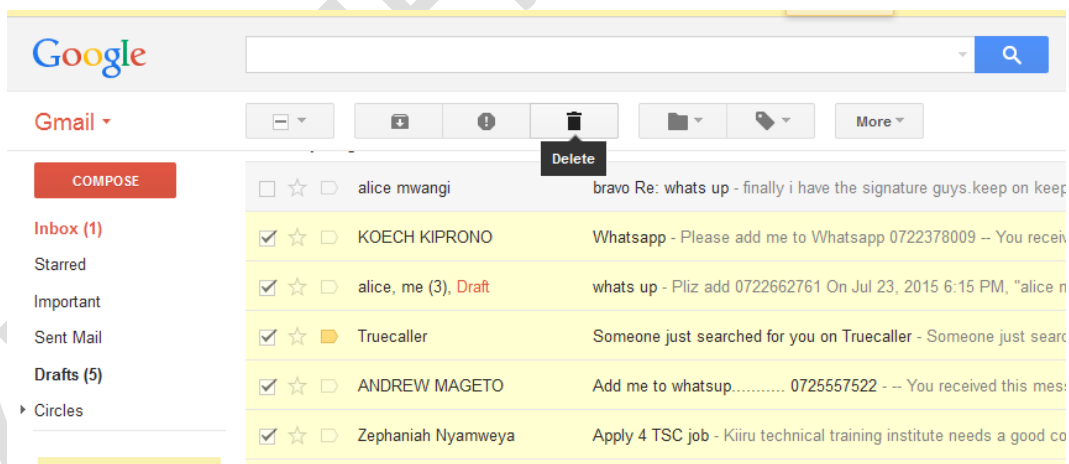


Picture 5. 14: Moving an Email from one folder to another

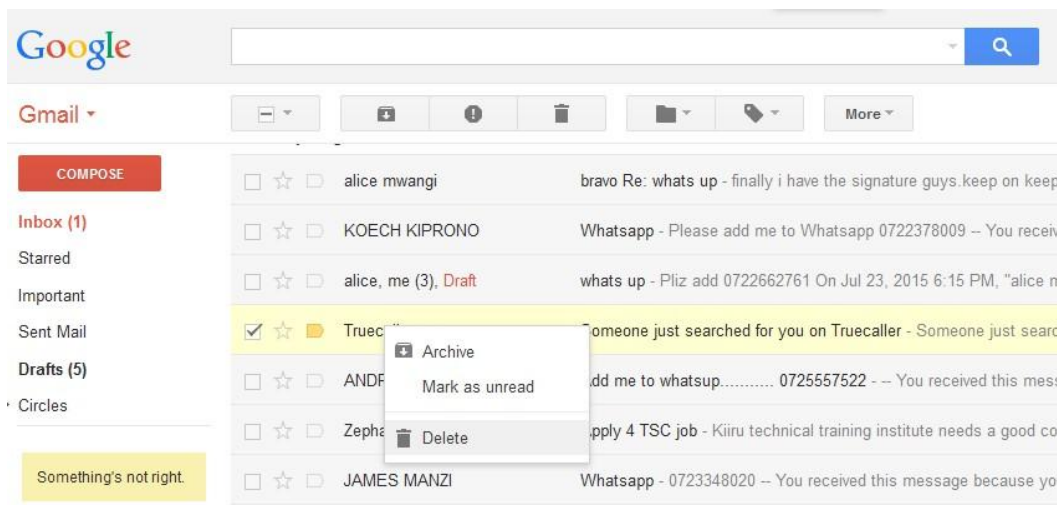
ii. Delete

An email can be erased from the account using either of the following procedures:

Click on the email and then click the **Delete** button OR Right-click on the Email message and click **Delete** from the pop-up menu.



Picture 5. 15: Deleting an email using the Delete button



Picture 5. 16. Deleting an Email by right-clicking it

e) Add or switch folders

- On the left side of the page, click More to expand the labels list.
- Click Create new label at the bottom of the labels list
- Type the name of the new label.
- Click Create.
- Gmail does not use folders; instead, it uses labels to help users organize emails more effectively

Picture 5. 17: Gmail uses labels to help the user organise the mails effectively

Comparison between labels and folders:

Labels	Folders
A conversation can have more than one label.	You can only put a message in one folder.

A conversation can be in several locations (Inbox, All Mail, Sent Mail, etc.) At the same time, making it easier to find later.	You have to remember where you filed a message to retrieve it.
You can search conversations by label.	You can't always do folder-specific searches.

Table 5. 2: Comparison between labels and folders

e) Setting up *an* address book

It is an online contact manager used to store all contact information and it makes it easy to keep track of names, e-mail addresses, phone numbers, fax numbers and a lot more. It is possible to add a contact to the address book from an e-mail message.

To add names to the address book:

- Click on **Add contact** option. A dialog box is displayed.
- Type the details of the contact requested.
- Click **Add** on the dialog box. The window is automatically closed.

Using the address book

- Once a contact exists in the address book, when the user starts typing the e-mail address, a list of all the possible contacts that match it are displayed.
- Select the desired contact and press the **Enter** key.

f) Uploading and downloading files

It is possible to attach one or more files to an e-mail address. The attachment can be a Microsoft Word document file, a spreadsheet, a sound file, a video file, or a picture file among others.

To attach a file do the following:

- Click on the **Attach** command button. A dialog box appears.
- Select the file to be attached then click **Open** button. The attachment procedure begins.
- Once the procedure is complete, click the **Send** button.

Note: Once a message is sent, a copy is stored in the **Sent** folder.

Downloading is the process of getting information from the internet. This information can be viewed, backed up (saved) or printed. Any attachment sent in an e-mail needs to be downloaded.

To download a file, do the following:

- Click **File** on the page to be downloaded.

- Select **Save As** option and follow the procedure for saving a document.

Note that for most computers, by default the downloaded files are stored in the **Downloads** folder.

To print the page without saving, do the following:

- Click **File** on the page to be downloaded.
- Select the **Print** option and follow the procedure for printing a document.

Application activity 5.5

- 1) Fill in the blanks
 - a) E-mail messages are stored in the _____ folder.
 - b) CC means _____.
 - c) BCC hides the list of _____.
 - d) Files sent with an e-mail are called _____.
- 2) List the steps for creating an e-mail account.
- 3) Describe how to send an e-mail message.
- 4) State two ways of organizing e-mails.
- 5) Write an email message with an attached file to your colleague. The message would be about sending to her/him a student book of any subject.

5.6. End of unit assessment

5.6.1. Project: My school's Network

Your school wants to connect its computers so that staff share different resources such as printers using that network. That network must access the internet so that staff can send emails and do research. Draw a diagram of that network including the different devices that will be needed such as computers (clients), a server, a router, a switch, a modem, network cables (Ethernet), a shared printer, and an internet connection.

Guidance:

- Survey your schools' room and identify where each network device will be put
- Identify how the cables will be connected from those different devices
- Identify where the central devices (switches, routers, servers) will be placed
- Draw a diagram of that idea on how the network will be organized

Questions on the project:

- 1) a) Why does the school need to have a local network?
b) Why is it necessary to connect that local network to the internet
- 2) In a table, list all the devices needed to set up this network and their role
- 3) Draw a network diagram showing all devices and how they are connected
- 4) Explain how your network works to transmit data among the local computers and the internet.

5.6.2. Assessment questions

- 1) Answer with True or False.
 - a) The internet is a worldwide network of computers.
 - b) A web browser is used to view web pages.
 - c) Google is an example of a web browser.
 - d) E-mail messages can be sent to many recipients at once.
- 2) Which of the following is a transmission medium?
 - a) Printer
 - b) Cable
 - c) Monitor
 - d) Keyboard
- 3) Which network covers a city or town?
 - a) LAN
 - b) PAN
 - c) MAN
 - d) WAN
- 4) Which part of a URL comes first?
 - a) Server name
 - b) Resource ID
 - c) Domain name
 - d) Protocol
- 5) Which of the following is not a web browser?
 - a) Google Chrome
 - b) Mozilla Firefox
 - c) Yahoo
 - d) Opera
- 6) Which option hides recipients' addresses in an e-mail?
 - a) To
 - b) CC
 - c) BCC
 - d) Subject
- 7) Write two examples of search engines.
- 8) Outline the steps used to search for information using a search engine.
- 9) State two responsible ways of using the internet.
- 10) Why is it important to use strong passwords for e-mail accounts?
- 11) Give two dangers of improper use of the internet.

EXPERIMENTAL VERSION

UNIT 6: MUSIC SPRITE PROGRAMMING



Key unit competence

Create a suitable music sprite using Scratch and Audacity

Learning objectives

At the end of this unit, you should be able to:

Knowledge and understanding

- Identify different tools in Scratch that enable to produce a melody.
- Identify different sound blocks and commands and their uses
- Explain the use of variables in controlling sprite actions
- Describe steps to create a song or tune using the Sound Toolkit.
- Explain the use of Audacity software in creating and editing sound clips.

Skills

- Manipulate variables for controlling sprite actions
- Compose a short tune or song using the Sound Toolkit.
- Record, edit, and export a sound clip using Audacity.

Attitudes and values

- Pay attention on how to use commands and to produce music.
- Develop patience and persistence in debugging and refining sprite and sound scripts.
- Demonstrate responsibility and respect for digital work, including proper use of others' audio or images.

6.0. Introduction

The purpose of this unit is to introduce you to basic programming concepts using Scratch. In this unit, you will learn how to create and import sprites, use different tools in the Scratch interface, and understand how sounds and music can be added to sprites to make programs more interactive.

You will also be introduced to the use of variables in Scratch and how they can be manipulated to control actions such as switching sound on and off. In addition, You will learn how to create tunes using the Sound Tool Kit and how to record and edit sound clips using Audacity before importing them into Scratch.

Introductory activity

Read the text below and answer the related questions:

During an ICT lesson, KABAYIZA uses Scratch to create a simple animation project. He selects sprites in different ways, including choosing from the sprite library, drawing with the Paint editor, and importing images from the computer. To enhance the project, he adds sounds using the Sound Tool Kit and controls how music plays by creating a variable called “Switch” with ON and OFF costumes. He also records a sound using Audacity, saves it as a WAV file, and imports it into Scratch to make the animation more interactive.

- 1) How did KABAYIZA use different tools in Scratch to create and customize sprites for his animation project?
- 2) Explain how sounds were added and controlled in the Scratch project to make it more interactive.
- 3) Why did KABAYIZA create a variable called “Switch,” and how did it help control the music in his project?
- 4) Describe the role of Audacity in the project and why the WAV file format was important for Scratch.
- 5) In your opinion, how do sprites, sounds, and variables work together to improve the quality of a Scratch animation?

6.1. Creating and importing a sprite

Activity 6.1

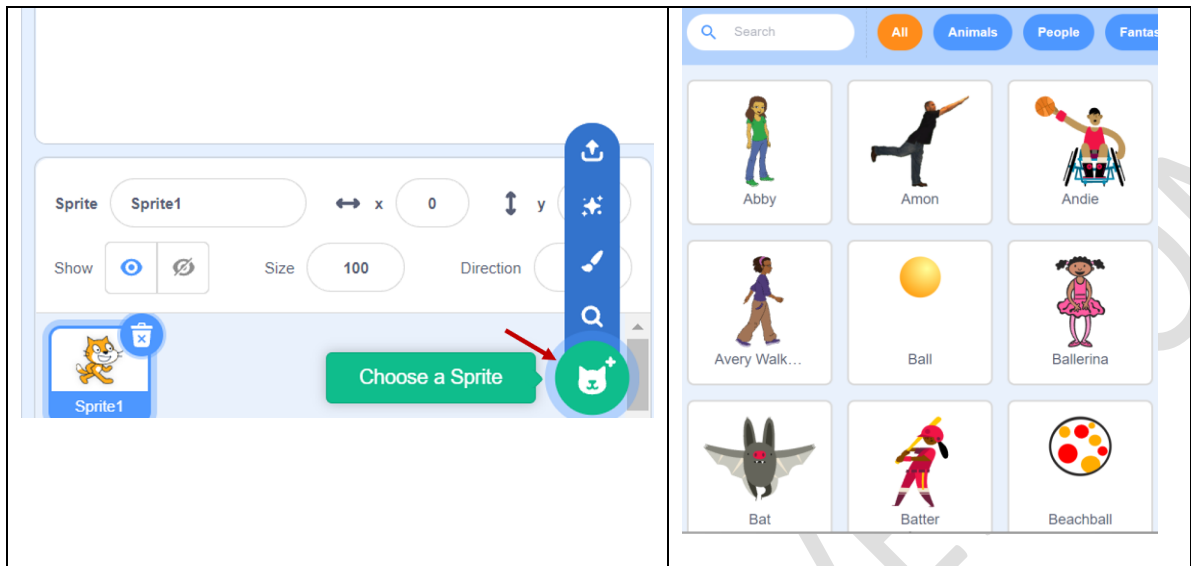
- 1) Define the term importing as used in Scratch.
- 2) What is a sprite?
- 3) Name three buttons used to create or import a sprite in Scratch.
- 4) Mention two examples of items that can be imported into Scratch besides sprites.

Importing refers to the processes of transferring an existing image into a program. The user can import sprites, backdrops, sounds, projects, costumes, scripts, and lists among others.

6.1.1. New sprite and importing the existing image

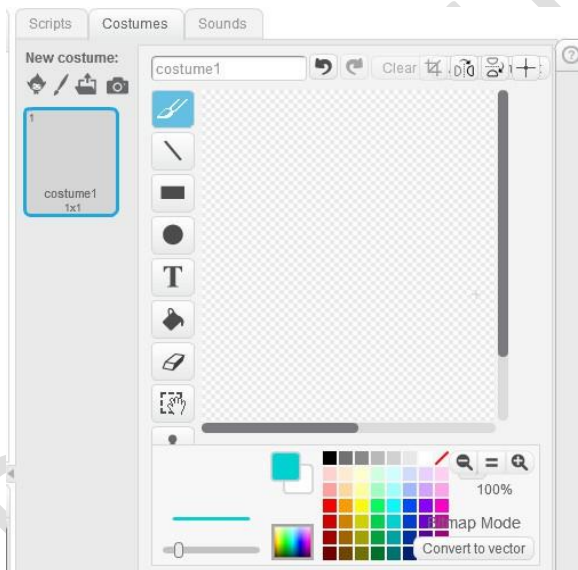
To create a sprite, do the following:

- (i) Click on the **Choose Sprite** button. A sprite library window appears as shown in picture below.



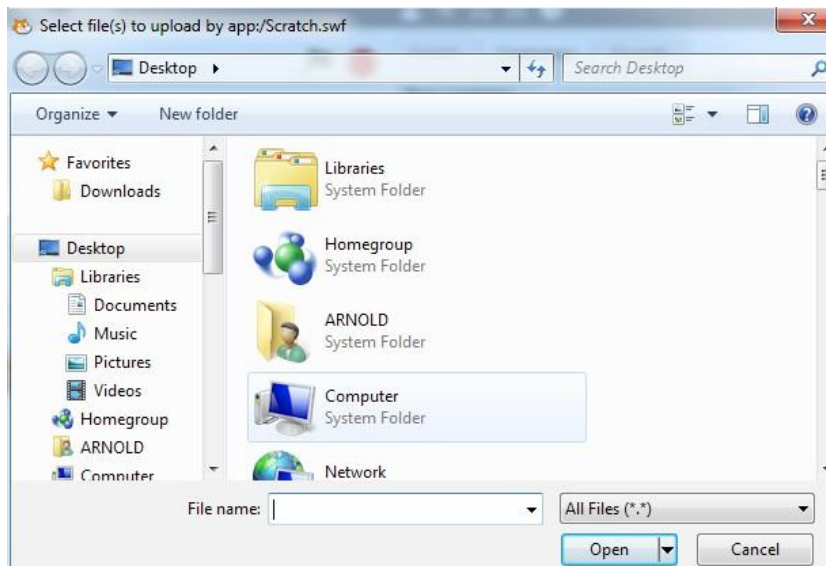
Picture 6. 1. Left to right: Choose a Sprite tab and the Sprite library

- (ii) Double-click on the desired sprite or select the desired sprite and click **OK**. The selected sprite appears on the stage area.
- B. The **Paintbrush** button opens the Paint editor and is used to manually draw the sprite with colour tools provided by Scratch. The picture below shows a Paint editor in a Scratch window.



Picture 6. 2: Part of Paint editor Window

- C. The **Folder** button opens a file browser and is used to upload a sprite or image. Select the location from where to upload the sprite, select the file name then click **Open**.



Picture 6. 3: File browser to upload sprite

D. The **Camera** button opens a camera which would be used to create a new sprite. A computer with a Web camera can be used to capture an image. When the **Camera** button in the Scratch program is clicked, the web camera is activated ready to capture A photograph. The picture below shows a computer with a web camera that can be used to capture images for use as sprites. A dialog box is displayed for saving the captured image.



Picture 6. 4: A web camera

Application activity 6.2.

- 1) Use the Choose a Sprite button to select a sprite from the sprite library and add it to the stage.
- 2) Use the Paintbrush button to draw your own sprite using the Paint editor.
- 3) Use the Folder button to import an image from your computer and add it as a sprite.
- 4) Which button did you use to select a sprite from the library?
- 5) Which button allows you to draw your own sprite?

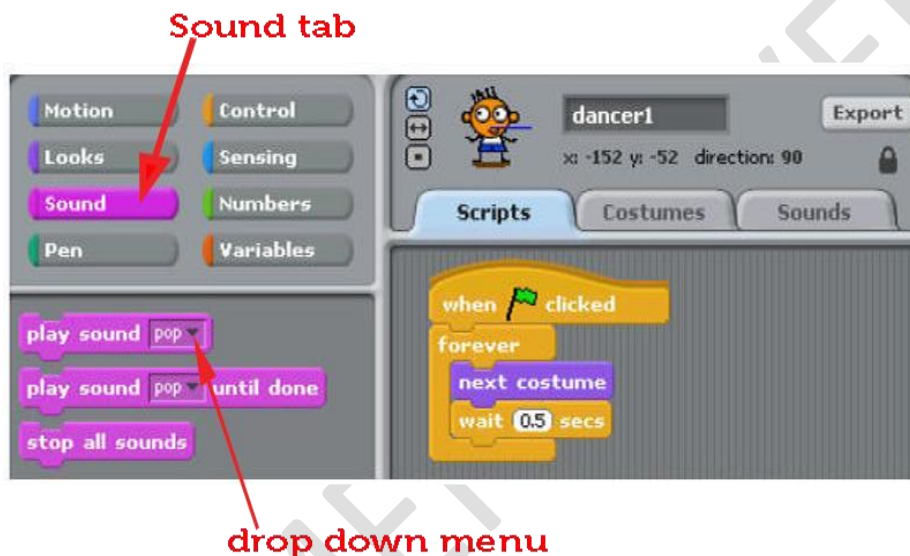
6.2. Using combo blocks to play sounds

Activity 6.2

- 1) What is the purpose of the Sound tab in Scratch?
- 2) How can you choose or change a sound in the Play Sound... Until done block?
- 3) What is the function of the Forever block when combined with sound blocks?
- 4) How do you start playing a sound script in Scratch?

Combo sprite blocks are formed by combining single sprite blocks. The default sound in the **play sound... until done** block is **meow**. Below is the procedure on how to play sound using combo blocks:

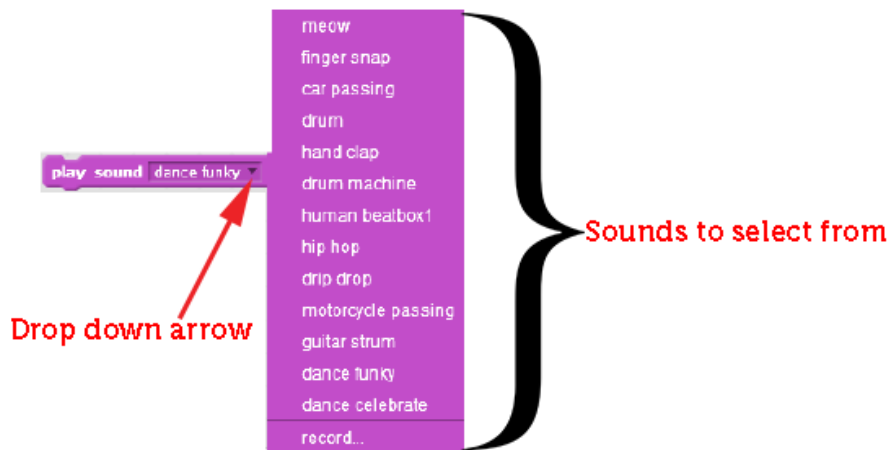
- (i) Click on **Sounds** tab to display the **Sound Library**. Select the preferred sound from the library or any other source.



Picture 6. 5. The sounds tab

- (ii) Double-click the preferred sound or click on the sound then click **OK**. The selected sound appears as a thumbnail under **Sounds** tab or in the **Play Sound Block** under the **Scripts** tab.

To select any other sound, click the drop-down arrow in the play sound block as shown in picture below:



Picture 6. 6: Sounds available in Play Sound block

- (iii) Drag the blocks under the Sound tool kit and click the drop-down arrow to select the sound. The picture below shows how you can combine sound combo blocks for playing the different sounds as follows:
- Playing the sound of a car passing until done.
 - Playing a drum until done.
 - Playing dance celebrate until done.
 - Playing dance funky until done.



Picture 6. 7: Sound combo block

- (iv) In the Control Tool Kit, drag Forever block to enclose the Sound Blocks.
 (v) Click the Events Tool Kit, drag the **When () clicked** to the top of **Forever Block**.
 (vi) Click the **Green Flag** to play the sound.

Note: To play a script, the user can click on the Green flag button or on the script.

For choosing the sprite especially from the things folder, the **Things** folder refers to the folders in Scratch that contain Sprites. They can be created by the user and are also inbuilt. These folders ensure that Sprites are organised in groups.

The **Sprite Library** contains various folders with different names, for example,

Animals, People, and Things among others. To choose a Sprite from the things folder:

- Click on the **Choose a Sprite** button to open the sprite library.
- Click on the folder to open. A window opens and displays the various items in the folder as shown in picture below.
- Select the desired Sprite

Application activity 6.2

Your school is organizing a small digital entertainment show. You are asked to design a simple Scratch program that plays different sounds continuously when the Green Flag is clicked.

Add When Green Flag clicked block and Forever block. Inside the Forever block, combine at least two play sound... until done blocks. Run your program by clicking the Green Flag.

- 1) Which sprite did you choose from the Things folder?
- 2) Which two sounds did you combine in your script?
- 3) What happened when you clicked the Green Flag?
- 4) What is the role of the Forever block in your project?
- 5) What would happen if you removed the Forever block?

6.3. Scratch variables with Switch ON/OFF

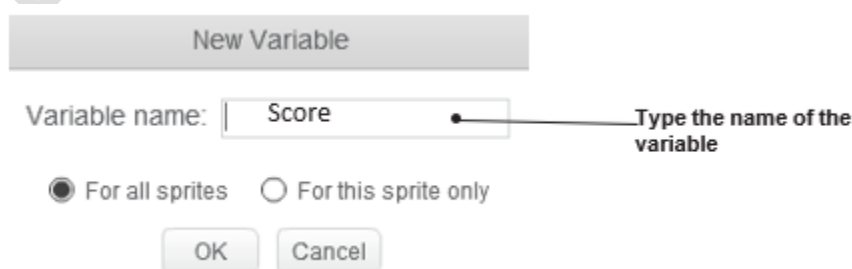
Activity 6.3

- 1) What is a variable in a computer program?
- 2) Can a value stored in a variable change?
- 3) What happens when you turn a real switch ON or OFF?
- 4) In Scratch, where do we click to create a new variable?

A. A variable

A variable is a changeable value that can only hold one value at a time. The value can be either a number or string (text). Variables enable one to see and manipulate numbers and text in the program. It is created from the **Data blocks** on the **Script Tab** using the **Variable block**.

Clicking on this button displays the new variable dialog box that enables the user to name the variable and specify its scope as shown in picture below.



Picture 6. 9: Creating a Global variable named score

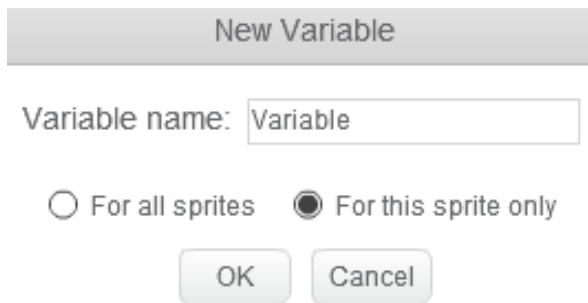
B. Types of variables

There are three types of variables in Scratch namely **local** (private), **global** (public) and **cloud**.

- **Local variable:** This is a variable that can only be changed by the sprite that owns it. It can be read by other sprites using the **Change () by ()** block but they cannot change it.

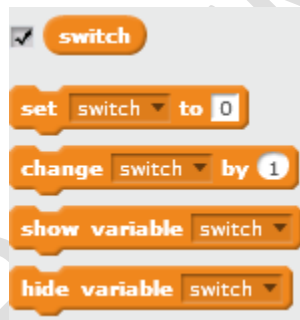
For example:  block.

A local variable is created by selecting **For this sprite only** option button in the New variable dialog box.



Picture 6. 10: Creating a local variable

- **Global variable:** This is a variable that can be read and changed by any sprite in the project. By default, **for all sprites** option button is selected meaning that it is a **global variable**.
- **Cloud variable:** This is a variable that is stored in the cloud (on the Scratch server). It can be accessed by all users viewing the project shared on the Scratch website.



Picture 6. 11: Variable blocks

- **Variable name block:** It contains the name of the variable and shows or hides the variable's monitor on the stage.
- **Set block:** It sets the variable to a specified value.
- **Change block:** It changes the variable by a specified amount either positive or negative.
- **Show variable block:** It shows the variable's monitor on the stage while the script is running.
- **Hide variable block:** It makes the variable's monitor invisible on the stage while the script is running.

To manipulate variables to program switch on and switch off, a virtual switch is

Created. A virtual switch is like a real switch with on and off buttons. The virtual switch is simply a sprite and must provide two costumes, one to represent when switch is on, and the other to represent when the switch is off.

It works by combining blocks from different tool kits available in Scratch program and may be useful when making a project that has music or sound that can be stopped or muted.

Example: A sprite can be programmed such that clicking on it turns the switch on and off. Two costumes for the sprite are needed such that one costume represents **on (1)** and the other represents **off (0)**. The switch is on when the music is playing and the switch is off when the music is off.

The following steps are followed when creating a Switch as shown in the picture below:

- Create a new sprite and add two costumes to the sprite. Costume1 represents on and Costume2 represents off.
- Under Scripts tab click Data tool kit.
- Click the Make a Variable button and name the variable as Switch.

This example uses blocks from six tool kits namely Sound, Looks, Data, Operators, Events and Control tool kits. Select and combine blocks from the tool kits to create the scripts as shown in the picture below.



The image shows two Scratch scripts for a virtual switch. The first script, labeled 'Music keeps playing when Switch is on', starts with a 'when green flag clicked' block followed by a 'forever' loop containing four 'play sound' blocks: 'drum set1 until done', 'drum set2 until done', 'drum until done', and 'pop until done'. The second script, labeled 'The first costume turns the Switch on (1) and sets volume to 100% and the second costume turns the Switch off (0) and sets volume to 0%', starts with a 'when green flag clicked' block followed by a 'next costume' block. It then has an 'if' block with the condition 'costume # = 2'. If true, it sets volume to 0% and sets the 'Switch' variable to 0. If false, it sets volume to 100% and sets the 'Switch' variable to 1.

Music keeps playing when Switch is on

The first costume turns the Switch on (1) and sets volume to 100% and the second costume turns the Switch off (0) and sets volume to 0%

Picture 6. 12. Switch that turns music on and off

Application activity 6.3

1) Create a new sprite and add two costumes ON and OFF. Create a variable named Switch. Program the sprite so that when the sprite is clicked: If Switch = 0 then change it to 1 and play a sound; If Switch = 1 then change it to 0 and stop the sound.

- What happens when Switch = 1?
- What happens when Switch = 0?
- Which blocks did you use from the Data toolkit?
- Why did you use two costumes for the sprite?

6.4. Creating a tune using the sound tool kit

Activity 6.4

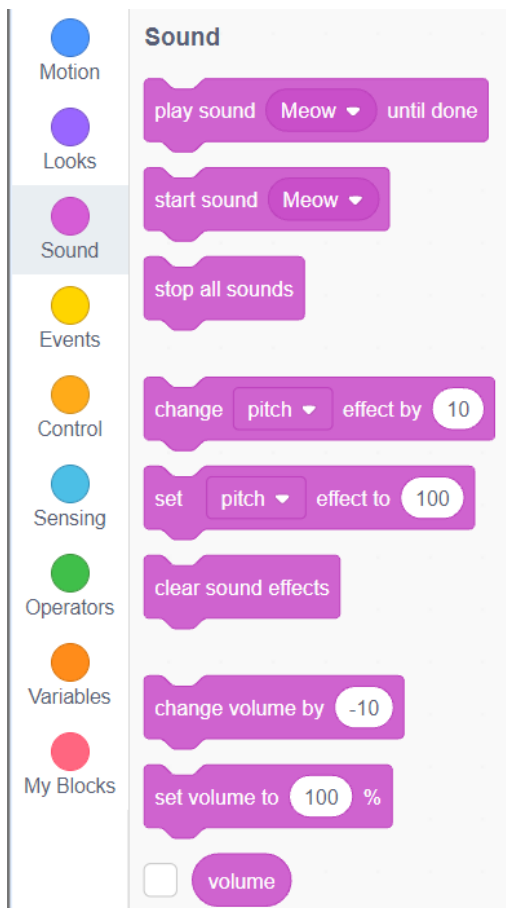
- What is the use of the Sound Tool Kit in Scratch?
- Which tab do you click to find sound blocks like play note?
- What block is used to add a pause between musical notes?
- How can you make a sprite produce different musical notes in Scratch?

The Sound toolkit is displayed by clicking on the **Sound** block under the **Scripts** tab. This tool kit is used to add a music script to a sprite.

6.4.1. Add a music script to a sprite

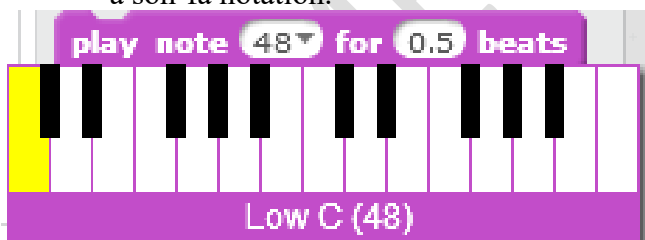
In this practical activity, you will learn how to create a tune using the Sound Tool Kit. To **add a music script to a sprite**, do the following:

- Select the **Sprite** to which you want to add music.
- Click the Sounds Tab block under Scripts tab.
- Add the desired music using the blocks under the Sound block as shown in picture below.



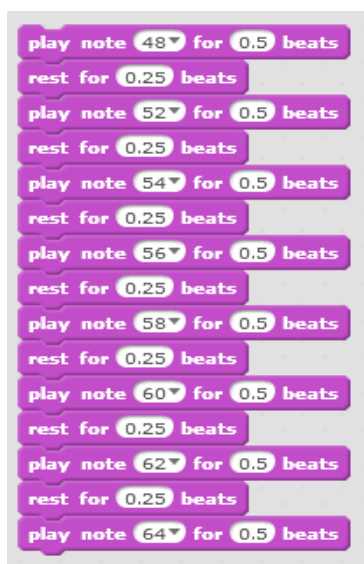
Picture 6. 13. Sound blocks

- You could, for example, drag the block shown in picture below to the scripts area to play a soh-fa notation.



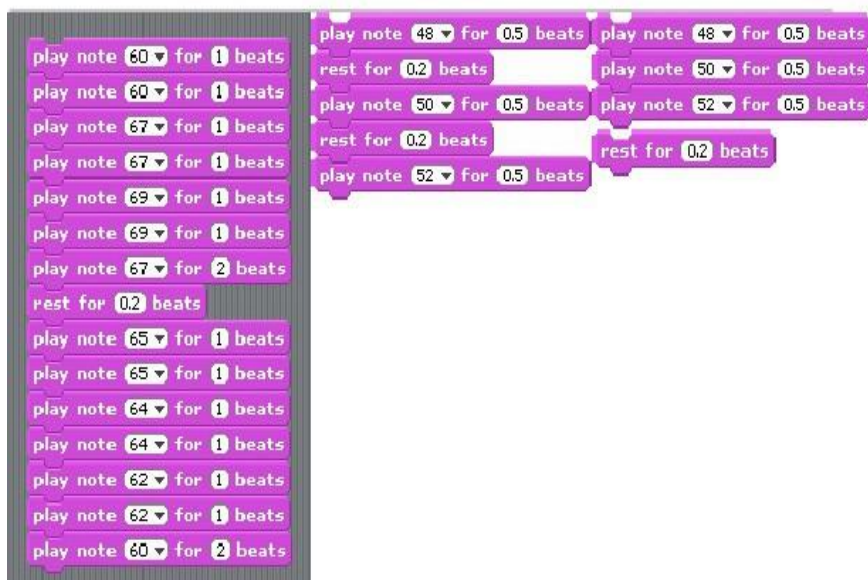
Picture 6. 14: Keyboard Sol-fa Notation Do

- Click the drop-down arrow on the block to display the keyboard.
- To create a tune using the Sound tool kit that will play the tune is Do, Re, Mi, Fa, So, La, Ti, Do and pauses for 0.25 seconds between the notations, make a script combo block as shown in the picture below. Double click on the blocks to play the sounds.



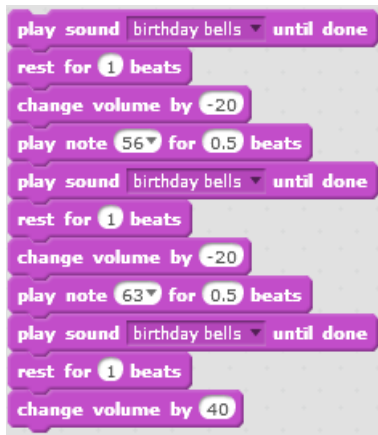
Picture 6. 15: Script block for the Soh-fa notation

To add pause between notes, use the block that reads, “**Rest for 0.2 beats.**”



Picture 6. 16: Adding pause between notes

The script shown below plays the birthday bells sound in a sprite. The tune plays 3 times, rests for 1 beat between them, and reduces volume by -20% until it stops then resets volume to 100%.



Picture 6. 17: Birthday bells Script block

Application activity 6.4

- 1) Which sprite is used to add music script?
- 2) Which musical can be used to create a tune?
- 3) How are pauses between the notes added?
- 4) How to make the tune repeat more than once?

6.5. Creating a sound clip using Audacity

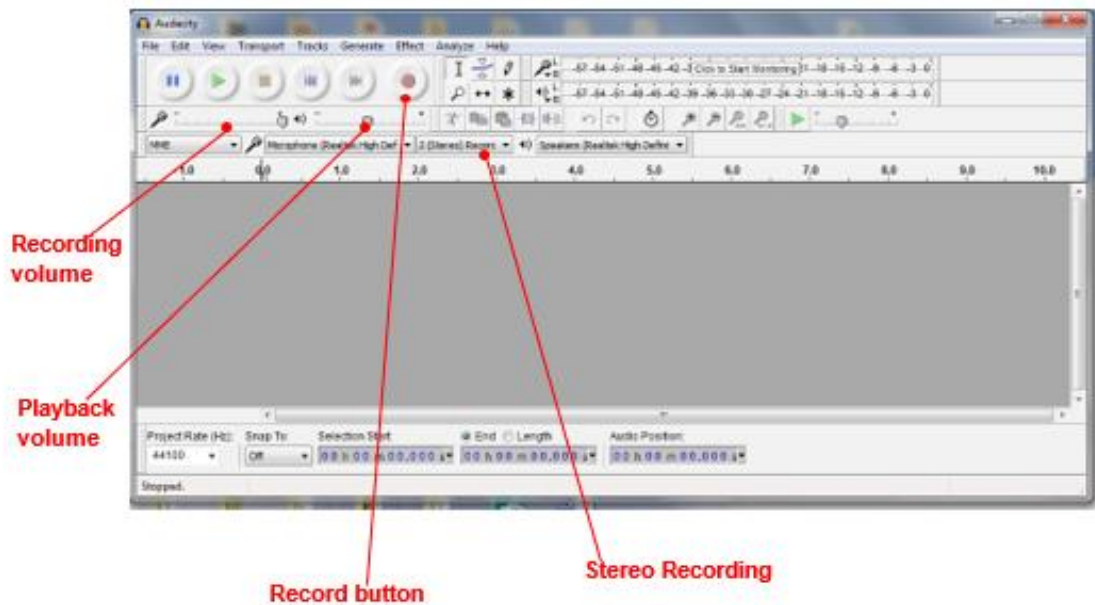
Activity 6.5

- 1) Mention one software that can be used to record sound on a computer.
- 2) What happens when you click the Record and Stop buttons in a sound recording program?
- 3) Why recorded sound files need to be saved before using them in another program like Scratch?

Audacity is a free, open source software that is used for recording and editing sounds. In Scratch, creating a sound clip or song using Sound Tool Kit can be time-consuming. The Audacity software provides the Scratch user with several features that can be used to create Scratch sound clips or songs. This program must be installed in the computer for it to be used.

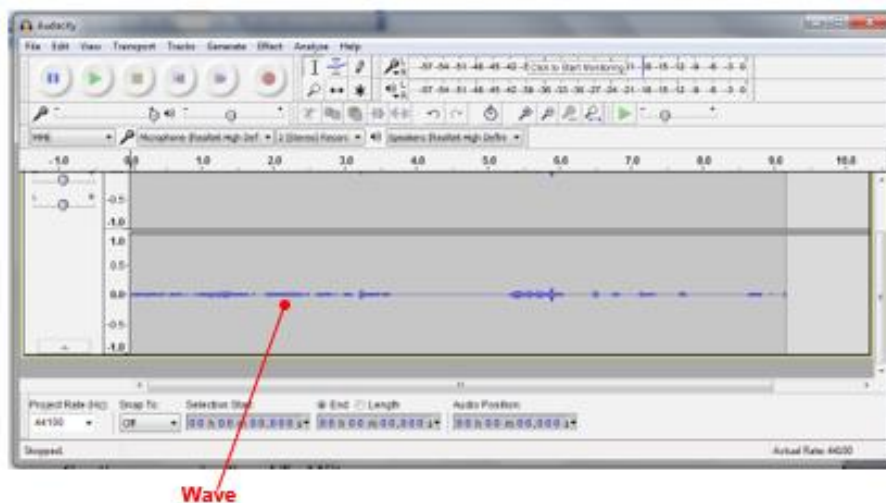
To use the Audacity program, do the following:

- Open the Audacity program. The window shown in the picture below is displayed.



Picture 6. 18: Birthday bells Script block

- Start playing the preferred sound clip or song using any of the players on the computer.
- Select **Stereo Recording**.
- Click the **Record** button  to start recording. This displays sound waves as shown in picture below.



Picture 6. 19: Audacity window showing a wave

- Click the **Stop** button  to stop recording.
- To save the recorded sound clip or song, click **File** then select **Export Audio**. Save the file as type WAV, for example, **rwandanationalanthem.wav**.
- To import the recorded sound clip or song to a sprite, select the **Sounds** tab and click **Import** button to upload sound from file. The block below shows a sound clip created using Audacity.



Picture 6. 20.A sound clip created using Audacity

6.5.1. Creating a song using Sound Tool Kit

To create a song, do the following:

- Click **Sounds** tab.
- Create a new sound (s) by using the Sounds tab.
- The created sound is added to the **Play sound** and **Play Sound... Until Done** blocks.
- Click on **Scripts** tab and select **Sound**.
- Drag **Play Sound** or **Play Sound... Until Done** blocks into the scripts area.
- Click the drop-down arrow on the block and select the preferred song.

Application activity 6.5

- 1) Using Audacity record a short sound clip (for example: your voice saying “Welcome to our Scratch project” or a short song). Stop the recording and check the sound waves then export the file as a WAV file and save it on the computer.
 - a) What type of file format did you use when exporting your sound?
 - b) What happened on the screen when you clicked the Record button?
- 2) Open Scratch and import the recorded sound into a sprite using the Sounds tab and add a script so that the sound plays when the Green Flag is clicked.
 - a) How did you import the sound into Scratch?
 - b) Which block did you use to make the sound play?
 - c) What challenge did you face while importing the sound?

6.6. End of unit assessment

6.6.1. Project Title: Dancing Robot

Guidance:

In Scratch software:

- Create or import a sprite.
- Program the sprite to move.
- Add sound or music to the sprite.
- Control sound ON and OFF.

Project questions

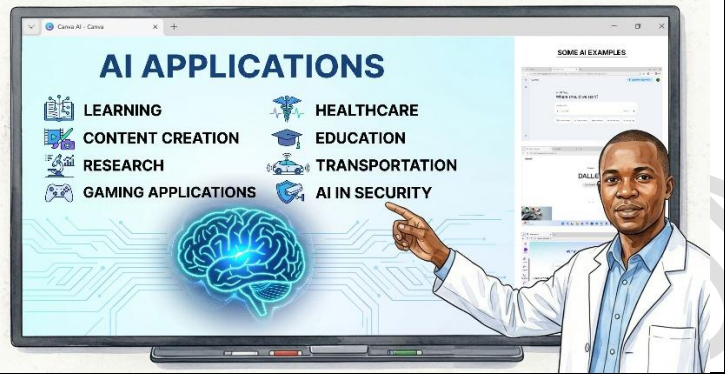
- 1) How do you import or create a sprite in Scratch?
- 2) How can you program a sprite to move in rhythm with music?
- 3) How do you test and correct errors in your animation?
- 4) How does creativity help in designing animations?

6.6.2. Assessment questions

- 1) Answer with True or False.

- a) Importing means transferring an existing image into a program.
- b) A sprite is created only using the Paintbrush button.
- c) The default sound in the “Play Sound Until Done” block is meow.
- 2) Choose the correct answer.
 - a) Which button is used to upload a sprite from a computer?
 - i) Paintbrush button ii) Folder button iii) Camera button
 - b) Which Scratch tab is used to add and manage sounds?
 - i) Scripts ii) Sounds iii) Costumes iv) Control
- 3) State three types of variables in Scratch.
- 4) Give two uses of the Sound Tool Kit.
- 5) Explain how to upload a sprite using the Folder button.
- 6) State the steps used to create a tune using the Sound Tool Kit.
- 7) Explain how to import a sound clip into Scratch.
- 8) Why is it important to organize sprites and sounds properly in Scratch projects?
- 9) Mention two good practices when creating Scratch projects with sound.

UNIT 7: INTRODUCTION TO AI



Key unit competence

Explain the concepts of Artificial Intelligence & Machine Learning and their application in real life

Learning objectives

At the end of this unit, you will be able to:

Knowledge and understanding

- Explain basic concepts, characteristics, and types of AI
- Differentiate between human intelligence and machine intelligence
- Describe Machine Learning and how machines learn from data
- Identify different types of Machine Learning and their applications

Skills

- Analyze how AI systems learn from data to make predictions and suggestions.
- Compare human intelligence and machine intelligence using examples from daily life.
- Classify different types of AI and machine learning models

Attitudes and values

- Appreciate the importance of AI in modern society
- Use AI ethically and responsibly

7.0. Introduction

Artificial Intelligence (AI) is transforming how humans work, learn, and solve problems. From smartphones and chatbots to self-driving cars and medical diagnosis, AI systems help machines perform tasks that normally require human intelligence. Behind many AI systems is Machine Learning, which allows machines to learn from data and become more intelligent.

In this unit, you will become more aware of AI and how you can use it appropriately.

Introductory activity

Akaliza and Irasubiza, Senior One students, visit a Smart City during their school holidays. They observe how computers and intelligent systems are used in daily life, such as face-recognition at city entrances, computer-assisted diagnosis in hospitals, learning assistants in schools, secure banking systems, automatic traffic lights, and personalized mobile phone suggestions. However there are concerns about privacy, fairness, and decision-making when AI powered tools are utilized.

- 1) a) Which tasks in the Smart City are performed by machines?
b) Which ones still require human intelligence?
- 2) Can machines learn from experience? If yes, explain.
- 3) What role does data play in helping machines make decisions?
- 4) What risks may arise when machines or computer systems make decisions for people?
- 5) When an intelligent computer system makes a wrong decision, who should be held responsible?

7.1. Basic concepts of AI

Activity 7.1

At the school library, a new digital system is installed to help students find books faster. When a student types “plants”, the system suggests books about biology, agriculture, and environmental science. If a student borrows many history books, the system later recommends more history materials. The system can also read the title of a book using a camera and identify whether it is available or already borrowed.

- 1) Which actions of the library system show that it is intelligent?
- 2) How does the system learn the students’ reading preferences?
- 3) Does the system think like a human librarian? Explain?
- 4) What role do data and patterns play in making the system intelligent?

AI is a branch of computer science that enables machines to perform tasks that normally require human intelligence, such as learning, problem-solving, understanding language, and making decisions. Unlike ordinary computers, AI systems can learn from data and improve over time using algorithms. AI allows machines to think and act like humans without human emotions.

Examples of AI in daily life include Voice assistants (Google Assistant, Siri), Chatbots and face recognition on smartphones.

7.1.1. Characteristics of AI systems

AI systems are designed to perform tasks that usually require human intelligence. Unlike ordinary computer programs that follow fixed instructions, AI systems have special features that make them more flexible and intelligent. Some of the main characteristics include:

- **Learning ability:** AI systems can learn from data, experience, or their environment.
- **Reasoning and problem solving:** AI can analyze situations and make decisions based on logic or rules.
- **Adaptability:** AI systems can adjust their behavior when conditions change.
- **Perception:** AI can interpret sensory data such as images, sounds, and speech.
- **Autonomy:** AI systems can operate without human intervention once programmed or trained.
- **Interaction and communication:** Some AI systems can interact with humans using natural language.
- **Problem generalization:** AI can apply knowledge from one problem to similar problems.

7.1.2. Human intelligence vs machine intelligence

Human intelligence is natural, flexible, and influenced by experience and emotions, while machine intelligence is artificial, depends on data and algorithms, and is limited to tasks it has been programmed or trained to do.

Below is a comparison:

Human intelligence	Machine intelligence
Learns from experience and emotions	Learns from data and algorithms
Uses common sense and creativity	Uses patterns and calculations
Can understand emotions	Cannot feel emotions
Flexible thinking	Limited to programmed or trained tasks

Machines can be very fast and accurate but still depend on humans to design, train, and guide them.

7.1.3. Importance of AI

Artificial Intelligence is one of the most important modern technologies that is transforming how people live, work, and learn. It enables machines and software to perform tasks that normally require human intelligence, such as learning, problem-solving, decision-making, and recognizing patterns.

Some of the importance of AI are outlined below:

- **Automating tasks:** AI can do repetitive or dangerous tasks which would otherwise be done by humans

- **Improving decision making:** AI can analyze very large data quickly and help make better decisions. This can be done in banks where they can track the transactions and track fraudulent ones.
- **Increasing accuracy and efficiency:** Machines using AI do tasks rapidly with more accuracy. They don't make as many mistakes as humans if they were trained correctly.
- **They are available all the time:** AI systems can work without fatigue, unlike humans. Example: a Chatbots can provide customer support at any time even during the nights.
- **Support learning:** AI is used in learning whereby a student can use AI tools and get explanation about different subjects.

7.1.4. Types of Artificial Intelligence

AI comes in different types depending on what it can do and how it functions. Basing on the level of intelligence and ability to perform tasks, these are the types of AI:

- **Narrow AI:** It performs specific tasks only. Examples include Voice assistants (Siri, Google Assistant), chatbots, recommendation systems. Generative AI is a type of narrow AI which can generate content such as text, images, music.
- **General AI:** It can perform any intellectual task that a human can do. It can reason, learn, and solve problems in different areas.
- **Super AI:** It is a kind of AI that is far beyond human intelligence in all areas. They can make decisions, create, and solve problems better than humans.

Note that General AI and Super AI are still hypothetical and are not yet developed.

7.1.5. AI tools in action

Artificial Intelligence tools are widely used to help people perform tasks more easily and efficiently. Some of the examples include chatbot and ChatGPT.

A. Chatboats

Chatbots are computer programs that use AI to simulate human conversation. They can understand questions, provide answers, and assist users in real time. It uses technologies such as Natural Language Processing (NLP) and machine learning to understand and respond to user inputs.

How a chatbot works:

When interacting with a chatbot, the user begins by typing or speaking a question. The chatbot then analyzes the input using artificial intelligence algorithms to understand the meaning of the request. After that, it searches its knowledge base or the data it has learned from to find relevant information. Finally, the chatbot generates and provides the most suitable response to the user.

Examples of chatbots in daily life include customer service chatbots on websites and mobile apps, virtual assistants like google assistant and Siri and educational chatbots that help students learn or revise topics.

B. ChatGPT

ChatGPT is an AI chatbot developed by OpenAI. It can answer questions, explain concepts, help students learn different subjects, and generate text based on the user's request. The chatbot works using machine learning and Natural Language Processing (NLP) technologies to understand the user's input and provide appropriate responses. These technologies allow the system to analyze language, learn from data, and interact with users in a conversational way.

Application activity 7.1

- 1) Differentiate human intelligence and machine intelligence.
- 2) Give two examples of Narrow AI tools used in daily life.
- 3) List and describe at least four main characteristics of AI systems.
- 4) Give two advantages and two disadvantages of AI in real life.
- 5) Explain what a chatbot is and describe how it works.

7.2. Applications of AI

Learning activity 7.2

You have been hired as a junior analyst at Futuretech company, a multinational organization that develops and implements AI solutions for different industries. Your supervisor has assigned your team to investigate how AI technologies affect people's lives, businesses, and the environment.

Your team's goal is to prepare a report for the management showing how AI is applied, what benefits it brings, what challenges exist, and how to use AI responsibly.

- 1) Identify two AI applications in agriculture.
- 2) State one positive impact of AI in your sector.
- 3) Discuss a social or ethical issue that might arise from using AI in this sector.

7.2.1. AI in various sectors

AI is widely used in many sectors to improve efficiency, accuracy, and decision-making. By analyzing large amounts of data and performing tasks automatically, AI helps organizations and individuals solve problems faster and more effectively. Below are some key sectors where AI is commonly applied.

- **AI in education:** AI is used in education to help students learn better. It supports personalized learning, where lessons match the needs of each student. AI can also create educational chatbots that help students revise their lessons by answering their questions. Some AI tools can do automatic grading and assessment, which helps teachers check students' work quickly.
- **AI in healthcare:** AI helps doctors diagnose diseases more easily. AI can analyze medical images such as x-rays and scans to find health problems. It also helps hospitals manage patient records and hospital systems.
- **AI in business and banking:** AI is also used in finance and business. It can detect fraud in financial transactions and help keep money safe. AI chatbots provide customer service support by answering questions from customers. AI also helps companies make better decisions by analyzing data.

- **AI in transportation:** AI is used in self-driving and smart vehicles to help them move safely. AI can help manage traffic systems and reduce traffic jams on the roads. It also improves route planning and navigation to help people reach their destinations more easily.
- **AI in agriculture:** AI can help predict the weather and crop yield for farmers. It can also detect plant diseases early, which helps protect crops and support smart irrigation.
- **AI in security:** AI can use facial recognition systems to identify people. It can also work in surveillance systems to detect suspicious activities and help keep places safe. In addition, AI is used in cybersecurity systems to prevent online attacks and protect computer systems and data.
- **AI in entertainment and media:** AI can recommend movies, music, and videos that people may like. AI also makes video games more fun by creating intelligent characters.

7.2.2. Impacts of AI

Artificial Intelligence has a strong impact on society by changing the way people live, work, learn, and communicate. AI brings many benefits, but it also creates some challenges. Understanding both positive and negative impacts helps people use AI responsibly and effectively.

A. Positive impacts of AI

AI improves efficiency and productivity by automating tasks, supports personalized learning and tutoring, enhances healthcare through better diagnosis and treatment, increases accuracy in data analysis and decision-making and makes transportation and security safer.

B. Negative impacts of AI

AI may lead to job loss because of automation, reduce human interaction and raise privacy and data security concerns. It is expensive to develop and maintain, and depend on quality data and human guidance.

Application activity 7.2

- 1) Give one example of how AI is used in education.
- 2) Mention one way AI can help students learn better.
- 3) Explain why AI can cause job loss in some sectors.
- 4) Explain how bias in AI systems can affect people.
- 5) Explain how AI can both improve and affect human life.
- 6) Your community wants to introduce AI in agriculture.
 - a) Identify two AI applications that can be used.
 - b) Explain how they would benefit farmers.
- 7) Explain how AI can be used to promote inclusiveness in the community.

7.3. AI Ethics

Activity 7.3

- 1) An AI system rejects students unfairly because it was trained using biased data. Who should be held responsible for this mistake?
- 2) Should AI systems be allowed to make decisions on their own without human supervision? Explain.

7.3.1. Human role in AI

AI systems do not work on their own without human involvement. Humans play a key role in creating, controlling, and guiding AI systems. AI tools are designed to support human activities, not to replace human responsibility or decision-making.

Humans are responsible for:

- **Designing AI systems:** Humans create the programs, models, and rules that AI follows.
- **Training AI with data:** Humans provide data so that AI can learn and improve its performance.
- **Monitoring AI decisions:** Humans check and control AI outputs to ensure accuracy, fairness, and safety.

7.3.2. Ethical and social concerns of AI

AI ethics are guiding principles used by engineers, companies, and governments to ensure that AI is developed and used responsibly, safely, fairly, and in an environmentally friendly way. They focus on avoiding bias, protecting user privacy, and reducing environmental risks. AI ethics are applied through company codes of conduct and government regulations.

- **Bias in AI:** AI systems may favor certain groups unfairly if the data used to train them is biased. This can lead to discrimination in areas such as education, employment, and security.
- **Privacy and surveillance:** AI can collect, store, and analyze large amounts of personal data and this may happen without the user's full knowledge or consent, raising privacy concerns.

7.3.3. Responsible and fair use of AI

Responsible and fair use of AI means applying it in a way that is ethical, safe, and beneficial to everyone. Since AI systems are created and guided by humans, people must ensure that AI is used correctly and does not cause harm or unfair treatment.

Key practices for responsible and fair use of AI:

- **Use AI transparently:** Be clear about when and how AI is being used.

- **Protect personal data:** Ensure privacy and keep users' information secure.
- **Always verify AI outputs:** Check AI results before making important decisions.
- **Avoid discrimination and misuse:** Use AI fairly and prevent bias or harmful applications.

Application activity 7.3

- 1) What is bias in AI?
- 2) Give one example of unethical use of AI.
- 3) A company uses an AI tool to filter job applications. The AI consistently rejects candidates from a specific geographic area.
 - a) Which ethical concern is being raised here?
 - b) What should the human role be in fixing this?

7.4. Introduction to Machine Learning

Activity 7.4

- 1) How does youtube know which video you like?
- 2) How does a phone recognize your face?

7.4.1. Machine Learning

Machine Learning is a branch of AI that allows computers to learn from data and improve their performance automatically. Unlike traditional programming, where programmers explicitly write instructions for the computer to follow, ML systems learn patterns from data and make predictions or decisions without being explicitly programmed for each task.

Machine Learning allows computers to solve complex problems, analyze large datasets, and adapt to new situations. Examples include email spam filters, recommendation systems, and self-driving cars.

7.4.2.Importance of Machine Learning

Machine Learning has become an essential technology due to its ability to enhance efficiency, accuracy, and personalization across various sectors. Its importance includes:

- **Automation:** ML automates tasks that would otherwise require human intelligence, saving time and effort.
- **Accuracy:** ML systems can analyze large datasets to make precise predictions and decisions.
- **Personalization:** ML powers recommendation systems in youtube, Netflix, and e-commerce platforms by suggesting content which is adapted to the users' preferences.
- **Efficiency:** ML improves performance in healthcare, finance, education, and transportation by supporting quick and informed decisions.
- **Problem solving:** ML is capable of solving complex problems that traditional programming cannot handle easily.

7.4.3.Traditional programming vs Machine Learning

Traditional programming is a way of giving a computer fixed instructions (rules) written by a programmer so that the computer can perform a specific task while with Machine Learning a computer learns from data and improves its performance without being given fixed rules.

Traditional programming	Machine Learning
Programmer provides explicit rules and instructions.	System learns patterns and rules from data.
Output is predictable and fixed.	Output improves as the system learns from new data.
Cannot adapt to new situations without reprogramming.	Can adapt automatically to new situations.
Example: Calculator performing addition.	Example: Email spam filter detecting new spam emails.

7.4.4.Types of Machine Learning

The function of a machine learning system can be descriptive, meaning that the system uses the data to explain what happened; predictive, meaning the system uses the data to predict what will happen; or prescriptive, meaning the system will use the data to make suggestions about what action to take,” the researchers wrote.

There are three subcategories of machine learning which are supervised, unsupervised and reinforced machine learning.

- **Supervised:** Machine learning models are trained with labeled data sets, which allow the models to learn and grow more accurate over time. For example, an algorithm would be trained with pictures of dogs and other things, all labeled by humans, and the machine would learn ways to identify pictures of dogs on its own. Supervised machine learning is the most common type used today.
- **Unsupervised:** Machine learning, a program looks for patterns in unlabeled data. Unsupervised machine learning can find patterns or trends that people aren’t explicitly looking for. For example, an unsupervised machine learning program could look through online sales data and identify different types of clients making purchases.
- **Reinforcement:** Machine learning trains machines through trial and error to take the best action by establishing a reward system. Reinforcement learning can train models to play games or train autonomous vehicles to drive by telling the machine when it made the right decisions, which helps it learn over time what actions it should take.

Application activity 7.4

- 1) State one example of Machine Learning used in daily life.
- 2) Which technology helps youtube recommend videos?
- 3) What is the main difference between Machine Learning and traditional programming?
- 4) Mention one way Machine Learning can help students in learning.

7.5. Data in Machine Learning

Activity 7.5

Holy Wisdom Academy School plans to introduce a smart system to improve learning and administration. This system can predict students' academic performance based on past grades and attendance, suggest topics for which students need more practice, help teachers identify students at risk of falling behind and recommend learning resources tailored to each student's needs.

- 1) How does the system know which student needs help?
- 2) What information does the system use to make predictions?
- 3) Can a machine improve its recommendations over time? Explain.
- 4) How is this system different from a traditional grading system that only uses fixed rules?

7.5.1. Machines learning through data

Data is the foundation of Machine Learning. Machine Learning systems depend on data to learn patterns, make predictions, and improve their performance over time. Unlike traditional programming, where rules are explicitly written by humans, ML systems learn from examples provided in data.

Without data, Machine Learning cannot function, due to the following reasons:

- Data provides examples for learning
- Data helps models identify patterns and relationships
- More and better-quality data usually leads to better accuracy
- ML systems use past data to predict future outcomes

a. Types of data in Machine Learning

Machine Learning uses different types of data depending on the problem to be solved. The main types are structured data, unstructured data, and semi-structured data.

- **Structured data** is well organized and stored in a fixed format, usually in tables with rows and columns. They are characterized by being stored in databases and spreadsheets organized in rows and columns with clearly defined data types. This makes them easy to search, process and analyze.

Examples: Student grades stored in a school database, sales figures in an Excel sheet.

Structured data is commonly used in Machine Learning because it is easy to clean, easy to analyze, and works well with many ML algorithms.

- **Unstructured data** does not follow a predefined format or structure. It is more complex but contains valuable information. They have no fixed format and therefore difficult to store in tables. Unstructured data is very important in areas such as Natural Language Processing (NLP), image and face recognition, speech recognition and video analysis.
- **Semi-structured data** is data that is partially organized. It does not fit neatly into tables but contains tags or markers that provide structure. They are used in web applications, data exchange between systems in API and cloud services.

b. Importance of data in Machine Learning

Machine Learning systems learn by analyzing data, not by following fixed rules written by programmers. Without data, a machine cannot learn, improve, or make intelligent decisions.

The amount and quality of data are very important for machine learning. If there is not enough data or if the data is bad, the machine cannot find the correct patterns, and its results will be wrong and unreliable.

7.5.2. Training and testing data.

Training data: Data used to teach a machine learning model how to perform a task by allowing it to learn patterns and relationships.

Testing data: Data used to evaluate the model's performance and check how accurately it can make predictions on new, unseen data.

Data is essential for training machine learning models, which is the process of teaching a machine how to perform a task. During training, the machine is provided with input data, such as images, numbers, or text, so it can learn from examples and improve its ability to make accurate predictions or decisions.

Training and testing can be done on the following categories of data:

- **Images:** A model can be trained with thousands of pictures of animals to learn how to recognize different species.
- **Numbers:** A model can use past sales data to predict future sales for a store.
- **Text:** A model can be trained on customer reviews to identify whether the feedback is positive or negative.
- **Audio:** A voice assistant can be trained with recordings of speech to understand and respond to spoken commands.

i. Improving accuracy with more and better data

This means making a machine learning system's predictions and decisions more correct and reliable. The accuracy of a model shows how well it can make the right choices based on the data it has learned from. In general, having more data and higher-quality data allows the model to understand patterns better, reduce errors, and perform tasks more effectively. The better the data, the better the system can learn and make accurate predictions.

ii. Making decisions and predictions using data

Machine Learning systems use data to make informed decisions and predictions. These decisions help individuals, businesses, and organizations solve problems and plan for the future.

For example in education, Machine Learning systems can analyze students' attendance, grades, and participation to predict academic performance and suggest interventions.

All these decisions depend on data. Without data, the system would only guess, leading to unreliable outcomes.

iii. Data preparation in Machine Learning

Before data be used in Machine Learning (ML), it must be properly prepared. Data preparation is an important step because the quality and format of data directly affect how well a model can learn and make predictions. The process involves the following steps:

- **Data collection:** The first step is to gather data from different sources. The sources of data include sensors, surveys, databases, websites, or files. The more relevant and diverse the data, the better the model can learn.
- **Data cleaning:** After collection, data must be cleaned. Cleaning involves removing errors, duplicates, missing values, or irrelevant information. Clean data ensures that the machine learning model learns correctly and avoids mistakes caused by poor quality data.
- **Data transformation:** Sometimes data is not in a form that a machine learning model can use. Transformation converts the data into a suitable format. This can include changing text into numbers, scaling numerical values, or encoding categories so the model can process them easily.
- **Data splitting:** Data is split into two main parts namely the training data which is used to teach the model how to perform a task and the testing data which is used to check how accurate and reliable the model is after training.

7.5.3 Machine learning in action

Machines learn in a way similar to students learning from textbooks and examples. By analyzing large amounts of data, machines can recognize patterns and relationships that humans may not notice.

- **Learning from examples:** Machines are provided with data such as images, numbers, or text. By studying these examples, they learn to recognize patterns and make predictions.
- **Improving over time:** The more data a machine sees, the better it becomes at making correct predictions. For example, a model trained on thousands of cat and dog images can identify new images correctly.

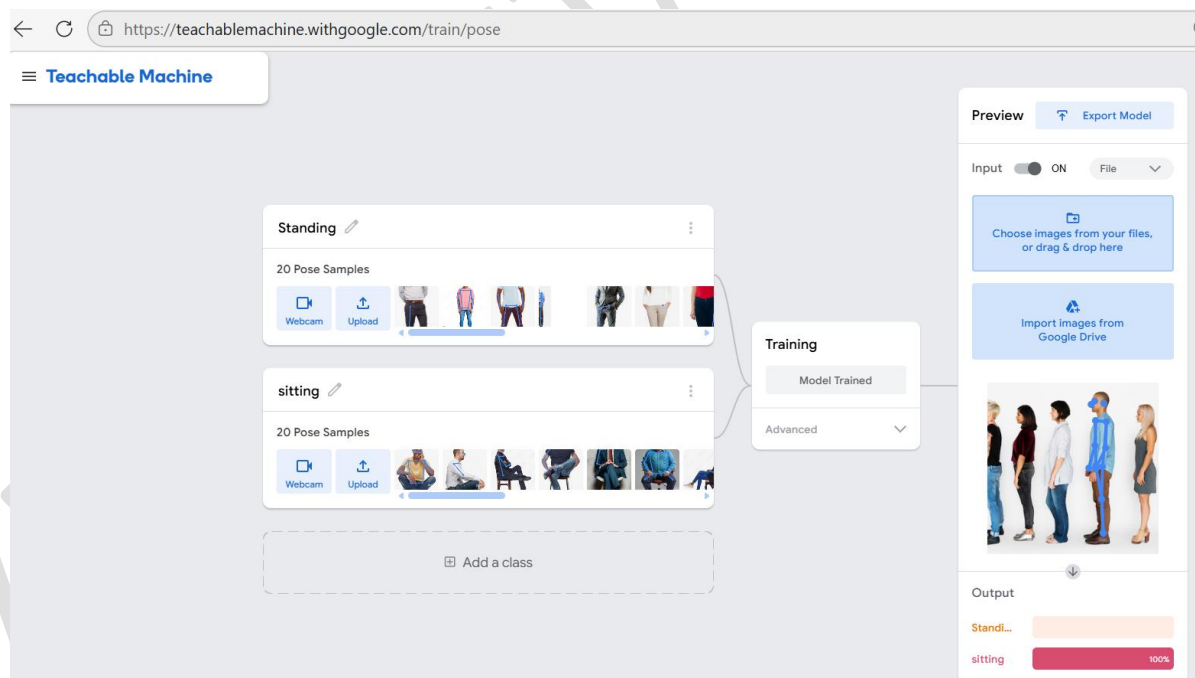
- **Making predictions and decisions:** Once trained, machines can make predictions about new data or provide recommendations. For instance, a music app can suggest songs based on a user's listening history.

a. Google teachable machine

This is a simple online tool that is used to build machine learning models. Such models can be like a model to recognize images, sounds, or poses by giving it examples from which to learn. After learning from these examples the computer can make predictions when it sees something new.

Follow these steps for using the google teachable machine to train a model which recognizes standing and sitting postures:

- Open Google Teachable Machine by going to teachablemachine.withgoogle.com and select Image Project.
- Create classes by making two categories “**Standing**” and “**Sitting**”.
- Collect data by use the webcam to take several pictures or upload images from your computer. Images should be between 20 and 30.
- After uploading the images train the model by clicking on **Train Model**
- Test the model by uploading the images or use the computer webcam to test different postures and see if the computer correctly predicts the posture.



Picture 7. 1. Google teachable machine

- After testing the model export it so that you may use later in an application, website, or device. The model above has recognized successfully that the uploaded image is in a standing position.

Application activity 7.5

- 1) Explain why data is important for machine learning systems.
- 2) Name the three main types of data used in machine learning.
- 3) Differentiate between training data and testing data.
- 4) Give two examples of how machines learn from data in daily life.
- 5) Explain how more and better-quality data improves the accuracy of a machine learning system.

7.6. Impact of Machine Learning

Activity 7.6

A school has recently introduced a Machine Learning (ML) system to improve how it operates and supports learning. The system can:

- Analyze student marks, attendance, and participation to predict performance and suggest personalized exercises
 - Help teachers grade assignments automatically, saving time and reducing errors
 - Identify students who are struggling and provide extra support
 - Improve school safety by monitoring entrances using face recognition
 - Help students learn faster using smart devices and voice assistants
- 1) How ML system can improve learning outcomes for students?
 - 2) How could ML help teachers make better decisions?
 - 3) In what ways does ML make the school safer?
 - 4) How can ML systems impact hospitals, businesses, or farms?

The impact of Machine Learning refers to the effects, changes, and influence that Machine Learning has on individuals, organizations, and society as a whole.

Machine Learning is applied in many different fields each benefitting from ML in unique ways depending on the type of data used and the problems being solved. Below are areas where ML is applied.

7.6.1. Education

In education, Machine Learning is used to improve teaching and learning processes. Online learning platforms use ML to analyze students' performance and learning behavior. Based on this analysis, the system can recommend suitable learning materials, exercises, or revision topics for each student. Machine Learning also supports:

- Automatic grading of quizzes and assignments,
- Personalized learning paths for students,
- Detection of students who are struggling and need academic support,
- Intelligent tutoring systems that provide instant feedback.

7.6.2. Healthcare

ML systems analyze medical data such as patient records, laboratory results, and medical images to support doctors and healthcare professionals.

Applications of ML in healthcare include:

- Disease diagnosis and prediction
- Analysis of X-rays, CT scans, and MRI images
- Predicting patient recovery or risk of complications
- Drug discovery and development
- Managing hospital operations and patient records

7.6.3. Business and finance

Businesses and financial institutions use ML to improve decision-making and efficiency. ML systems analyze customer data, transaction records, and market trends to provide useful insights. By using ML, businesses can reduce risks, increase profits, and provide better services to customers.

Examples include fraud detection in banking and online payments, credit scoring and loan approval systems, customer behavior analysis and market prediction and personalized advertising and marketing.

7.6.4. Agriculture

Agriculture has greatly benefited from Machine Learning, especially in improving food production and resource management. ML systems analyze data from weather forecasts, soil sensors, and satellite images.

Machine Learning is used to predict weather and climate patterns, detect crop diseases and pests, estimate crop yield and optimize irrigation and fertilizer use

7.6.5. Transportation

Transportation systems rely on Machine Learning to improve safety, efficiency, and planning. ML is used in traffic prediction, navigation systems such as Google Maps, research and development of self-driving vehicles and in accident detection and prevention systems

7.6.6. Security and surveillance

Security systems use Machine Learning to detect threats and protect people and information. ML can analyze large amounts of data in real time to identify unusual or suspicious activities.

Applications include:

- Face recognition and fingerprint identification
- Video surveillance and monitoring systems
- Cybersecurity and detection of hacking attempts
- Spam and malware detection
- Identity verification systems

7.6.7. Smartphones

Smartphones use Machine Learning in many features such as face unlock and fingerprint recognition, voice to text typing, photo organization and enhancement. These features make smartphones smarter, more secure and easier to use.

Application activity 7.6

- 1) Explain the impact of ML on individuals, organizations and society.
- 2) List four areas where ML is applied.
- 3) How does ML improve teaching and learning in schools? Give two examples.
- 4) How does ML help in business?

End of unit assessment 7

- 1) State two characteristics of AI systems.
- 2) Explain the role of data in machine learning.
- 3) Identify three types of AI based on capabilities.
- 4) Classify the following examples as supervised, unsupervised, or reinforcement learning:
 - a) Predicting student marks using past results
 - b) Grouping customers based on buying behavior
 - c) Training a robot to avoid obstacles using rewards
- 5) Give a situation where AI gives incorrect results and explain two actions humans should take to correct the system.
- 6) Using Google Teachable Machine create a ML model that can recognize two different objects.
- 7) Explain why responsible use of AI is important in society.
- 8) How can bias in AI systems affect individuals or communities?
- 9) Explain why humans should monitor AI decisions instead of allowing AI to work alone.